



**Generation Interconnection
System Impact Study Report
for
Queue Project AF1-222
OCEANVIEW WIND 2 230 KV
140.25 MW Capacity / 510 MW Energy**

August 2020

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1 Introduction

This System Impact Study has been prepared in accordance with the PJM Open Access Transmission Tariff, 205, as well as the System Impact Study Agreement between the Interconnection Customer (IC), and PJM Interconnection, LLC (PJM), Transmission Provider (TP). The Interconnected Transmission Owner (ITO) is Jersey Central Power & Light Company (JCPL).

2 Preface

The intent of the System Impact Study is to determine a plan, with approximate cost and construction time estimates, to connect the subject generation interconnection project to the PJM network at a location specified by the Interconnection Customer. As a requirement for interconnection, the Interconnection Customer may be responsible for the cost of constructing: Network Upgrades, which are facility additions, or upgrades to existing facilities, that are needed to maintain the reliability of the PJM system. All facilities required for interconnection of a generation interconnection project must be designed to meet the technical specifications (on PJM web site) for the appropriate transmission owner.

In some instances an Interconnection Customer may not be responsible for 100% of the identified network upgrade cost because other transmission network uses, e.g. another generation interconnection or merchant transmission upgrade, may also contribute to the need for the same network reinforcement. The possibility of sharing the reinforcement costs with other projects may be identified in the Feasibility Study, but the actual allocation will be deferred until the System Impact Study is performed.

The System Impact Study estimates do not include the feasibility, cost, or time required to obtain property rights and permits for construction of the required facilities. The project developer is responsible for the right of way, real estate, and construction permit issues. For properties currently owned by Transmission Owners, the costs may be included in the study.

The Interconnection Customer seeking to interconnect a wind or solar generation facility shall maintain meteorological data facilities as well as provide that meteorological data which is required per Schedule H to the Interconnection Service Agreement and Section 8 of Manual 14D.

An Interconnection Customer with a proposed new Customer Facility that has a Maximum Facility Output equal to or greater than 100 MW shall install and maintain, at its expense, phasor measurement units (PMUs). See Section 8.5.3 of Appendix 2 to the Interconnection Service Agreement as well as section 4.3 of PJM Manual 14D for additional information.

3 General

The Interconnection Customer (IC) has proposed an uprate to a planned/existing Offshore Wind generating facility located in Monmouth, New Jersey. This project is an increase to the Interconnection Customer's AE1-238 project, which will share the same point of interconnection. The AF1-222 queue position is a 510 MW uprate (140.25 MW Capacity uprate) to the previous project. The total installed facilities will have a capability of 1326 MW with 365.25 MW of this output being recognized by PJM as Capacity. The proposed in-service date for this uprate project is December 30, 2025. This study does not imply a TO commitment to this in-service date.

Queue Number	AF1-222
Project Name	OCEANVIEW WIND 2 230 KV
State	New Jersey
County	Monmouth
Transmission Owner	JCPL
MFO	1326
MWE	510
MWC	140.25
Fuel	Offshore Wind
Basecase Study Year	2023

Any new service customers who can feasibly be commercially operable prior to June 1st of the basecase study year are required to request interim deliverability analysis.

4 Point of Interconnection

The interconnection of the project at the Primary POI will be accomplished by utilizing the AE1-238 direct connection to the Oceanview 230 kV Substation in the JCPL area. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct the associated Attachment facilities. The project will also require non-direct connection upgrades at Oceanview Substation.

Attachment 1 shows a one-line diagram of the proposed primary direct connection facilities for the AF1-222 generation project to connect to the FirstEnergy ("FE") transmission system. IC will be responsible for constructing all of the facilities on its side of the POI, including the Attachment facilities which connect the generator to the FE transmission system.

5 Cost Summary

The AF1-222 project will be responsible for the following costs:

Description	Total Cost
Total Physical Interconnection Costs	\$141,500
Allocation towards System Network Upgrade Costs* (Summer Peak)	\$130,224,243
Allocation towards System Network Upgrade Costs* (Light Load)	\$1,190,924
Total Costs	\$131,556,667 ¹

*As your project progresses through the study process and other projects modify their request or withdraw, then your cost allocation could change.

This cost excludes a Federal Income Tax Gross Up charges. This tax may or may not be charged based on whether this project meets the eligibility requirements of IRS Notice 88-129. If at a future date it is determined that the Federal Income Tax Gross charge is required, the Transmission Owner shall be reimbursed by the Interconnection Customer for such taxes.

Note 1: PJM Open Access Transmission Tariff (OATT) section 217.3A outline cost allocation rules. The rules are further clarified in PJM Manual 14A Attachment B. The allocation of costs for a network upgrade will start with the first Queue project to cause the need for the upgrade. Later queue projects will receive cost allocation contingent on their contribution to the violation and are allocated to the queues that have not

¹ This project currently causes and contributes to overloads of the Transmission System (see Summer Peak and Light Load Load Flow Analysis section below) and therefore has potential to have cost allocation for the system reinforcements listed in the report. The results may vary with queue customers withdrawing from the queue and other generators deactivating over time. If a customer is the first to cause the need for a project (causes loading to exceed 100% of rating), then the customer is responsible. If a customer's MW contribution to a facility that is already overloaded by a prior queue is less than 5 MW, then they will likely not receive cost allocation (See PJM Manual 14A, Attachment B for Cost Allocation Rules).

closed less than 5 years following the execution of the first Interconnection Service Agreement which identifies the need for this upgrade.

Note 2: For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

6 Transmission Owner Scope of Work

The interconnection of the project at the Primary POI will be accomplished by utilizing the AE1-238 direct connection to the Oceanview 230 kV Substation in the JCPL area. The IC will be responsible for acquiring all easements, properties, and permits that may be required to construct the associated Attachment facilities. The project will also require non-direct connection upgrades at Oceanview Substation.

The total physical interconnection costs is given in the table below:

Description	Total Cost
Relay setting changes on Equinor Wind US LLC line terminal at Oceanview Substation	\$107,800
Project Management	\$33,700
Total Physical Interconnection Costs	\$141,500

7 Schedule

Based on the scope of work for the interconnection facilities, it is expected to take a minimum of **6 months** after the signing of an Interconnection Construction Service Agreement and construction kickoff call to complete the installation. This assumes that there will be no environmental issues with any of the new properties associated with this project, that there will be no delays in acquiring the necessary permits for implementing the defined work and that all system outages will be allowed when requested.

The schedule for any required Network Impact Reinforcements will be more clearly identified in future study phases. The estimated time to complete each of the required reinforcements is identified in the “System Reinforcements” section of the report.

8 Transmission Owner Analysis

FE performed an analysis of its underlying transmission <100 kV system. The AF1-222 project did not contribute to any overloads on the FE transmission <100 kV system.

9 Interconnection Customer Requirements

9.1 System Protection

The IC must design its Customer Facilities in accordance with all applicable standards, including the standards in FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

Preliminary Protection requirements will be provided as part of the Facilities Study. Detailed Protection Requirements will be provided once the project enters the construction phase.

The IC has requested a non-standard GSU transformer winding configuration. This transformer is in violation of section 14.2.6 of FE’s “Requirements for Transmission Connected Facilities” document and will not be accepted. The GSU transformer must have a grounded wye connection on the high (utility) side and a delta connection on the low (generator) side.

9.2 Compliance Issues and Interconnection Customer Requirements

The proposed Customer Facilities must be designed in accordance with FE’s “Requirements for Transmission Connected Facilities” document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>. In particular, the IC is responsible for the following:

1. The purchase and installation of a fully rated 230 kV circuit breaker to protect the AF1-222 generator lead line. A single circuit breaker must be used to protect this line; if the project has several GSU transformers, the individual GSU transformer breakers cannot be used to protect this line.

2. The purchase and installation of the minimum required FE generation interconnection relaying and control facilities. This includes over/under voltage protection, over/under frequency protection, and zero sequence voltage protection relays.
3. The purchase and installation of supervisory control and data acquisition (“SCADA”) equipment to provide information in a compatible format to the FE Transmission System Control Center.
4. Compliance with the FE and PJM generator power factor and voltage control requirements.
5. The execution of a back-up service agreement to serve the customer load supplied from the AF1-222 generation project metering point when the units are out-of-service. This assumes the intent of the IC is to net the generation with the load.

The IC will also be required to meet all PJM, ReliabilityFirst, and NERC reliability criteria and operating procedures for standards compliance. For example, the IC will need to properly locate and report the over and under voltage and over and under frequency system protection elements for its units as well as the submission of the generator model and protection data required to satisfy the PJM and ReliabilityFirst audits. Failure to comply with these requirements may result in a disconnection of service if the violation is found to compromise the reliability of the FE system.

9.3 Power Factor Requirements

The IC shall design its Wind or non-synchronous Customer Facility with the ability to maintain a power factor of at least 0.95 leading (absorbing VARs) to 0.95 lagging (supplying VARs) measured at the high-side of the facility substation transformer(s) connected to the FE transmission system.

10 Revenue Metering and SCADA Requirements

10.1 PJM Requirements

The Interconnection Customer will be required to install equipment necessary to provide Revenue Metering (KWH, KVARH) and real time data (KW, KVAR) for IC's generating Resource. See PJM Manuals M-01 and M-14D, and PJM Tariff Section 8 of Attachment O.

10.2 Interconnected Transmission Owner Requirements

The IC will be required to comply with all FE revenue metering requirements for generation interconnection customers which can be found in FE's "Requirements for Transmission Connected Facilities" document located at: <http://www.pjm.com/planning/design-engineering/to-tech-standards/private-firstenergy.aspx>.

11 Summer Peak Analysis

The Queue Project AF1-222 was evaluated as a 510.0 MW (Capacity 140.2 MW) injection as an uprate to AE1-238 at the Oceanview 230 kV substation in the JCPL area. Project AF1-222 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-222 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

11.1 Generation Deliverability

(Single or N-1 contingencies for the Capacity portion only of the interconnection)

None

11.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies for the full energy output)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPACT
42768576	206294	28LARRABEE	230.0	JCP&L	206318	28VANHISV L	230.0	JCP&L	1	JC-P7-1-JCC-230-11A	tower	869.0	97.4	106.39	AC	79.53
42237121	206300	28OCEANV W	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	Y	JC-P2-3-JCC-230-47B	breaker	869.0	65.1	120.02	AC	490.69
42237122	206300	28OCEANV W	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	Y	JC-P2-3-JCC-230-47C	breaker	869.0	81.79	119.8	AC	339.92
42237180	206300	28OCEANV W	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	X	JC-P2-3-JCC-230-47E	breaker	869.0	81.58	119.58	AC	339.69
42237181	206300	28OCEANV W	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	X	JC-P2-3-JCC-230-47D	breaker	869.0	80.27	118.19	AC	339.13
42768372	206300	28OCEANV W	230.0	JCP&L	206294	28LARRABEE	230.0	JCP&L	1	JC-P7-1-JCC-230-15	tower	869.0	60.93	112.22	AC	455.72
42237457	206411	28R11RING A	230.0	JCP&L	206410	28R11RING B	230.0	JCP&L	1	JC-P2-3-JCC-230-85	breaker	999.0	88.91	100.68	AC	117.94

11.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC D C	MW IMPACT
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ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJE C T LOADIN G %	POST PROJE C T LOADIN G %	AC D C	MW IMPAC T
42236940	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	2	JC-P2-3-JCC-230-15F	breaker	813.0	149.95	161.05	AC	105.7
42236941	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	2	JC-P2-3-JCC-230-13A	breaker	813.0	148.67	159.84	AC	106.38
42236945	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	1	JC-P2-3-JCC-230-15G	breaker	817.0	149.22	160.26	AC	105.7
42236946	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	1	JC-P2-3-JCC-230-15A	breaker	817.0	145.16	156.28	AC	106.39
41034227	206302	28OYSTER C	230.0	JCP&L	227955	CEDAR	230.0	AE	1	PS_P7-1_1LINE+2LIN E	tower	564.0	130.3	138.28	AC	54.02
41632856	206302	28OYSTER C	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P2-3-JCC-230-15E	breaker	564.0	119.45	126.55	AC	48.11
41632857	206302	28OYSTER C	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P2-3-JCC-230-17A	breaker	564.0	119.29	126.4	AC	48.16
49863000	206302	28OYSTER C	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P7-1-JCC-230-7A	tower	564.0	134.26	142.72	AC	57.28
41632989	206305	28RAR RVR	230.0	JCP&L	218332	KILMER_W	230.0	PSE&G	1	PS_P2-3_LNEPS_P1-2_1_LT	breaker	817.0	131.56	142.43	AC	102.12
42768396	206309	28SMITHBR G	230.0	JCP&L	206326	28E WINDSR	230.0	JCP&L	1	PS_P7-1_1LINE+2LIN E	tower	1394.0	103.85	119.49	AC	220.05
42768397	206309	28SMITHBR G	230.0	JCP&L	206326	28E WINDSR	230.0	JCP&L	1	PS_P7-1_1I023+GBK-LN_LT	tower	1394.0	100.43	116.05	AC	220.02
42236930	206314	28RED OAKA	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P2-3-JCC-230-26B	breaker	869.0	164.83	182.26	AC	151.97
42236931	206314	28RED OAKA	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P2-3-JCC-230-026A	breaker	869.0	151.74	169.16	AC	152.34
42236925	206315	28RED OAKB	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P2-3-JCC-230-26D	breaker	869.0	168.48	185.84	AC	151.26
42236926	206315	28RED OAKB	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P2-3-JCC-230-026C	breaker	869.0	155.42	172.77	AC	151.62
41034357	206316	28WINDSO R	230.0	JCP&L	219752	CLRKSVLL_1	230.0	PSE&G	1	PS_P7-1_1LINE+2LIN E	tower	813.0	126.48	136.74	AC	97.95
41034358	206316	28WINDSO R	230.0	JCP&L	219752	CLRKSVLL_1	230.0	PSE&G	1	PS_P7-1_1I023+GBK-LN_LT	tower	813.0	123.67	133.92	AC	97.86
41633249	206316	28WINDSO R	230.0	JCP&L	219752	CLRKSVLL_1	230.0	PSE&G	1	JC-P2-3-JCC-230-80	breaker	813.0	102.17	112.54	AC	86.36
42237333	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSO R	230.0	JCP&L	1	JC-P2-3-JCC-230-80	breaker	869.0	105.4	114.73	AC	82.67
42768367	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSO R	230.0	JCP&L	1	PS_P7-1_1LINE+2LIN E	tower	869.0	126.0	135.89	AC	87.12
42768368	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSO R	230.0	JCP&L	1	PS_P7-1_1I023+GBK-LN_LT	tower	869.0	123.55	133.41	AC	87.03
46869219	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSO R	230.0	JCP&L	1	PJM500_PS_P2-3_DEAN5_5-6	breaker	869.0	103.02	110.47	AC	65.95
46869220	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSO R	230.0	JCP&L	1	PJM_P4_P484B	breaker	869.0	103.01	110.46	AC	65.95
42237050	206410	28R11RING B	230.0	JCP&L	206315	28RED OAKB	230.0	JCP&L	1	JC-P2-3-JCC-230-26D	breaker	869.0	124.97	142.46	AC	152.36
42237051	206410	28R11RING B	230.0	JCP&L	206315	28RED OAKB	230.0	JCP&L	1	JC-P2-3-JCC-230-026C	breaker	869.0	112.64	130.15	AC	152.72
42237041	206411	28R11RING A	230.0	JCP&L	206314	28RED OAKA	230.0	JCP&L	1	JC-P2-3-JCC-230-26B	breaker	869.0	129.08	146.65	AC	152.89
42237042	206411	28R11RING A	230.0	JCP&L	206314	28RED OAKA	230.0	JCP&L	1	JC-P2-3-JCC-230-026A	breaker	869.0	116.4	134.0	AC	153.26
42768447	219104	CLRKSVLL_2	230.0	PSE&G	217150	LAWRENCE	230.0	PSE&G	1	PS_P7-1_1LINE+2LIN E	tower	869.0	110.18	119.76	AC	97.95

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS LOADIN G %	POST PROJECTIONS LOADIN G %	AC/D C	MW IMPACT
42768448	219104	CLRKSVLL_2	230.0	PSE&G	217150	LAWRENCE	230.0	PSE&G	1	PS_P7-1_11023+GBK-LN_LT	tower	869.0	107.88	117.43	AC	97.86

11.4 Steady-State Voltage Requirements

To be determined

11.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS LOADIN G %	POST PROJECTIONS LOADIN G %	AC/D C	MW IMPACT
42541346	206294	28LARRABEE	230.0	JCP&L	206309	28SMITHBRG	230.0	JCP&L	2	JC-P1-2-JCC-230-017	operation	813.0	147.97	159.08	AC	105.71
42541347	206294	28LARRABEE	230.0	JCP&L	206309	28SMITHBRG	230.0	JCP&L	2	Base Case	operation	678.0	116.6	125.31	AC	69.12
42541472	206294	28LARRABEE	230.0	JCP&L	206309	28SMITHBRG	230.0	JCP&L	1	JC-P1-2-JCC-230-018	operation	817.0	134.89	144.43	AC	91.34
42541473	206294	28LARRABEE	230.0	JCP&L	206309	28SMITHBRG	230.0	JCP&L	1	Base Case	operation	650.0	121.62	130.7	AC	69.12
42541756	206300	28OCEANVW	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	Y	JC-P1-2-JCC-230-003	operation	869.0	75.13	112.77	AC	336.61
42541757	206300	28OCEANVW	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	X	JC-P1-2-JCC-230-004	operation	869.0	75.13	112.77	AC	336.61
41314210	206302	28OYSTER C	230.0	JCP&L	227955	CEDAR	230.0	AE	1	Base Case	operation	464.0	128.82	136.04	AC	40.2
41314211	206302	28OYSTER C	230.0	JCP&L	227955	CEDAR	230.0	AE	1	JC-P1-2-JCC-230-031	operation	564.0	118.69	125.79	AC	48.09
41314275	206305	28RAR RVR	230.0	JCP&L	218331	KILMER_I	230.0	PSE&G	1	PS_P1-2_#2LINE	operation	869.0	147.89	161.73	AC	138.2
41314280	206305	28RAR RVR	230.0	JCP&L	218331	KILMER_I	230.0	PSE&G	1	Base Case	operation	709.0	125.1	137.85	AC	103.94
41314450	206305	28RAR RVR	230.0	JCP&L	218332	KILMER_W	230.0	PSE&G	1	PS_P1-2_#1LINE	operation	817.0	130.69	141.62	AC	102.57
41314455	206305	28RAR RVR	230.0	JCP&L	218332	KILMER_W	230.0	PSE&G	1	Base Case	operation	650.0	101.86	109.55	AC	57.6
42541408	206314	28RED OAKA	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P1-2-JCC-230-027	operation	869.0	146.28	163.69	AC	152.25
42541409	206314	28RED OAKA	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	Base Case	operation	709.0	116.92	128.47	AC	81.8
42541372	206315	28RED OAKB	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P1-2-JCC-230-028	operation	869.0	149.99	167.34	AC	151.53
42541373	206315	28RED OAKB	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	Base Case	operation	709.0	105.53	116.66	AC	78.59

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS LOADIN G %	POST PROJECTIONS LOADIN G %	AC/D C	MW IMPACT
41314677	206316	28WINDSOR	230.0	JCP&L	219752	CLRKSPLL_1	230.0	PSE&G	1	JC-P1-2-JCC-230-042T	operation	813.0	102.49	111.63	AC	76.14
41314680	206316	28WINDSOR	230.0	JCP&L	219752	CLRKSPLL_1	230.0	PSE&G	1	Base Case	operation	678.0	101.06	110.07	AC	62.87
42541854	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSOR	230.0	JCP&L	1	Base Case	operation	706.0	108.49	116.5	AC	58.0
49598742	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSOR	230.0	JCP&L	1	PJM500_PS_P1-2_5022	operation	869.0	101.66	109.1	AC	65.91
42541727	206410	28R11RINGB	230.0	JCP&L	206315	28RED OAKB	230.0	JCP&L	1	JC-P1-2-JCC-230-028	operation	869.0	107.88	125.37	AC	152.63
42541679	206411	28R11RINGA	230.0	JCP&L	206314	28RED OAKA	230.0	JCP&L	1	JC-P1-2-JCC-230-027	operation	869.0	111.79	129.37	AC	153.17
41314953	218313	GREENBK_W	230.0	PSE&G	206239	28GILLET W	230.0	JCP&L	1	JC-P1-2-JCC-230-042T	operation	802.0	94.25	103.32	AC	83.68
42541450	218331	KILMER_I	230.0	PSE&G	218333	LNELSN_I	230.0	PSE&G	1	JC-P1-2-JCC-230-043T	operation	805.0	152.76	167.7	AC	138.2
42541455	218331	KILMER_I	230.0	PSE&G	218333	LNELSN_I	230.0	PSE&G	1	Base Case	operation	650.0	131.47	145.32	AC	103.94
42541477	218332	KILMER_W	230.0	PSE&G	218334	LNELSN_W	230.0	PSE&G	1	PS_P1-2_I-1023	operation	679.0	148.74	161.83	AC	102.12
42541482	218332	KILMER_W	230.0	PSE&G	218334	LNELSN_W	230.0	PSE&G	1	Base Case	operation	523.0	117.67	127.22	AC	57.6
42541609	218333	LNELSN_I	230.0	PSE&G	218301	MIDDLESEX_I	230.0	PSE&G	1	JC-P1-2-JCC-230-043T	operation	819.0	130.1	143.69	AC	128.34
42541614	218333	LNELSN_I	230.0	PSE&G	218301	MIDDLESEX_I	230.0	PSE&G	1	Base Case	operation	709.0	105.65	106.11	AC	98.43
42541840	218334	LNELSN_W	230.0	PSE&G	218522	MIDDLESEX_W	230.0	PSE&G	1	JC-P1-2-JCC-230-042T	operation	887.0	110.53	120.57	AC	102.35
42541834	218522	MIDDLESEX_W	230.0	PSE&G	218313	GREENBK_W	230.0	PSE&G	1	JC-P1-2-JCC-230-042T	operation	887.0	110.53	120.57	AC	102.35

11.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number
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ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																			
410342 27,416 32857, 416328 56,498 63000	9	28OYSTER C 230.0 kV - CEDAR 230.0 kV Ckt 1	JCPL: Reconductor the Oyster Creek - Cedar 230 kV line (JCP&L portion only ~0.1 miles. AE portion ~14 miles) Upgrade terminal equipment at Oyster Creek Project Type : FAC Cost : \$469,866 New Ratings : 913.0/1147.0/1147.0 Time Estimate : 12 Months Queue Project AF1-222 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-222 could receive cost allocation. Note 2: Although Queue Project AF1-222 may not have cost responsibility for this upgrade, Queue Project AF1-222 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-222 comes into service prior to completion of the upgrade, Queue Project AF1-222 will need an interim study. AEC: To mitigate the (ACE) Cedar Oyster Creek 230 kV line (from bus 227955 to bus 206302 ckt 1) overload, it will require increasing the emergency rating of the Cedar to Oyster Creek 230 kV line by rebuilding the circuit. The rebuild will include the installation of new poles, foundations, insulators, and conductor. In addition, various terminal reinforcements are required at Cedar. Project Type : FAC Cost : \$27,000,000 New Ratings : 1195/1195/1195 Time Estimate : 36-60 Months The Cost allocation table is below: <table> <tr> <th>Queue</th><th>MW contribution</th><th>% of Cost</th><th>Cost</th><th>SP/L</th><th>Contingency</th><th>Contingency Type</th></tr> <tr> <td>AE2-000</td><td>123.84</td><td>0.17</td><td>\$ 4,582,769</td><td>SP</td><td>JC-P7-1-JCC-230-13</td><td>Tower</td></tr> <tr> <td>AF1-101</td><td>531.89</td><td>0.73</td><td>\$ 19,682,890</td><td>SP</td><td>JC-P7-1-JCC-230-13</td><td>Tower</td></tr> <tr> <td>AF1-126</td><td>16.61</td><td>0.02</td><td>\$ 614,662</td><td>SP</td><td>JC-P7-1-JCC-230-10A</td><td>Tower</td></tr> <tr> <td>AF1-222</td><td>57.28</td><td>0.08</td><td>\$ 2,119,679</td><td>SP</td><td>JC-P7-1-JCC-230-7A</td><td>Tower</td></tr> </table>	Queue	MW contribution	% of Cost	Cost	SP/L	Contingency	Contingency Type	AE2-000	123.84	0.17	\$ 4,582,769	SP	JC-P7-1-JCC-230-13	Tower	AF1-101	531.89	0.73	\$ 19,682,890	SP	JC-P7-1-JCC-230-13	Tower	AF1-126	16.61	0.02	\$ 614,662	SP	JC-P7-1-JCC-230-10A	Tower	AF1-222	57.28	0.08	\$ 2,119,679	SP	JC-P7-1-JCC-230-7A	Tower	\$27,469,866	\$2,119,679	N6587 & N6728
Queue	MW contribution	% of Cost	Cost	SP/L	Contingency	Contingency Type																																			
AE2-000	123.84	0.17	\$ 4,582,769	SP	JC-P7-1-JCC-230-13	Tower																																			
AF1-101	531.89	0.73	\$ 19,682,890	SP	JC-P7-1-JCC-230-13	Tower																																			
AF1-126	16.61	0.02	\$ 614,662	SP	JC-P7-1-JCC-230-10A	Tower																																			
AF1-222	57.28	0.08	\$ 2,119,679	SP	JC-P7-1-JCC-230-7A	Tower																																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number
422374 57	5	28R11RINGA 230.0 kV - 28R11RINGB 230.0 kV Ckt 1	JCPL: Reconductor e R11 Ring Bus A to R11 Ring Bus B 230 kV transmission line. Upgrade terminal equipment as required. Project Type : FAC New ratings : 1156.0/1334.0/1334.0 Cost : \$1,342,475 Time Estimate : 12 Months AF1-222 is the driver for this upgrade	\$1,342,475	\$1,342,475	N6751

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																																																																				
422369 45,422 36946	8	28LARRABEE 230.0 kV - 28SMITHBRG 230.0 kV Ckt 1	JCPL Upgrade no.1: "Replace substation conductor at Larrabee Reconductor the Larrabee - Smithburg 230 kV line (~11.9 miles). Replace wave trap at Larrabee" Project Type : FAC Cost : \$56,249,703 New Ratings : 913/1147/1147 Time Estimate : 54 Months The cost allocation table is below: <table> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> <tr> <td>AE2-025</td><td>125.2</td><td>17.32%</td><td>\$ 9,744,656</td><td>breaker</td><td>SP</td><td>'JC-P2-3-JCC-230-13G'</td></tr> <tr> <td>AE2-222</td><td>27.5</td><td>3.81%</td><td>\$ 2,140,400</td><td>breaker</td><td>SP</td><td>'JC-P2-3-JCC-230-13G'</td></tr> <tr> <td>AE2-232</td><td>120.1</td><td>16.62%</td><td>\$ 9,347,709</td><td>breaker</td><td>SP</td><td>'JC-P2-3-JCC-230-13G'</td></tr> <tr> <td>AE2-251</td><td>119</td><td>16.47%</td><td>\$ 9,262,093</td><td>breaker</td><td>SP</td><td>'JC-P2-3-JCC-230-13G'</td></tr> <tr> <td>AF1-101</td><td>173.05</td><td>23.94%</td><td>\$ 13,468,951</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15G</td></tr> <tr> <td>AF1-126</td><td>52.15</td><td>7.22%</td><td>\$ 4,058,976</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15G</td></tr> <tr> <td>AF1-222</td><td>105.7</td><td>14.63%</td><td>\$ 8,226,918</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15G</td></tr> </table> Upgrade no.2: Replace disconnect switch at Larrabee substation. Project Type : FAC Cost : \$234,933 New Ratings : 1222.0/1385.0/1385.0 Time Estimate : 12 Months The cost allocation table is below: <table> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> <tr> <td>AF1-101</td><td>173.05</td><td>52.30%</td><td>\$ 122,862</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15G</td></tr> <tr> <td>AF1-126</td><td>52.15</td><td>15.76%</td><td>\$ 37,026</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15G</td></tr> <tr> <td>AF1-222</td><td>105.7</td><td>31.94%</td><td>\$ 75,045</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15G</td></tr> </table>	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AE2-025	125.2	17.32%	\$ 9,744,656	breaker	SP	'JC-P2-3-JCC-230-13G'	AE2-222	27.5	3.81%	\$ 2,140,400	breaker	SP	'JC-P2-3-JCC-230-13G'	AE2-232	120.1	16.62%	\$ 9,347,709	breaker	SP	'JC-P2-3-JCC-230-13G'	AE2-251	119	16.47%	\$ 9,262,093	breaker	SP	'JC-P2-3-JCC-230-13G'	AF1-101	173.05	23.94%	\$ 13,468,951	breaker	SP	JC-P2-3-JCC-230-15G	AF1-126	52.15	7.22%	\$ 4,058,976	breaker	SP	JC-P2-3-JCC-230-15G	AF1-222	105.7	14.63%	\$ 8,226,918	breaker	SP	JC-P2-3-JCC-230-15G	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AF1-101	173.05	52.30%	\$ 122,862	breaker	SP	JC-P2-3-JCC-230-15G	AF1-126	52.15	15.76%	\$ 37,026	breaker	SP	JC-P2-3-JCC-230-15G	AF1-222	105.7	31.94%	\$ 75,045	breaker	SP	JC-P2-3-JCC-230-15G	\$56,484,636	\$8,301,963	N6487 & N6488
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																																																																				
AE2-025	125.2	17.32%	\$ 9,744,656	breaker	SP	'JC-P2-3-JCC-230-13G'																																																																																				
AE2-222	27.5	3.81%	\$ 2,140,400	breaker	SP	'JC-P2-3-JCC-230-13G'																																																																																				
AE2-232	120.1	16.62%	\$ 9,347,709	breaker	SP	'JC-P2-3-JCC-230-13G'																																																																																				
AE2-251	119	16.47%	\$ 9,262,093	breaker	SP	'JC-P2-3-JCC-230-13G'																																																																																				
AF1-101	173.05	23.94%	\$ 13,468,951	breaker	SP	JC-P2-3-JCC-230-15G																																																																																				
AF1-126	52.15	7.22%	\$ 4,058,976	breaker	SP	JC-P2-3-JCC-230-15G																																																																																				
AF1-222	105.7	14.63%	\$ 8,226,918	breaker	SP	JC-P2-3-JCC-230-15G																																																																																				
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																																																																				
AF1-101	173.05	52.30%	\$ 122,862	breaker	SP	JC-P2-3-JCC-230-15G																																																																																				
AF1-126	52.15	15.76%	\$ 37,026	breaker	SP	JC-P2-3-JCC-230-15G																																																																																				
AF1-222	105.7	31.94%	\$ 75,045	breaker	SP	JC-P2-3-JCC-230-15G																																																																																				

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																			
410343 58,410 34357, 416332 49	14	28WINDSOR 230.0 kV - CLRKSVLL_1 230.0 kV Ckt 1	JCPL: Replace substation conductor at Windsor. Replace current transformers at Windsor. Reconductor the Clariksville (PS) - Windsor 230 kV transmission line (~7.45 miles). Replace wave trap at Windsor Replace circuit breaker at Windsor Replace disconnect switches at Windsor. Replace relaying, metering, and fault switch at Windsor". Project Type : FAC Cost : \$38,495,471 New Ratings : 1140.0/1334.0/1334.0 Time Estimate : 42 Months The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AE2 Queue</td><td>129.82</td><td>38.31%</td><td>\$ 14,746,620</td><td>tower</td><td>SP</td><td>PS_P7-1_1LINE+2 LINE</td></tr> <tr> <td>AF1-101</td><td>86.29</td><td>25.46%</td><td>\$ 9,801,924</td><td>tower</td><td>SP</td><td>PS_P7-1_1LINE+2 LINE</td></tr> <tr> <td>AF1-126</td><td>24.83</td><td>7.33%</td><td>\$ 2,820,510</td><td>tower</td><td>SP</td><td>PS_P7-1_1LINE+2 LINE</td></tr> <tr> <td>AF1-222</td><td>97.95</td><td>28.90%</td><td>\$ 11,126,417</td><td>tower</td><td>SP</td><td>PS_P7-1_1LINE+2 LINE</td></tr> </tbody> </table> PSEG : No upgrade is required.	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AE2 Queue	129.82	38.31%	\$ 14,746,620	tower	SP	PS_P7-1_1LINE+2 LINE	AF1-101	86.29	25.46%	\$ 9,801,924	tower	SP	PS_P7-1_1LINE+2 LINE	AF1-126	24.83	7.33%	\$ 2,820,510	tower	SP	PS_P7-1_1LINE+2 LINE	AF1-222	97.95	28.90%	\$ 11,126,417	tower	SP	PS_P7-1_1LINE+2 LINE	\$38,495,471	\$11,126,417	N6768
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																			
AE2 Queue	129.82	38.31%	\$ 14,746,620	tower	SP	PS_P7-1_1LINE+2 LINE																																			
AF1-101	86.29	25.46%	\$ 9,801,924	tower	SP	PS_P7-1_1LINE+2 LINE																																			
AF1-126	24.83	7.33%	\$ 2,820,510	tower	SP	PS_P7-1_1LINE+2 LINE																																			
AF1-222	97.95	28.90%	\$ 11,126,417	tower	SP	PS_P7-1_1LINE+2 LINE																																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																												
427683 97,427 68396	11	28SMITHBRG 230.0 kV - 28E WINDSR 230.0 kV Ckt 1	JCPL: Upgrade no. 1 "Replace Gas Insulated Bus terminal equipment at Smithburg. Replace fault switches at East Windsor substation." Project Type : FAC Cost : \$1,879,465 New Ratings: 1274.0/1528.0/1528.0 Time Estimate : 12 Months The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AF1-101</td><td>194.02</td><td>4113.0 0%</td><td>\$ 773,063</td><td>tower</td><td>SP</td><td>PS_P7- 1_1LINE+2LI NE</td></tr> <tr> <td>AF1-126</td><td>57.63</td><td>1222.0 0%</td><td>\$ 229,624</td><td>tower</td><td>SP</td><td>PS_P7- 1_1LINE+2LI NE</td></tr> <tr> <td>AF1-222</td><td>220.05</td><td>4665.0 0%</td><td>\$ 876,778</td><td>tower</td><td>SP</td><td>PS_P7- 1_1LINE+2LI NE</td></tr> </tbody> </table> Upgrade no. 2 Replace substation conductor at East Windsor and Smithburg substations. Project Type : FAC Cost : \$167,809 New Ratings: 1370/1720/1720 Time Estimate : 12 Months AF1-222 is the driver for this upgrade.	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AF1-101	194.02	4113.0 0%	\$ 773,063	tower	SP	PS_P7- 1_1LINE+2LI NE	AF1-126	57.63	1222.0 0%	\$ 229,624	tower	SP	PS_P7- 1_1LINE+2LI NE	AF1-222	220.05	4665.0 0%	\$ 876,778	tower	SP	PS_P7- 1_1LINE+2LI NE	\$ 2,047,274	\$ 1,044,587	N6762 & N6762.1
Queue	MW	Cost %	Cost \$	Type	Study	ContName																												
AF1-101	194.02	4113.0 0%	\$ 773,063	tower	SP	PS_P7- 1_1LINE+2LI NE																												
AF1-126	57.63	1222.0 0%	\$ 229,624	tower	SP	PS_P7- 1_1LINE+2LI NE																												
AF1-222	220.05	4665.0 0%	\$ 876,778	tower	SP	PS_P7- 1_1LINE+2LI NE																												

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number
427683 72	4	28OCEANVW 230.0 kV - 28LARRABEE 230.0 kV Ckt 1	JCPL: Reconductor Larrabee - Oceanview 230 kV line. Type : FAC Total Cost : \$22,694,540 Time Estimate : 36.0 Months Ratings : 1114.0/1285.0/1285.0 AF1-222 is the driver for this upgrade.	\$22,694,540	\$22,694,540	N6755
422370 51,422 37050	16	28R11RINGB 230.0 kV - 28RED OAKB 230.0 kV Ckt 1	JCPL: "Reconductor R11 Ring bus to Red Oak B 230 kV transmission line (~0.6 miles). Upgrade terminals as required." Project Type : FAC Cost : \$2,819,198 New Ratings : 1156.0/1334.0/1334.0 Time Estimate : 18-24 Months Queue Project AF1-222 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-222 could receive cost allocation. Note 2: Although Queue Project AF1-222 may not have cost responsibility for this upgrade, Queue Project AF1-222 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-222 comes into service prior to completion of the upgrade, Queue Project AF1-222 will need an interim study.	\$2,819,198	\$0	N6596

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																			
422369 40,422 36941	7	28LARRABEE 230.0 kV - 28SMITHBRG 230.0 kV Ckt 2	JCPL: Upgrade no. 1: Replace wave traps at Larrabee and Smithburg Type : FAC Total Cost : \$55,779,836 Time Estimate : 54.0 Months New Ratings : 1245.0/1272.0/1272.0 The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AE2-251</td><td>48.5</td><td>12.78%</td><td>\$ 7,130,527</td><td>Single</td><td>LL</td><td>'JC-P1-2-JCC-230-017'</td></tr> <tr> <td>AF1-101</td><td>173.05</td><td>45.61%</td><td>\$ 25,442,015</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15F</td></tr> <tr> <td>AF1-126</td><td>52.15</td><td>13.75%</td><td>\$ 7,667,155</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15F</td></tr> <tr> <td>AF1-222</td><td>105.7</td><td>27.86%</td><td>\$ 15,540,139</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-15F</td></tr> </tbody> </table> Upgrade no. 2: Replace gas insulated bus at Smithburg Type : FAC Total Cost : \$1,342,475 Time Estimate : 12.0 Months New Ratings : 1660.0/1977.0/1977.0 AF1-222 is the driver for this upgrade.	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AE2-251	48.5	12.78%	\$ 7,130,527	Single	LL	'JC-P1-2-JCC-230-017'	AF1-101	173.05	45.61%	\$ 25,442,015	breaker	SP	JC-P2-3-JCC-230-15F	AF1-126	52.15	13.75%	\$ 7,667,155	breaker	SP	JC-P2-3-JCC-230-15F	AF1-222	105.7	27.86%	\$ 15,540,139	breaker	SP	JC-P2-3-JCC-230-15F	\$57,122,311	\$16,882,614	N6426 & N6426.1
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																			
AE2-251	48.5	12.78%	\$ 7,130,527	Single	LL	'JC-P1-2-JCC-230-017'																																			
AF1-101	173.05	45.61%	\$ 25,442,015	breaker	SP	JC-P2-3-JCC-230-15F																																			
AF1-126	52.15	13.75%	\$ 7,667,155	breaker	SP	JC-P2-3-JCC-230-15F																																			
AF1-222	105.7	27.86%	\$ 15,540,139	breaker	SP	JC-P2-3-JCC-230-15F																																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																										
422369 26,422 36925	13	28RED OAKB 230.0 kV - 28RAR RVR 230.0 kV Ckt 1	JCPL: Upgrade no. 1: Reconductor existing sections of 1590 ACSR 45/7 (Segment 3, Position 2; Segment 7, Position 1) with 1590 ACSS 45/7. Project Type : FAC Cost : \$3,900,000 Time Estimate : 30 Months New Ratings:: 1156/1334/1334 Queue Project AF1-222 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-222 could receive cost allocation. Note 2: Although Queue Project AF1-222 may not have cost responsibility for this upgrade, Queue Project AF1-222 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-222 comes into service prior to completion of the upgrade, Queue Project AF1-222 will need an interim study. Upgrade no. 2: Reconductor the Raritan River - Red Oak B 230 kV transmission line (~2.58 miles). Project Type : FAC Cost : \$12,122,549 Time Estimate : 30 Months New Ratings:: 1162.0/1519.0/1519.0 The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AE2-025</td><td>63.65</td><td>16.12%</td><td>\$ 1,954,102</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26D</td></tr> <tr> <td>AE2-232</td><td>62.3</td><td>15.78%</td><td>\$ 1,912,667</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26D</td></tr> <tr> <td>AF1-101</td><td>89.32</td><td>22.62%</td><td>\$ 2,742,205</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26D</td></tr> <tr> <td>AF1-126</td><td>28.33</td><td>7.17%</td><td>\$ 869,757</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26D</td></tr> <tr> <td>AF1-222</td><td>151.26</td><td>38.31%</td><td>\$ 4,643,819</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26D</td></tr> </tbody> </table> Upgrade no. 3: "Replace substation conductor at Raritan River.	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AE2-025	63.65	16.12%	\$ 1,954,102	breaker	SP	JC-P2-3-JCC-230-26D	AE2-232	62.3	15.78%	\$ 1,912,667	breaker	SP	JC-P2-3-JCC-230-26D	AF1-101	89.32	22.62%	\$ 2,742,205	breaker	SP	JC-P2-3-JCC-230-26D	AF1-126	28.33	7.17%	\$ 869,757	breaker	SP	JC-P2-3-JCC-230-26D	AF1-222	151.26	38.31%	\$ 4,643,819	breaker	SP	JC-P2-3-JCC-230-26D	\$17,667,081	\$6,288,351	N6204 & N6550 & N6550.1
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																										
AE2-025	63.65	16.12%	\$ 1,954,102	breaker	SP	JC-P2-3-JCC-230-26D																																										
AE2-232	62.3	15.78%	\$ 1,912,667	breaker	SP	JC-P2-3-JCC-230-26D																																										
AF1-101	89.32	22.62%	\$ 2,742,205	breaker	SP	JC-P2-3-JCC-230-26D																																										
AF1-126	28.33	7.17%	\$ 869,757	breaker	SP	JC-P2-3-JCC-230-26D																																										
AF1-222	151.26	38.31%	\$ 4,643,819	breaker	SP	JC-P2-3-JCC-230-26D																																										

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																																								
41632989	10	28RAR RVR 230.0 kV - KILMER_W 230.0 kV Ckt 1	JCPL: Replace substation conductor at Raritan River. Reconductor the Kilmer (W1037) - Raritan River 230 kV transmission line (~5.92 miles) Project Type : FAC Cost : \$27,983,891 New Ratings: 1156.0/1334.0/1334.0 Time Estimate : 42 Months The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AE2-000</td><td>68.85</td><td>14.05%</td><td>\$ 3,930,498</td><td>breaker</td><td>SP</td><td>'PS_P2-3_LNEPS_P1-2_1_LT'</td></tr> <tr> <td>AE2-024</td><td>122.74</td><td>25.04%</td><td>\$ 7,006,962</td><td>breaker</td><td>SP</td><td>'PS_P2-3_LNEPS_P1-2_1_LT'</td></tr> <tr> <td>AE2-025</td><td>61.95</td><td>12.64%</td><td>\$ 3,536,592</td><td>breaker</td><td>SP</td><td>'PS_P2-3_LNEPS_P1-2_1_LT'</td></tr> <tr> <td>AE2-232</td><td>46.47</td><td>9.48%</td><td>\$ 2,652,872</td><td>breaker</td><td>SP</td><td>'PS_P2-3_LNEPS_P1-2_1_LT'</td></tr> <tr> <td>AF1-101</td><td>66.57</td><td>13.58%</td><td>\$ 3,800,338</td><td>breaker</td><td>SP</td><td>'PS_P2-3_LNEPS_P1-2_1_LT'</td></tr> <tr> <td>AF1-126</td><td>21.49</td><td>4.38%</td><td>\$ 1,226,818</td><td>breaker</td><td>SP</td><td>'PS_P2-3_LNEPS_P1-2_1_LT'</td></tr> <tr> <td>AF1-222</td><td>102.12</td><td>20.83%</td><td>\$ 5,829,811</td><td>breaker</td><td>SP</td><td>'PS_P2-3_LNEPS_P1-2_1_LT'</td></tr> </tbody> </table> PSEG: No upgrade required.	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AE2-000	68.85	14.05%	\$ 3,930,498	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'	AE2-024	122.74	25.04%	\$ 7,006,962	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'	AE2-025	61.95	12.64%	\$ 3,536,592	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'	AE2-232	46.47	9.48%	\$ 2,652,872	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'	AF1-101	66.57	13.58%	\$ 3,800,338	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'	AF1-126	21.49	4.38%	\$ 1,226,818	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'	AF1-222	102.12	20.83%	\$ 5,829,811	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'	\$27,983,891	\$5,829,811	N6202
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																																								
AE2-000	68.85	14.05%	\$ 3,930,498	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'																																																								
AE2-024	122.74	25.04%	\$ 7,006,962	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'																																																								
AE2-025	61.95	12.64%	\$ 3,536,592	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'																																																								
AE2-232	46.47	9.48%	\$ 2,652,872	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'																																																								
AF1-101	66.57	13.58%	\$ 3,800,338	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'																																																								
AF1-126	21.49	4.38%	\$ 1,226,818	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'																																																								
AF1-222	102.12	20.83%	\$ 5,829,811	breaker	SP	'PS_P2-3_LNEPS_P1-2_1_LT'																																																								

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																			
427683 68,468 69220, 422373 33,468 69219, 427683 67	15	28E WINDSR 230.0 kV - 28WINDSOR 230.0 kV Ckt 1	JCPL: Upgrade no. 1 Reconductor the East Windsor - Windsor 230 kV line (~2.5 miles). Replace wave trap at Windsor. Replace circuit breaker at Windsor. Replace metering, relaying, and fault switch at Windsor. Project Type : FAC Cost : \$13,693,245 Ratings : 913.0/1147.0/1147.0 Time Estimate : 30 Months <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AE2 Queue</td><td>161.84</td><td>46.57%</td><td>\$ 6,376,575</td><td>tower</td><td>SP</td><td>PS_P7-1_1LINE+2LINE</td></tr> <tr> <td>AF1-101</td><td>78.2</td><td>22.50%</td><td>\$ 3,081,118</td><td>tower</td><td>SP</td><td>PS_P7-1_1LINE+2LINE</td></tr> <tr> <td>AF1-126</td><td>20.38</td><td>5.86%</td><td>\$ 802,982</td><td>tower</td><td>SP</td><td>PS_P7-1_1LINE+2LINE</td></tr> <tr> <td>AF1-222</td><td>87.12</td><td>25.07%</td><td>\$ 3,432,570</td><td>tower</td><td>SP</td><td>PS_P7-1_1LINE+2LINE</td></tr> </tbody> </table> Upgrade no. 2 Replace Disconnect Switches at Windsor Project Type : FAC Cost : \$ 1,174,666 Ratings : 1222/1355/1355 Time Estimate : 12 Months AF1-222 is the driver for this upgrade.	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AE2 Queue	161.84	46.57%	\$ 6,376,575	tower	SP	PS_P7-1_1LINE+2LINE	AF1-101	78.2	22.50%	\$ 3,081,118	tower	SP	PS_P7-1_1LINE+2LINE	AF1-126	20.38	5.86%	\$ 802,982	tower	SP	PS_P7-1_1LINE+2LINE	AF1-222	87.12	25.07%	\$ 3,432,570	tower	SP	PS_P7-1_1LINE+2LINE	\$14,867,911	\$4,607,236	N6774.1 & N6774.2
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																			
AE2 Queue	161.84	46.57%	\$ 6,376,575	tower	SP	PS_P7-1_1LINE+2LINE																																			
AF1-101	78.2	22.50%	\$ 3,081,118	tower	SP	PS_P7-1_1LINE+2LINE																																			
AF1-126	20.38	5.86%	\$ 802,982	tower	SP	PS_P7-1_1LINE+2LINE																																			
AF1-222	87.12	25.07%	\$ 3,432,570	tower	SP	PS_P7-1_1LINE+2LINE																																			

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number
422371 81,422 37180	3	28OCEANVW 230.0 kV - 28ATLANTIC 230.0 kV Ckt X	JCPL: Upgrade no. 1: Reconductor the Oceanview - Atlantic X 230 kV line (~4.6 miles) Project Type : FAC Cost : \$21,613,848 New Ratings : 731.0/917.0/917.0 Time Estimate : 36 Months AF1-222 is the driver for this upgrade. Upgrade no. 2: Replace disconnect switch at Atlantic Project Type : FAC Cost : \$234,933 New Ratings : 1306.0/1697.0/1697.0 Time Estimate : 12 Months AF1-222 is the driver for this upgrade.	\$21,848,780	\$21,848,780	N6763.1 & N6763.2

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																																															
42768576	1	28LARRABEE 230.0 kV - 28VANHISVL 230.0 kV Ckt 1	JCPL: "Reconductor the Larrabee - Van Hiseville 230 kV line (~7.5 miles). Replace wave trap at Larrabee" Project Type : FAC Cost : \$35,407,778 New Ratings : 956.0/956.0/956.0 Time Estimate : 42 Months The cost allocation table is below: <table> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> <tr> <td>AE2-023</td><td>45.81</td><td>5.86%</td><td>\$ 2,076,279</td><td>tower</td><td>SP</td><td>'JC-P7-1-JCC-230-10A'</td></tr> <tr> <td>AE2-222</td><td>39.3</td><td>5.03%</td><td>\$ 1,781,197</td><td>tower</td><td>SP</td><td>'JC-P7-1-JCC-230-10A'</td></tr> <tr> <td>AE2-232</td><td>148.5</td><td>19.01%</td><td>\$ 6,730,477</td><td>tower</td><td>SP</td><td>'JC-P7-1-JCC-230-10A'</td></tr> <tr> <td>AE2-251</td><td>167.62</td><td>21.46%</td><td>\$ 7,597,055</td><td>tower</td><td>SP</td><td>'JC-P7-1-JCC-230-10A'</td></tr> <tr> <td>AE2-314</td><td>10.06</td><td>1.29%</td><td>\$ 455,950</td><td>tower</td><td>SP</td><td>'JC-P7-1-JCC-230-10A'</td></tr> <tr> <td>AF1-101</td><td>213.84</td><td>27.37%</td><td>\$ 9,691,887</td><td>tower</td><td>SP</td><td>JC-P7-1-JCC-230-10A</td></tr> <tr> <td>AF1-126</td><td>99.57</td><td>12.75%</td><td>\$ 4,512,819</td><td>tower</td><td>SP</td><td>JC-P7-1-JCC-230-10A</td></tr> <tr> <td>AF1-222</td><td>56.53</td><td>7.24%</td><td>\$ 2,562,114</td><td>tower</td><td>SP</td><td>'JC-P7-1-JCC-230-11A'</td></tr> </table>	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AE2-023	45.81	5.86%	\$ 2,076,279	tower	SP	'JC-P7-1-JCC-230-10A'	AE2-222	39.3	5.03%	\$ 1,781,197	tower	SP	'JC-P7-1-JCC-230-10A'	AE2-232	148.5	19.01%	\$ 6,730,477	tower	SP	'JC-P7-1-JCC-230-10A'	AE2-251	167.62	21.46%	\$ 7,597,055	tower	SP	'JC-P7-1-JCC-230-10A'	AE2-314	10.06	1.29%	\$ 455,950	tower	SP	'JC-P7-1-JCC-230-10A'	AF1-101	213.84	27.37%	\$ 9,691,887	tower	SP	JC-P7-1-JCC-230-10A	AF1-126	99.57	12.75%	\$ 4,512,819	tower	SP	JC-P7-1-JCC-230-10A	AF1-222	56.53	7.24%	\$ 2,562,114	tower	SP	'JC-P7-1-JCC-230-11A'	\$35,407,778	\$2,562,114	N6537
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																																															
AE2-023	45.81	5.86%	\$ 2,076,279	tower	SP	'JC-P7-1-JCC-230-10A'																																																															
AE2-222	39.3	5.03%	\$ 1,781,197	tower	SP	'JC-P7-1-JCC-230-10A'																																																															
AE2-232	148.5	19.01%	\$ 6,730,477	tower	SP	'JC-P7-1-JCC-230-10A'																																																															
AE2-251	167.62	21.46%	\$ 7,597,055	tower	SP	'JC-P7-1-JCC-230-10A'																																																															
AE2-314	10.06	1.29%	\$ 455,950	tower	SP	'JC-P7-1-JCC-230-10A'																																																															
AF1-101	213.84	27.37%	\$ 9,691,887	tower	SP	JC-P7-1-JCC-230-10A																																																															
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AF1-222	56.53	7.24%	\$ 2,562,114	tower	SP	'JC-P7-1-JCC-230-11A'																																																															

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																																																																													
422369 31,422 36930	12	28RED OAKA 230.0 kV - 28RAR RVR 230.0 kV Ckt 1	JCPL: Reconductor the Raritan River - Red Oak A 230 kV transmission line (~2.58 miles). Project Type : FAC Cost : \$12,122,549 New Ratings : 1306.0/1593.0/1593.0 Time Estimate : 30 Months The cost allocation table is below: <table border="1"> <thead> <tr> <th>Queue</th><th>MW</th><th>Cost %</th><th>Cost \$</th><th>Type</th><th>Study</th><th>ContName</th></tr> </thead> <tbody> <tr> <td>AE1-238</td><td>238.34</td><td>23.30%</td><td>\$ 2,824,142</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26B'</td></tr> <tr> <td>AE2-000</td><td>127.8</td><td>12.49%</td><td>\$ 1,514,325</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26B'</td></tr> <tr> <td>AE2-024</td><td>166.2</td><td>16.25%</td><td>\$ 1,969,334</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26B'</td></tr> <tr> <td>AE2-025</td><td>83.87</td><td>8.20%</td><td>\$ 993,791</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26B'</td></tr> <tr> <td>AE2-222</td><td>14.1</td><td>1.38%</td><td>\$ 167,073</td><td>single</td><td>LL</td><td>'JC-P1-2-JCC-230-027'</td></tr> <tr> <td>AE2-232</td><td>62.6</td><td>6.12%</td><td>\$ 741,759</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26B'</td></tr> <tr> <td>AE2-251</td><td>59.9</td><td>5.85%</td><td>\$ 709,766</td><td>single</td><td>LL</td><td>'JC-P1-2-JCC-230-027'</td></tr> <tr> <td>AF1-101</td><td>89.79</td><td>8.78%</td><td>\$ 1,063,938</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26B</td></tr> <tr> <td>AF1-126</td><td>28.5</td><td>2.79%</td><td>\$ 337,702</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26B</td></tr> <tr> <td>AF1-222</td><td>151.97</td><td>14.85%</td><td>\$ 1,800,720</td><td>breaker</td><td>SP</td><td>JC-P2-3-JCC-230-26B</td></tr> </tbody> </table>	Queue	MW	Cost %	Cost \$	Type	Study	ContName	AE1-238	238.34	23.30%	\$ 2,824,142	breaker	SP	JC-P2-3-JCC-230-26B'	AE2-000	127.8	12.49%	\$ 1,514,325	breaker	SP	JC-P2-3-JCC-230-26B'	AE2-024	166.2	16.25%	\$ 1,969,334	breaker	SP	JC-P2-3-JCC-230-26B'	AE2-025	83.87	8.20%	\$ 993,791	breaker	SP	JC-P2-3-JCC-230-26B'	AE2-222	14.1	1.38%	\$ 167,073	single	LL	'JC-P1-2-JCC-230-027'	AE2-232	62.6	6.12%	\$ 741,759	breaker	SP	JC-P2-3-JCC-230-26B'	AE2-251	59.9	5.85%	\$ 709,766	single	LL	'JC-P1-2-JCC-230-027'	AF1-101	89.79	8.78%	\$ 1,063,938	breaker	SP	JC-P2-3-JCC-230-26B	AF1-126	28.5	2.79%	\$ 337,702	breaker	SP	JC-P2-3-JCC-230-26B	AF1-222	151.97	14.85%	\$ 1,800,720	breaker	SP	JC-P2-3-JCC-230-26B	\$12,122,549	\$1,800,720	N5870
Queue	MW	Cost %	Cost \$	Type	Study	ContName																																																																													
AE1-238	238.34	23.30%	\$ 2,824,142	breaker	SP	JC-P2-3-JCC-230-26B'																																																																													
AE2-000	127.8	12.49%	\$ 1,514,325	breaker	SP	JC-P2-3-JCC-230-26B'																																																																													
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AF1-126	28.5	2.79%	\$ 337,702	breaker	SP	JC-P2-3-JCC-230-26B																																																																													
AF1-222	151.97	14.85%	\$ 1,800,720	breaker	SP	JC-P2-3-JCC-230-26B																																																																													

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																								
427684 48,427 68447	6	CLRKSVLL_2 230.0 kV - LAWRENCE 230.0 kV Ckt 1	JCPL: Reconductor the Clariksville (PS) - Lawrence (PS) 230 kV line. (~0.8 miles) Project Type : FAC Cost : \$3,759,000 New Ratings : 1140/1387/1387 Time Estimate : 24 Months The cost allocation table is below: <table><tr><th>Queue</th><th>MW contri butio n</th><th>% of Cost</th><th>Cost</th><th>Contingency</th><th>Conti ngen cy Type</th></tr><tr><td>AF1-101</td><td>73.3</td><td>38.37%</td><td>\$1,442,280</td><td>'PS_P7-1_1LINE+2LINE'</td><td>towe r</td></tr><tr><td>AF1-126</td><td>19.9</td><td>10.39%</td><td>\$390,545</td><td>'PS_P7-1_1LINE+2LINE'</td><td>towe r</td></tr><tr><td>AF1-222</td><td>98.0</td><td>51.24%</td><td>\$1,926,176</td><td>'PS_P7-1_1LINE+2LINE'</td><td>towe r</td></tr></table>	Queue	MW contri butio n	% of Cost	Cost	Contingency	Conti ngen cy Type	AF1-101	73.3	38.37%	\$1,442,280	'PS_P7-1_1LINE+2LINE'	towe r	AF1-126	19.9	10.39%	\$390,545	'PS_P7-1_1LINE+2LINE'	towe r	AF1-222	98.0	51.24%	\$1,926,176	'PS_P7-1_1LINE+2LINE'	towe r	\$3,759,000	\$1,926,176	N6793
Queue	MW contri butio n	% of Cost	Cost	Contingency	Conti ngen cy Type																									
AF1-101	73.3	38.37%	\$1,442,280	'PS_P7-1_1LINE+2LINE'	towe r																									
AF1-126	19.9	10.39%	\$390,545	'PS_P7-1_1LINE+2LINE'	towe r																									
AF1-222	98.0	51.24%	\$1,926,176	'PS_P7-1_1LINE+2LINE'	towe r																									

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number
422371 21,422 37122	2	28OCEANVW 230.0 kV - 28ATLANTIC 230.0 kV Ckt Y	JCPL: Upgrade no.1: Reconductor the Oceanview - Atlantic Y 230 kV line (~4.6 miles) Project Type : FAC Cost : \$ 21,613,847 New Ratings : 731.0/917.0/917.0 Time Estimate : 36 Months AF1-222 is the driver for this upgrade. Upgrade no.2: Replace disconnect switch at Atlantic Project Type : FAC Cost : \$ 234,933 New Ratings : 1306.0/1697.0/1697.0 Time Estimate : 12 Months AF1-222 is the driver for this upgrade.	\$21,848,780	\$21,848,780	N6767.1 & N6767.2

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number
422370 42,422 37041	17	28R11RINGA 230.0 kV - 28RED OAKA 230.0 kV Ckt 1	JCPL: "Reconductor R11 Ring bus to Red Oak A 230 kV transmission line (~0.6 miles). Upgrade terminals as required." Project Type : FAC Cost : \$2,987,006 New Ratings: 1156.0/1334.0/1334.0 Time Estimate : 12 Months Queue Project AF1-222 presently does not receive cost allocation for this upgrade. Note 1: as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, Queue Project AF1-222 could receive cost allocation. Note 2: Although Queue Project AF1-222 may not have cost responsibility for this upgrade, Queue Project AF1-222 may need this upgrade in-service to be deliverable to the PJM system. If Queue Project AF1-222 comes into service prior to completion of the upgrade, Queue Project AF1-222 will need an interim study.	\$2,987,006	\$0	N6597
			TOTAL COST	\$366,968,547	\$130,224,243 ¹	N/A

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

11.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

11.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42768576	206294	28LARRABEE	JCP&L	206318	28VANHISVL	JCP&L	1	JC-P7-1-JCC-230-11A	tower	869.0	97.4	106.39	AC	79.53

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206201	28JERSVGEN	0.0435	50/50	0.0435
206271	28MCRC/REC	0.9134	50/50	0.9134
206306	28LKWD G1	2.9436	50/50	2.9436
206308	28LKWD G2	2.9436	50/50	2.9436
206312	28LKWD G3	2.1580	50/50	2.1580
206366	28LKWD CT1	5.9331	50/50	5.9331
206367	28LKWD CT2	5.9331	50/50	5.9331
206368	28MDLSEXCO	0.3827	50/50	0.3827
206370	LKWD CT3	39.8560	50/50	39.8560
206412	28R11	37.6805	Adder	44.33
206414	28X4-005E	1.5739	Adder	1.85
206416	28MLH_W1-032	0.0230	50/50	0.0230
207143	28BYD_X4-031	0.2663	50/50	0.2663
207144	28HOW_X1-037	0.1440	50/50	0.1440
207145	28FRN_Y2-051	0.0353	50/50	0.0353
207148	28MDF_W3-079	0.0569	50/50	0.0569
207169	28W4-060COL	0.0775	50/50	0.0775
207204	28HOL_W1-112	0.0359	50/50	0.0359
207205	28MAN_W1-024	0.0302	50/50	0.0302
207206	28TIN_W1-124	0.1623	50/50	0.1623
207401	AA1-060 BAT	4.8964	50/50	4.8964
207419	AB2-139 E	0.0341	Adder	0.04
207420	AC1-207 C	0.4228	Adder	0.5
227928	V4-067E	-0.1854	Adder	-0.22
901032	W1-024E OP1	0.2808	50/50	0.2808
901112	W1-032 E OP1	0.2191	50/50	0.2191
901422	W1-113 E	0.3593	Adder	0.42
901912	W1-112E OP1	0.3307	50/50	0.3307
902032	W1-124E	1.5037	50/50	1.5037
902322	W2-019 E	-1.0042	Adder	-1.18
902912	W2-078 E	0.4023	Adder	0.47
903982	W3-079 E	0.5318	50/50	0.5318
905502	W4-060 E OP1	0.7139	50/50	0.7139
907082	X1-037 E	1.3265	50/50	1.3265
914092	Y2-051 E	0.3250	50/50	0.3250
919662	AA2-048 E	0.8461	50/50	0.8461
920732	AA2-184 E	1.6501	50/50	1.6501
923791	AB2-014	0.2859	50/50	0.2859
924701	AB2-122 C	-0.0801	Adder	-0.09

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
927132	AC1-207 E	0.7076	Adder	0.83
934351	AD1-059	0.5259	50/50	0.5259
934841	AD1-113 (Suspended)	5.9020	Adder	6.94
938423	AE1-061 BAT	0.4930	Merchant Transmission	0.4930
939121	AE1-142 C O1	-2.7307	Adder	-3.21
939303	AE1-161 BAT	3.9545	Merchant Transmission	3.9545
939981	AE1-238 C	35.0865	50/50	35.0865
939982	AE1-238 E	92.1605	50/50	92.1605
940401	AE2-024 C O1	30.9389	50/50	30.9389
940402	AE2-024 E O1	144.8525	50/50	144.8525
940411	AE2-025 C	15.6179	50/50	15.6179
940412	AE2-025 E	73.1149	50/50	73.1149
940691	AE2-056 C	0.2009	50/50	0.2009
940692	AE2-056 E	0.1758	50/50	0.1758
940701	AE2-057 C	0.0613	50/50	0.0613
940702	AE2-057 E	0.0613	50/50	0.0613
940921	AE2-081 C	0.2135	50/50	0.2135
940922	AE2-081 E	0.2197	50/50	0.2197
940931	AE2-082 C	0.2983	Adder	0.35
940932	AE2-082 E	0.3159	Adder	0.37
943521	AF1-023	1.1949	50/50	1.1949
944613	AF1-126 BAT	99.9225	50/50	99.9225
945201	AF1-185 1	0.3541	Adder	0.42
945211	AF1-185 2	0.0787	Adder	0.09
945571	AF1-222 C	21.8706	50/50	21.8706
945572	AF1-222 E	57.6588	50/50	57.6588
945733	AF1-238 BAT	10.5991	Merchant Transmission	10.5991
945743	AF1-239 BAT	2.4939	Merchant Transmission	2.4939
945951	AF1-260	0.1209	50/50	0.1209
999905	MARINGEN 2	-0.4480	Adder	-0.53
999906	PVILLEG 2	-0.1926	Adder	-0.23
NEWTON	NEWTON	0.0484	Confirmed LTF	0.0484
FARMERCITY	FARMERCITY	0.0027	Confirmed LTF	0.0027
G-007A	G-007A	22.1961	Confirmed LTF	22.1961
VFT	VFT	23.1749	Confirmed LTF	23.1749
DUCKCREEK /* 35% REVERSE 3023630 4273103 TURNED OFF IN CASE FOR DEACTIVATION	DUCKCREEK /* 35% REVERSE 3023630 4273103 TURNED OFF IN CASE FOR DEACTIVATION	0.0000	LTF	0.0000
PRAIRIE	PRAIRIE	0.1266	Confirmed LTF	0.1266
COFFEEN	COFFEEN	0.0090	Confirmed LTF	0.0090
CHEOAH	CHEOAH	0.0375	Confirmed LTF	0.0375
EDWARDS	EDWARDS	0.0144	Confirmed LTF	0.0144
TILTON	TILTON	0.0258	Confirmed LTF	0.0258
GIBSON	GIBSON	0.0240	Confirmed LTF	0.0240
CALDERWOOD	CALDERWOOD	0.0368	Confirmed LTF	0.0368
BLUEG	BLUEG	0.0764	Confirmed LTF	0.0764
TRIMBLE	TRIMBLE	0.0239	Confirmed LTF	0.0239
CATAWBA	CATAWBA	0.0346	Confirmed LTF	0.0346

11.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237121	206300	28OCEANVW	JCP&L	206286	28ATLANTIC	JCP&L	Y	JC-P2-3-JCC-230-47B	breaker	869.0	65.1	120.02	AC	490.69

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	0.7593	50/50	0.7593
207143	28BYD_X4-031	0.1290	Adder	0.15
207206	28TIN_W1-124	0.2482	50/50	0.2482
902032	W1-124E	2.2988	50/50	2.2988
903982	W3-079 E	0.2389	Adder	0.28
919662	AA2-048 E	0.3800	Adder	0.45
920732	AA2-184 E	1.2279	50/50	1.2279
939981	AE1-238 C	216.4793	50/50	216.4793
939982	AE1-238 E	568.6188	50/50	568.6188
943521	AF1-023	0.8891	50/50	0.8891
945571	AF1-222 C	134.9387	50/50	134.9387
945572	AF1-222 E	355.7476	50/50	355.7476
945951	AF1-260	0.0543	Adder	0.06
NY	NY	0.0116	Confirmed LTF	0.0116
O-066	O-066	0.2486	Confirmed LTF	0.2486
G-007	G-007	0.1747	Confirmed LTF	0.1747

11.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237180	206300	28OCEANVW	JCP&L	206286	28ATLANTIC	JCP&L	X	JC-P2-3-JCC-230-47E	breaker	869.0	81.58	119.58	AC	339.69

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	14.0826	Adder	16.57
206370	LKWD CT3	14.6234	Adder	17.2
206414	28X4-005E	-1.2079	Adder	-1.42
901982	W1-119 E	0.2683	Adder	0.32
901992	W1-120E	0.2683	Adder	0.32
902082	W1-129E	0.1425	Adder	0.17
902322	W2-019 E	0.1684	Adder	0.2
902952	W2-082 E OP1	0.4884	Adder	0.57
903982	W3-079 E	0.2953	Adder	0.35
905252	W4-025 E	0.2022	Adder	0.24
912102	X4-015 E	0.1432	Adder	0.17
914092	Y2-051 E	0.2255	Adder	0.27
917612	Z2-102 E	0.3956	Adder	0.47
919662	AA2-048 E	0.4698	Adder	0.55
923293	AB1-138 E	0.3956	Adder	0.47
923463	AB1-163 E	0.3012	Adder	0.35
939121	AE1-142 C O1	0.5547	Adder	0.65
939122	AE1-142 E O1	0.7982	Adder	0.94
939981	AE1-238 C	149.8613	50/50	149.8613
939982	AE1-238 E	393.6356	50/50	393.6356
940161	AE2-000 C O1	14.3645	Adder	16.9
940162	AE2-000 E O1	36.7538	Adder	43.24
940401	AE2-024 C O1	11.7893	Adder	13.87
940402	AE2-024 E O1	55.1964	Adder	64.94
940411	AE2-025 C	5.9512	Adder	7.0
940412	AE2-025 E	27.8606	Adder	32.78
944332	AF1-101 E O1	36.0334	Adder	42.39
944611	AF1-126 C O1	2.2504	Adder	2.65
944612	AF1-126 E O1	9.0015	Adder	10.59
945201	AF1-185 1	-0.2718	Adder	-0.32
945211	AF1-185 2	-0.0604	Adder	-0.07
945571	AF1-222 C	93.4135	50/50	93.4135
945572	AF1-222 E	246.2720	50/50	246.2720
945951	AF1-260	0.0671	Adder	0.08
WEC	WEC	0.0158	Confirmed LTF	0.0158
LGEE	LGEE	0.0291	Confirmed LTF	0.0291
CPL	CPL	0.0463	Confirmed LTF	0.0463
CBM-W2	CBM-W2	0.4423	Confirmed LTF	0.4423

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
NY	NY	0.3921	Confirmed LTF	0.3921
TVA	TVA	0.0798	Confirmed LTF	0.0798
O-066	O-066	8.8032	Confirmed LTF	8.8032
CBM-S2	CBM-S2	0.3815	Confirmed LTF	0.3815
CBM-S1	CBM-S1	0.4771	Confirmed LTF	0.4771
G-007	G-007	7.3902	Confirmed LTF	7.3902
MADISON	MADISON	0.0040	Confirmed LTF	0.0040
MEC	MEC	0.0826	Confirmed LTF	0.0826
CBM-W1	CBM-W1	0.6130	Confirmed LTF	0.6130

11.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42768372	206300	28OCEANVW	JCP&L	206294	28LARRABEE	JCP&L	1	JC-P7-1-JCC-230-15	tower	869.0	60.93	112.22	AC	455.72

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	0.7953	50/50	0.7953
207143	28BYD_X4-031	0.1253	Adder	0.15
207206	28TIN_W1-124	0.2449	50/50	0.2449
902032	W1-124E	2.2687	50/50	2.2687
920732	AA2-184 E	1.3164	50/50	1.3164
939981	AE1-238 C	201.0533	50/50	201.0533
939982	AE1-238 E	528.0999	50/50	528.0999
943521	AF1-023	0.9532	50/50	0.9532
945571	AF1-222 C	125.3232	50/50	125.3232
945572	AF1-222 E	330.3975	50/50	330.3975
NEWTON	NEWTON	0.0086	Confirmed LTF	0.0086
FARMERCITY	FARMERCITY	0.0004	Confirmed LTF	0.0004
G-007A	G-007A	2.3543	Confirmed LTF	2.3543
VFT	VFT	0.9223	Confirmed LTF	0.9223
PRAIRIE	PRAIRIE	0.0207	Confirmed LTF	0.0207
COFFEEN	COFFEEN	0.0016	Confirmed LTF	0.0016
CHEOAH	CHEOAH	0.0045	Confirmed LTF	0.0045
EDWARDS	EDWARDS	0.0028	Confirmed LTF	0.0028
TILTON	TILTON	0.0050	Confirmed LTF	0.0050
GIBSON	GIBSON	0.0044	Confirmed LTF	0.0044
CALDERWOOD	CALDERWOOD	0.0045	Confirmed LTF	0.0045
BLUEG	BLUEG	0.0139	Confirmed LTF	0.0139
TRIMBLE	TRIMBLE	0.0045	Confirmed LTF	0.0045
CATAWBA	CATAWBA	0.0035	Confirmed LTF	0.0035

11.7.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237457	206411	28R11RINGA	JCP&L	206410	28R11RINGB	JCP&L	1	JC-P2-3-JCC-230-85	breaker	999.0	88.91	100.68	AC	117.94

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	1.4099	50/50	1.4099
206306	28LKWD G1	2.0721	50/50	2.0721
206308	28LKWD G2	2.0721	50/50	2.0721
206325	28O C GEN (Deactivation : 17/09/2018)	23.1398	Adder	27.22
206327	28S RIV G1	8.9560	50/50	8.9560
206328	28S RIV G2	8.9560	50/50	8.9560
206329	28S RIV G3	11.9696	50/50	11.9696
206358	28PARLN1&2	1.9100	50/50	1.9100
206359	28PARLN3&4	2.3090	50/50	2.3090
206366	28LKWD CT1	4.1765	50/50	4.1765
206367	28LKWD CT2	4.1765	50/50	4.1765
206370	LKWD CT3	23.8476	Adder	28.06
206412	28R11	303.4284	50/50	303.4284
206414	28X4-005E	-3.6440	Adder	-4.29
207143	28BYD_X4-031	0.3677	50/50	0.3677
207144	28HOW_X1-037	0.1672	50/50	0.1672
207145	28FRN_Y2-051	0.0508	50/50	0.0508
207148	28MDF_W3-079	0.0722	50/50	0.0722
207204	28HOL_W1-112	0.0512	50/50	0.0512
207206	28TIN_W1-124	0.2491	50/50	0.2491
207400	28AA1-060 E	1.2523	Adder	1.47
901112	W1-032 E OP1	0.1277	Adder	0.15
901912	W1-112E OP1	0.4715	50/50	0.4715
901982	W1-119 E	0.4635	Adder	0.55
901992	W1-120E	0.4635	Adder	0.55
902032	W1-124E	2.3072	50/50	2.3072
902082	W1-129E	0.2489	Adder	0.29
902322	W2-019 E	0.2947	Adder	0.35
902952	W2-082 E OP1	0.8514	Adder	1.0
903982	W3-079 E	0.6747	50/50	0.6747
905252	W4-025 E	0.3533	Adder	0.42
907082	X1-037 E	1.5401	50/50	1.5401
912102	X4-015 E	0.2499	Adder	0.29
914092	Y2-051 E	0.4680	50/50	0.4680
917612	Z2-102 E	0.6833	Adder	0.8
919662	AA2-048 E	1.0734	50/50	1.0734
920732	AA2-184 E	2.5406	50/50	2.5406
923293	AB1-138 E	0.6756	Adder	0.79

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
923463	AB1-163 E	0.5469	Adder	0.64
939121	AE1-142 C O1	0.9122	Adder	1.07
939122	AE1-142 E O1	1.3127	Adder	1.54
939981	AE1-238 C	52.0335	50/50	52.0335
939982	AE1-238 E	136.6747	50/50	136.6747
940161	AE2-000 C O1	23.6030	Adder	27.77
940162	AE2-000 E O1	60.3920	Adder	71.05
940401	AE2-024 C O1	22.5596	50/50	22.5596
940402	AE2-024 E O1	105.6215	50/50	105.6215
940411	AE2-025 C	11.3881	50/50	11.3881
940412	AE2-025 E	53.3129	50/50	53.3129
940691	AE2-056 C	0.2314	50/50	0.2314
940692	AE2-056 E	0.2025	50/50	0.2025
940701	AE2-057 C	0.0546	Adder	0.06
940702	AE2-057 E	0.0546	Adder	0.06
940921	AE2-081 C	0.2458	50/50	0.2458
940922	AE2-081 E	0.2531	50/50	0.2531
943521	AF1-023	1.8398	50/50	1.8398
943561	AF1-027	0.1391	Adder	0.16
944332	AF1-101 E O1	59.2082	Adder	69.66
944611	AF1-126 C O1	3.7761	Adder	4.44
944612	AF1-126 E O1	15.1045	Adder	17.77
945201	AF1-185 1	-0.8199	Adder	-0.96
945211	AF1-185 2	-0.1822	Adder	-0.21
945571	AF1-222 C	32.4342	50/50	32.4342
945572	AF1-222 E	85.5084	50/50	85.5084
945951	AF1-260	0.1533	50/50	0.1533
WEC	WEC	0.0009	Confirmed LTF	0.0009
LGEE	LGEE	0.0034	Confirmed LTF	0.0034
CPL	CPL	0.0324	Confirmed LTF	0.0324
CBM-W2	CBM-W2	0.0983	Confirmed LTF	0.0983
NY	NY	1.0518	Confirmed LTF	1.0518
TVA	TVA	0.0252	Confirmed LTF	0.0252
O-066	O-066	24.0845	Confirmed LTF	24.0845
CBM-S2	CBM-S2	0.2312	Confirmed LTF	0.2312
CBM-S1	CBM-S1	0.1363	Confirmed LTF	0.1363
G-007	G-007	22.2934	Confirmed LTF	22.2934
MADISON	MADISON	0.0081	Confirmed LTF	0.0081
MEC	MEC	0.0095	Confirmed LTF	0.0095

11.7.6 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42768447	219104	CLRKSVLL_2	PSE&G	217150	LAWRENCE	PSE&G	1	PS_P7-1_1LINE+2LINE	tower	869.0	110.18	119.76	AC	97.95

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206201	28JERSVGEN	0.0713	50/50	0.0713
206271	28MCRC/REC	1.0626	Adder	1.25
206325	28O C GEN (Deactivation : 17/09/2018)	28.6639	Adder	33.72
206350	28RRCT1&2	2.8687	50/50	2.8687
206351	28RRCT3&4	2.6862	50/50	2.6862
206362	28RDOAKCT1	6.0788	50/50	6.0788
206365	28RDOAKST1	8.5243	50/50	8.5243
206368	28MDLSEXCO	0.7120	50/50	0.7120
206370	LKWD CT3	30.0220	Adder	35.32
206403	28W4-009	25.3899	50/50	25.3899
206412	28R11	73.7191	Adder	86.73
206414	28X4-005E	3.9546	50/50	3.9546
206416	28MLH_W1-032	0.0416	50/50	0.0416
207143	28BYD_X4-031	0.3252	Adder	0.38
207152	28NPK_W1-113	0.1326	50/50	0.1326
207162	28APLW2-078	0.1484	50/50	0.1484
207169	28W4-060COL	0.1341	50/50	0.1341
207205	28MAN_W1-024	0.0582	50/50	0.0582
207400	28AA1-060 E	2.4094	Adder	2.83
207419	AB2-139 E	0.0979	50/50	0.0979
207420	AC1-207 C	1.1006	50/50	1.1006
219278	CLARKSP1_C	0.0016	50/50	0.0016
219281	CLARKSP1_E	0.0148	50/50	0.0148
219283	CLARKSP2_C	0.0013	50/50	0.0013
219302	CLARKSP2_E	0.0122	50/50	0.0122
219327	CLARKSP3_C	0.0004	50/50	0.0004
219329	CLARKSP3_E	0.0031	50/50	0.0031
219330	CLARKSP4_C	0.0006	50/50	0.0006
219335	CLARKSP4_E	0.0061	50/50	0.0061
219679	NRGCENTER E	-0.3193	Adder	-0.38
901032	W1-024E OP1	0.5401	50/50	0.5401
901112	W1-032 E OP1	0.3973	50/50	0.3973
901422	W1-113 E	1.2314	50/50	1.2314
901912	W1-112E OP1	0.4117	Adder	0.48
901982	W1-119 E	0.6146	Adder	0.72
901992	W1-120E	0.6146	Adder	0.72
902032	W1-124E	1.7178	Adder	2.02
902082	W1-129E	0.3351	Adder	0.39
902322	W2-019 E	0.3977	Adder	0.47

Bus #	Bus	Gendoliv MW Impact	Type	Full MW Impact
902912	W2-078 E	1.3794	50/50	1.3794
902952	W2-082 E OP1	1.1433	Adder	1.35
903982	W3-079 E	0.6788	Adder	0.8
905252	W4-025 E	0.4757	Adder	0.56
905502	W4-060 E OP1	1.2352	50/50	1.2352
907082	X1-037 E	1.7604	Adder	2.07
907272	X1-085 E	0.4739	Adder	0.56
912102	X4-015 E	0.3361	Adder	0.4
914092	Y2-051 E	0.4813	Adder	0.57
917612	Z2-102 E	0.9064	Adder	1.07
919662	AA2-048 E	1.0799	Adder	1.27
919712	AA2-058 E	-0.1713	Adder	-0.2
919802	AA2-066 E	-0.3193	Adder	-0.38
920732	AA2-184 E	1.9000	Adder	2.24
923293	AB1-138 E	0.8425	Adder	0.99
923463	AB1-163 E	0.6828	Adder	0.8
927132	AC1-207 E	1.8419	50/50	1.8419
934841	AD1-113 (Suspended)	14.8298	50/50	14.8298
937523	AD2-213 BAT	1.8022	Merchant Transmission	1.8022
939121	AE1-142 C O1	1.1390	Adder	1.34
939122	AE1-142 E O1	1.6390	Adder	1.93
939981	AE1-238 C	36.7315	Adder	43.21
939982	AE1-238 E	96.4813	Adder	113.51
940161	AE2-000 C O1	29.2377	Adder	34.4
940162	AE2-000 E O1	74.8092	Adder	88.01
940401	AE2-024 C O1	24.2437	Adder	28.52
940402	AE2-024 E O1	113.5062	Adder	133.54
940411	AE2-025 C	12.2382	Adder	14.4
940412	AE2-025 E	57.2927	Adder	67.4
940691	AE2-056 C	0.2675	Adder	0.31
940692	AE2-056 E	0.2341	Adder	0.28
940701	AE2-057 C	0.1007	50/50	0.1007
940702	AE2-057 E	0.1007	50/50	0.1007
940921	AE2-081 C	0.2842	Adder	0.33
940922	AE2-081 E	0.2926	Adder	0.34
940931	AE2-082 C	0.7623	50/50	0.7623
940932	AE2-082 E	0.8072	50/50	0.8072
943521	AF1-023	1.3759	Adder	1.62
943561	AF1-027	0.2677	Adder	0.31
944332	AF1-101 E O1	73.3429	Adder	86.29
944432	AF1-108 BAT	1.8618	Merchant Transmission	1.8618
944442	AF1-109 BAT	5.4178	50/50	5.4178
944611	AF1-126 C O1	4.2215	Adder	4.97
944612	AF1-126 E O1	16.8861	Adder	19.87
945201	AF1-185 1	0.8898	50/50	0.8898
945211	AF1-185 2	0.1977	50/50	0.1977
945571	AF1-222 C	22.8960	Adder	26.94
945572	AF1-222 E	60.3621	Adder	71.01
945723	AF1-237 BAT	38.8140	Merchant Transmission	38.8140
945951	AF1-260	0.1543	Adder	0.18
NEWTON	NEWTON	0.2579	Confirmed LTF	0.2579
FARMERCITY	FARMERCITY	0.0134	Confirmed LTF	0.0134

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
G-007A	G-007A	47.4058	Confirmed LTF	47.4058
NY	NY	0.2124	Confirmed LTF	0.2124
PRAIRIE	PRAIRIE	0.6199	Confirmed LTF	0.6199
O-066	O-066	7.3315	Confirmed LTF	7.3315
COFFEEN	COFFEEN	0.0479	Confirmed LTF	0.0479
CHEOAH	CHEOAH	0.1196	Confirmed LTF	0.1196
EDWARDS	EDWARDS	0.0840	Confirmed LTF	0.0840
TILTON	TILTON	0.1512	Confirmed LTF	0.1512
GIBSON	GIBSON	0.1310	Confirmed LTF	0.1310
CALDERWOOD	CALDERWOOD	0.1188	Confirmed LTF	0.1188
BLUEG	BLUEG	0.4166	Confirmed LTF	0.4166
TRIMBLE	TRIMBLE	0.1336	Confirmed LTF	0.1336
CATAWBA	CATAWBA	0.0833	Confirmed LTF	0.0833

11.7.7 Index 7

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42236940	206294	28LARRABEE	JCP&L	206309	28SMITHBRG	JCP&L	2	JC-P2-3-JCC-230-15F	breaker	813.0	149.95	161.05	AC	105.7

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	1.0638	Adder	1.25
206280	28LAKEHURS	0.7862	50/50	0.7862
206306	28LKWD G1	5.2600	50/50	5.2600
206308	28LKWD G2	5.2600	50/50	5.2600
206312	28LKWD G3	3.8562	50/50	3.8562
206325	280 C GEN (Deactivation : 17/09/2018)	57.4883	Adder	67.63
206366	28LKWD CT1	10.6020	50/50	10.6020
206367	28LKWD CT2	10.6020	50/50	10.6020
206370	LKWD CT3	71.2200	50/50	71.2200
206412	28R11	31.3636	Adder	36.9
207143	28BYD_X4-031	0.3463	Adder	0.41
207145	28FRN_Y2-051	0.1059	50/50	0.1059
207148	28MDF_W3-079	0.1486	50/50	0.1486
207400	28AA1-060 E	2.3553	Adder	2.77
207419	AB2-139 E	-0.0275	Adder	-0.03
227928	V4-067E	0.1293	Adder	0.15
228102	BLE#2 ST (Deactivation : 30/04/2019)	9.9766	Adder	11.74
901912	W1-112E OP1	0.3071	Adder	0.36
901982	W1-119 E	1.0542	Adder	1.24
901992	W1-120E	1.0542	Adder	1.24
902032	W1-124E	1.7152	Adder	2.02
902082	W1-129E	0.5533	Adder	0.65
902322	W2-019 E	0.6526	Adder	0.77
902432	W2-030 E	0.4587	Adder	0.54
902952	W2-082 E OP1	1.9006	Adder	2.24
903982	W3-079 E	1.3875	50/50	1.3875
905252	W4-025 E	0.7853	Adder	0.92
905502	W4-060 E OP1	0.4651	Adder	0.55
907082	X1-037 E	1.2827	Adder	1.51
907272	X1-085 E	0.3063	Adder	0.36
912102	X4-015 E	0.5565	Adder	0.65
914092	Y2-051 E	0.9759	50/50	0.9759
917612	Z2-102 E	1.5535	Adder	1.83
919662	AA2-048 E	2.2073	50/50	2.2073
920732	AA2-184 E	1.8530	Adder	2.18
923292	AB1-138 C	0.2007	50/50	0.2007
923293	AB1-138 E	1.8890	50/50	1.8890

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
923463	AB1-163 E	1.2291	Adder	1.45
923791	AB2-014	0.5109	50/50	0.5109
924701	AB2-122 C	0.0559	Adder	0.07
924702	AB2-122 E	0.0958	Adder	0.11
930001	AB1-001 C	0.0824	Adder	0.1
930002	AB1-001 E	0.1355	Adder	0.16
933962	AD1-019 E	0.5933	Adder	0.7
934351	AD1-059	0.9397	50/50	0.9397
938781	AE1-104 C O1	8.0529	Adder	9.47
938782	AE1-104 E O1	20.6033	Adder	24.24
939121	AE1-142 C O1	2.6799	50/50	2.6799
939122	AE1-142 E O1	3.8565	50/50	3.8565
939981	AE1-238 C	39.6366	Adder	46.63
939982	AE1-238 E	104.1120	Adder	122.48
940161	AE2-000 C O1	58.6391	Adder	68.99
940162	AE2-000 E O1	150.0374	Adder	176.51
940361	AE2-020 C	8.9877	Adder	10.57
940362	AE2-020 E	42.0810	Adder	49.51
940371	AE2-021 C	8.9877	Adder	10.57
940372	AE2-021 E	42.0810	Adder	49.51
940381	AE2-022 C	5.2428	Adder	6.17
940382	AE2-022 E	24.5473	Adder	28.88
940401	AE2-024 C O1	57.6400	50/50	57.6400
940402	AE2-024 E O1	269.8643	50/50	269.8643
940411	AE2-025 C	29.0966	50/50	29.0966
940412	AE2-025 E	136.2150	50/50	136.2150
940691	AE2-056 C	0.1914	Adder	0.23
940692	AE2-056 E	0.1675	Adder	0.2
940701	AE2-057 C	0.0358	Adder	0.04
940702	AE2-057 E	0.0358	Adder	0.04
940921	AE2-081 C	0.2034	Adder	0.24
940922	AE2-081 E	0.2094	Adder	0.25
942101	AE2-222 C O1	6.7045	Adder	7.89
942102	AE2-222 E O1	16.8422	Adder	19.81
942381	AE2-251 C	28.4728	Adder	33.5
942382	AE2-251 E	72.8540	Adder	85.71
943521	AF1-023	1.3418	Adder	1.58
943561	AF1-027	0.2617	Adder	0.31
943732	AF1-041 E	0.1900	Adder	0.22
944332	AF1-101 E O1	147.0965	Adder	173.05
944611	AF1-126 C O1	8.8647	Adder	10.43
944612	AF1-126 E O1	35.4586	Adder	41.72
945571	AF1-222 C	24.7068	Adder	29.07
945572	AF1-222 E	65.1361	Adder	76.63
945951	AF1-260	0.3153	50/50	0.3153
999905	MARINGEN 2	0.3110	Adder	0.37
999906	PVILLEG 2	0.1341	Adder	0.16
NEWTON	NEWTON	0.3428	Confirmed LTF	0.3428
FARMERCITY	FARMERCITY	0.0178	Confirmed LTF	0.0178
G-007A	G-007A	16.1615	Confirmed LTF	16.1615
NY	NY	0.5563	Confirmed LTF	0.5563
PRAIRIE	PRAIRIE	0.8188	Confirmed LTF	0.8188

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
O-066	O-066	5.7389	Confirmed LTF	5.7389
COFFEEN	COFFEEN	0.0636	Confirmed LTF	0.0636
CHEOAH	CHEOAH	0.1537	Confirmed LTF	0.1537
EDWARDS	EDWARDS	0.1123	Confirmed LTF	0.1123
TILTON	TILTON	0.2022	Confirmed LTF	0.2022
MADISON	MADISON	0.0060	Confirmed LTF	0.0060
GIBSON	GIBSON	0.1742	Confirmed LTF	0.1742
CALDERWOOD	CALDERWOOD	0.1526	Confirmed LTF	0.1526
BLUEG	BLUEG	0.5538	Confirmed LTF	0.5538
TRIMBLE	TRIMBLE	0.1775	Confirmed LTF	0.1775
CATAWBA	CATAWBA	0.1040	Confirmed LTF	0.1040

11.7.8 Index 8

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42236945	206294	28LARRABEE	JCP&L	206309	28SMITHBRG	JCP&L	1	JC-P2-3-JCC-230-15G	breaker	817.0	149.22	160.26	AC	105.7

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	1.0638	Adder	1.25
206280	28LAKEHURS	0.7862	50/50	0.7862
206306	28LKWD G1	5.2600	50/50	5.2600
206308	28LKWD G2	5.2600	50/50	5.2600
206312	28LKWD G3	3.8562	50/50	3.8562
206325	280 C GEN (Deactivation : 17/09/2018)	57.4883	Adder	67.63
206366	28LKWD CT1	10.6020	50/50	10.6020
206367	28LKWD CT2	10.6020	50/50	10.6020
206370	LKWD CT3	71.2200	50/50	71.2200
206412	28R11	31.3636	Adder	36.9
207143	28BYD_X4-031	0.3463	Adder	0.41
207145	28FRN_Y2-051	0.1059	50/50	0.1059
207148	28MDF_W3-079	0.1486	50/50	0.1486
207400	28AA1-060 E	2.3553	Adder	2.77
207419	AB2-139 E	-0.0275	Adder	-0.03
227928	V4-067E	0.1293	Adder	0.15
228102	BLE#2 ST (Deactivation : 30/04/2019)	9.9766	Adder	11.74
901912	W1-112E OP1	0.3071	Adder	0.36
901982	W1-119 E	1.0542	Adder	1.24
901992	W1-120E	1.0542	Adder	1.24
902032	W1-124E	1.7152	Adder	2.02
902082	W1-129E	0.5533	Adder	0.65
902322	W2-019 E	0.6526	Adder	0.77
902432	W2-030 E	0.4587	Adder	0.54
902952	W2-082 E OP1	1.9006	Adder	2.24
903982	W3-079 E	1.3875	50/50	1.3875
905252	W4-025 E	0.7853	Adder	0.92
905502	W4-060 E OP1	0.4651	Adder	0.55
907082	X1-037 E	1.2827	Adder	1.51
907272	X1-085 E	0.3063	Adder	0.36
912102	X4-015 E	0.5565	Adder	0.65
914092	Y2-051 E	0.9759	50/50	0.9759
917612	Z2-102 E	1.5535	Adder	1.83
919662	AA2-048 E	2.2073	50/50	2.2073
920732	AA2-184 E	1.8530	Adder	2.18
923292	AB1-138 C	0.2007	50/50	0.2007
923293	AB1-138 E	1.8890	50/50	1.8890

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
923463	AB1-163 E	1.2291	Adder	1.45
923791	AB2-014	0.5109	50/50	0.5109
924701	AB2-122 C	0.0559	Adder	0.07
924702	AB2-122 E	0.0958	Adder	0.11
930001	AB1-001 C	0.0824	Adder	0.1
930002	AB1-001 E	0.1355	Adder	0.16
933962	AD1-019 E	0.5933	Adder	0.7
934351	AD1-059	0.9397	50/50	0.9397
938781	AE1-104 C O1	8.0529	Adder	9.47
938782	AE1-104 E O1	20.6033	Adder	24.24
939121	AE1-142 C O1	2.6799	50/50	2.6799
939122	AE1-142 E O1	3.8565	50/50	3.8565
939981	AE1-238 C	39.6366	Adder	46.63
939982	AE1-238 E	104.1120	Adder	122.48
940161	AE2-000 C O1	58.6391	Adder	68.99
940162	AE2-000 E O1	150.0374	Adder	176.51
940361	AE2-020 C	8.9877	Adder	10.57
940362	AE2-020 E	42.0810	Adder	49.51
940371	AE2-021 C	8.9877	Adder	10.57
940372	AE2-021 E	42.0810	Adder	49.51
940381	AE2-022 C	5.2428	Adder	6.17
940382	AE2-022 E	24.5473	Adder	28.88
940401	AE2-024 C O1	57.6400	50/50	57.6400
940402	AE2-024 E O1	269.8643	50/50	269.8643
940411	AE2-025 C	29.0966	50/50	29.0966
940412	AE2-025 E	136.2150	50/50	136.2150
940691	AE2-056 C	0.1914	Adder	0.23
940692	AE2-056 E	0.1675	Adder	0.2
940701	AE2-057 C	0.0358	Adder	0.04
940702	AE2-057 E	0.0358	Adder	0.04
940921	AE2-081 C	0.2034	Adder	0.24
940922	AE2-081 E	0.2094	Adder	0.25
942101	AE2-222 C O1	6.7045	Adder	7.89
942102	AE2-222 E O1	16.8422	Adder	19.81
942381	AE2-251 C	28.4728	Adder	33.5
942382	AE2-251 E	72.8540	Adder	85.71
943521	AF1-023	1.3418	Adder	1.58
943561	AF1-027	0.2617	Adder	0.31
943732	AF1-041 E	0.1900	Adder	0.22
944332	AF1-101 E O1	147.0965	Adder	173.05
944611	AF1-126 C O1	8.8647	Adder	10.43
944612	AF1-126 E O1	35.4586	Adder	41.72
945571	AF1-222 C	24.7068	Adder	29.07
945572	AF1-222 E	65.1361	Adder	76.63
945951	AF1-260	0.3153	50/50	0.3153
999905	MARINGEN 2	0.3110	Adder	0.37
999906	PVILLEG 2	0.1341	Adder	0.16
NEWTON	NEWTON	0.3428	Confirmed LTF	0.3428
FARMERCITY	FARMERCITY	0.0178	Confirmed LTF	0.0178
G-007A	G-007A	16.1615	Confirmed LTF	16.1615
NY	NY	0.5563	Confirmed LTF	0.5563
PRAIRIE	PRAIRIE	0.8188	Confirmed LTF	0.8188

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
O-066	O-066	5.7389	Confirmed LTF	5.7389
COFFEEN	COFFEEN	0.0636	Confirmed LTF	0.0636
CHEOAH	CHEOAH	0.1537	Confirmed LTF	0.1537
EDWARDS	EDWARDS	0.1123	Confirmed LTF	0.1123
TILTON	TILTON	0.2022	Confirmed LTF	0.2022
MADISON	MADISON	0.0060	Confirmed LTF	0.0060
GIBSON	GIBSON	0.1742	Confirmed LTF	0.1742
CALDERWOOD	CALDERWOOD	0.1526	Confirmed LTF	0.1526
BLUEG	BLUEG	0.5538	Confirmed LTF	0.5538
TRIMBLE	TRIMBLE	0.1775	Confirmed LTF	0.1775
CATAWBA	CATAWBA	0.1040	Confirmed LTF	0.1040

11.7.9 Index 9

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
49863000	206302	28OYSTER C	JCP&L	227955	CEDAR	AE	1	JC-P7-1-JCC-230-7A	tower	564.0	134.26	142.72	AC	57.28

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	0.6025	Adder	0.71
206280	28LAKEHURS	0.4626	50/50	0.4626
206308	28LKWD G2	2.6904	50/50	2.6904
206325	28O C GEN (Deactivation : 17/09/2018)	66.1407	50/50	66.1407
206360	28O CRK C1	1.7717	50/50	1.7717
206361	28O CRK C2	1.1857	50/50	1.1857
206366	28LKWD CT1	5.4228	50/50	5.4228
206367	28LKWD CT2	5.4228	50/50	5.4228
206370	LKWD CT3	30.9638	Adder	36.43
206412	28R11	24.5381	Adder	28.87
206414	28X4-005E	1.0025	Adder	1.18
207143	28BYD_X4-031	0.1950	Adder	0.23
207400	28AA1-060 E	1.3207	Adder	1.55
227928	V4-067E	-0.2727	Adder	-0.32
228261	V4-054E	-0.4096	Adder	-0.48
901912	W1-112E OP1	0.1949	Adder	0.23
901982	W1-119 E	0.5480	Adder	0.64
901992	W1-120E	0.5480	Adder	0.64
902032	W1-124E	0.9699	Adder	1.14
902082	W1-129E	0.2687	Adder	0.32
902322	W2-019 E	0.3134	Adder	0.37
902952	W2-082 E OP1	0.9350	Adder	1.1
903982	W3-079 E	0.5511	Adder	0.65
905252	W4-025 E	0.3814	Adder	0.45
905502	W4-060 E OP1	0.3906	Adder	0.46
907082	X1-037 E	0.9126	Adder	1.07
907272	X1-085 E	0.2038	Adder	0.24
912102	X4-015 E	0.2717	Adder	0.32
914092	Y2-051 E	0.4056	Adder	0.48
917612	Z2-102 E	0.8064	Adder	0.95
919662	AA2-048 E	0.8767	Adder	1.03
920732	AA2-184 E	1.0595	Adder	1.25
923292	AB1-138 C	0.1181	50/50	0.1181
923293	AB1-138 E	1.1114	50/50	1.1114
923463	AB1-163 E	0.6423	Adder	0.76
924701	AB2-122 C	-0.1178	Adder	-0.14
934841	AD1-113 (Suspended)	3.7593	Adder	4.42
938423	AE1-061 BAT	0.5371	Merchant Transmission	0.5371

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939121	AE1-142 C O1	1.8082	50/50	1.8082
939122	AE1-142 E O1	2.6020	50/50	2.6020
939303	AE1-161 BAT	3.7000	Merchant Transmission	3.7000
939981	AE1-238 C	21.4812	Adder	25.27
939982	AE1-238 E	56.4240	Adder	66.38
940161	AE2-000 C O1	67.4646	50/50	67.4646
940162	AE2-000 E O1	172.6189	50/50	172.6189
940401	AE2-024 C O1	20.9041	Adder	24.59
940402	AE2-024 E O1	97.8708	Adder	115.14
940411	AE2-025 C	10.5524	Adder	12.41
940412	AE2-025 E	49.4007	Adder	58.12
940691	AE2-056 C	0.1378	Adder	0.16
940692	AE2-056 E	0.1206	Adder	0.14
940701	AE2-057 C	0.0368	Adder	0.04
940702	AE2-057 E	0.0368	Adder	0.04
940921	AE2-081 C	0.1464	Adder	0.17
940922	AE2-081 E	0.1507	Adder	0.18
943521	AF1-023	0.7672	Adder	0.9
943561	AF1-027	0.1467	Adder	0.17
944332	AF1-101 E O1	169.2353	50/50	169.2353
944611	AF1-126 C O1	3.7396	Adder	4.4
944612	AF1-126 E O1	14.9583	Adder	17.6
945201	AF1-185 1	0.2256	Adder	0.27
945211	AF1-185 2	0.0501	Adder	0.06
945571	AF1-222 C	13.3899	Adder	15.75
945572	AF1-222 E	35.3008	Adder	41.53
945733	AF1-238 BAT	11.4240	Merchant Transmission	11.4240
945743	AF1-239 BAT	2.6880	Merchant Transmission	2.6880
945951	AF1-260	0.1252	Adder	0.15
999905	MARINGEN 2	-0.6526	Adder	-0.77
999906	PVILLEG 2	-0.2817	Adder	-0.33
NEWTON	NEWTON	0.0989	Confirmed LTF	0.0989
FARMERCITY	FARMERCITY	0.0053	Confirmed LTF	0.0053
G-007A	G-007A	14.1381	Confirmed LTF	14.1381
VFT	VFT	11.6100	Confirmed LTF	11.6100
PRAIRIE	PRAIRIE	0.2454	Confirmed LTF	0.2454
COFFEEN	COFFEEN	0.0184	Confirmed LTF	0.0184
CHEOAH	CHEOAH	0.0541	Confirmed LTF	0.0541
EDWARDS	EDWARDS	0.0315	Confirmed LTF	0.0315
TILTON	TILTON	0.0567	Confirmed LTF	0.0567
GIBSON	GIBSON	0.0502	Confirmed LTF	0.0502
CALDERWOOD	CALDERWOOD	0.0537	Confirmed LTF	0.0537
BLUEG	BLUEG	0.1597	Confirmed LTF	0.1597
TRIMBLE	TRIMBLE	0.0512	Confirmed LTF	0.0512
CATAWBA	CATAWBA	0.0424	Confirmed LTF	0.0424

11.7.10 Index 10

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41632989	206305	28RAR RVR	JCP&L	218332	KILMER_W	PSE&G	1	PS_P2-3_LNEPS_P1-2_1_LT	breaker	817.0	131.56	142.43	AC	102.12

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	1.1332	Adder	1.33
206325	28O C GEN (Deactivation : 17/09/2018)	22.1137	Adder	26.02
206329	28S RIV G3	6.1534	50/50	6.1534
206350	28RRCT1&2	5.4961	50/50	5.4961
206351	28RRCT3&4	5.1464	50/50	5.1464
206362	28RDOAKCT1	11.0885	50/50	11.0885
206363	28RDOAKCT2	11.0648	50/50	11.0648
206364	28RDOAKCT3	11.0648	50/50	11.0648
206365	28RDOAKST1	15.5494	50/50	15.5494
206370	LKWD CT3	22.7630	Adder	26.78
206403	28W4-009	48.6443	50/50	48.6443
206412	28R11	132.5905	Adder	155.99
206414	28X4-005E	7.5766	50/50	7.5766
207143	28BYD_X4-031	0.3133	Adder	0.37
207400	28AA1-060 E	1.4939	Adder	1.76
207419	AB2-139 E	0.0524	Adder	0.06
207420	AC1-207 C	0.6668	Adder	0.78
901032	W1-024E OP1	0.2947	Adder	0.35
901112	W1-032 E OP1	0.1784	Adder	0.21
901422	W1-113 E	0.6023	Adder	0.71
901912	W1-112E OP1	0.5187	Adder	0.61
901982	W1-119 E	0.4680	Adder	0.55
901992	W1-120E	0.4680	Adder	0.55
902032	W1-124E	1.8365	Adder	2.16
902082	W1-129E	0.2546	Adder	0.3
902322	W2-019 E	0.3020	Adder	0.36
902912	W2-078 E	0.6744	Adder	0.79
902952	W2-082 E OP1	0.8689	Adder	1.02
903982	W3-079 E	0.5518	Adder	0.65
905252	W4-025 E	0.3614	Adder	0.43
905502	W4-060 E OP1	1.4928	Adder	1.76
907082	X1-037 E	1.5482	Adder	1.82
907272	X1-085 E	0.2607	Adder	0.31
912102	X4-015 E	0.2554	Adder	0.3
914092	Y2-051 E	0.3833	Adder	0.45
917612	Z2-102 E	0.6902	Adder	0.81
919662	AA2-048 E	0.8778	Adder	1.03
920732	AA2-184 E	2.0445	Adder	2.41

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
923293	AB1-138 E	0.6512	Adder	0.77
923463	AB1-163 E	0.5335	Adder	0.63
927132	AC1-207 E	1.1158	Adder	1.31
934841	AD1-113 (Suspended)	28.4122	50/50	28.4122
937513	AD2-210 BAT (Suspended)	0.1766	Merchant Transmission	0.1766
939121	AE1-142 C O1	0.8725	Adder	1.03
939122	AE1-142 E O1	1.2556	Adder	1.48
939981	AE1-238 C	38.2940	Adder	45.05
939982	AE1-238 E	100.5855	Adder	118.34
940161	AE2-000 C O1	22.5564	Adder	26.54
940162	AE2-000 E O1	57.7140	Adder	67.9
940401	AE2-024 C O1	18.2942	Adder	21.52
940402	AE2-024 E O1	85.6517	Adder	100.77
940411	AE2-025 C	9.2349	Adder	10.86
940412	AE2-025 E	43.2330	Adder	50.86
940691	AE2-056 C	0.2344	Adder	0.28
940692	AE2-056 E	0.2051	Adder	0.24
940701	AE2-057 C	0.0735	Adder	0.09
940702	AE2-057 E	0.0735	Adder	0.09
940921	AE2-081 C	0.2491	Adder	0.29
940922	AE2-081 E	0.2564	Adder	0.3
940931	AE2-082 C	0.4347	Adder	0.51
940932	AE2-082 E	0.4603	Adder	0.54
943521	AF1-023	1.4805	Adder	1.74
943561	AF1-027	0.1660	Adder	0.2
944332	AF1-101 E O1	56.5827	Adder	66.57
944611	AF1-126 C O1	3.6525	Adder	4.3
944612	AF1-126 E O1	14.6098	Adder	17.19
945201	AF1-185 1	1.7047	50/50	1.7047
945211	AF1-185 2	0.3788	50/50	0.3788
945571	AF1-222 C	23.8699	Adder	28.08
945572	AF1-222 E	62.9298	Adder	74.04
945951	AF1-260	0.1254	Adder	0.15
NEWTON	NEWTON	0.4169	Confirmed LTF	0.4169
FARMERCITY	FARMERCITY	0.0216	Confirmed LTF	0.0216
G-007A	G-007A	90.8245	Confirmed LTF	90.8245
NY	NY	1.7895	Confirmed LTF	1.7895
PRAIRIE	PRAIRIE	0.9919	Confirmed LTF	0.9919
O-066	O-066	41.6438	Confirmed LTF	41.6438
COFFEEN	COFFEEN	0.0774	Confirmed LTF	0.0774
CHEOAH	CHEOAH	0.1827	Confirmed LTF	0.1827
EDWARDS	EDWARDS	0.1368	Confirmed LTF	0.1368
TILTON	TILTON	0.2463	Confirmed LTF	0.2463
MADISON	MADISON	0.0121	Confirmed LTF	0.0121
GIBSON	GIBSON	0.2124	Confirmed LTF	0.2124
CALDERWOOD	CALDERWOOD	0.1819	Confirmed LTF	0.1819
BLUEG	BLUEG	0.6736	Confirmed LTF	0.6736
TRIMBLE	TRIMBLE	0.2165	Confirmed LTF	0.2165
CATAWBA	CATAWBA	0.1215	Confirmed LTF	0.1215

11.7.11 Index 11

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42768396	206309	28SMITHBRG	JCP&L	206326	28E WINDSR	JCP&L	1	PS_P7-1_1LINE+2LINE	tower	1394.0	103.85	119.49	AC	220.05

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	2.3725	Adder	2.79
206325	28O C GEN (Deactivation : 17/09/2018)	64.4532	Adder	75.83
206327	28S RIV G1	5.5886	50/50	5.5886
206328	28S RIV G2	5.5886	50/50	5.5886
206329	28S RIV G3	7.4691	50/50	7.4691
206358	28PARLN1&2	3.5637	50/50	3.5637
206359	28PARLN3&4	4.3081	50/50	4.3081
206362	28RDOAKCT1	13.2577	50/50	13.2577
206363	28RDOAKCT2	13.2584	50/50	13.2584
206364	28RDOAKCT3	13.2584	50/50	13.2584
206365	28RDOAKST1	18.5913	50/50	18.5913
206370	LKWD CT3	68.1292	Adder	80.15
206403	28W4-009	55.1905	50/50	55.1905
206412	28R11	189.3408	50/50	189.3408
206414	28X4-005E	7.3068	Adder	8.6
206416	28MLH_W1-032	0.0943	50/50	0.0943
207143	28BYD_X4-031	0.7203	Adder	0.85
207204	28HOL_W1-112	0.1161	50/50	0.1161
207205	28MAN_W1-024	0.1175	50/50	0.1175
207206	28TIN_W1-124	0.4874	50/50	0.4874
207400	28AA1-060 E	2.9596	Adder	3.48
207419	AB2-139 E	0.1151	Adder	0.14
207420	AC1-207 C	1.5466	Adder	1.82
901032	W1-024E OP1	1.0916	50/50	1.0916
901112	W1-032 E OP1	0.9005	50/50	0.9005
901422	W1-113 E	1.0173	Adder	1.2
901912	W1-112E OP1	1.0695	50/50	1.0695
901982	W1-119 E	1.2147	Adder	1.43
901992	W1-120E	1.2147	Adder	1.43
902032	W1-124E	4.5152	50/50	4.5152
902082	W1-129E	0.6407	Adder	0.75
902322	W2-019 E	0.7562	Adder	0.89
902912	W2-078 E	1.1383	Adder	1.34
902952	W2-082 E OP1	2.1988	Adder	2.59
903982	W3-079 E	1.5246	Adder	1.79
905252	W4-025 E	0.9093	Adder	1.07
905502	W4-060 E OP1	2.2549	Adder	2.65
907082	X1-037 E	3.8094	Adder	4.48
907272	X1-085 E	0.4228	Adder	0.5

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
912102	X4-015 E	0.6442	Adder	0.76
914092	Y2-051 E	1.0861	Adder	1.28
917612	Z2-102 E	1.7903	Adder	2.11
919662	AA2-048 E	2.4255	Adder	2.85
920732	AA2-184 E	4.9894	50/50	4.9894
923293	AB1-138 E	1.8813	Adder	2.21
923463	AB1-163 E	1.4826	Adder	1.74
927132	AC1-207 E	2.5882	Adder	3.04
934841	AD1-113 (Suspended)	27.4004	Adder	32.24
939121	AE1-142 C O1	2.5715	Adder	3.03
939122	AE1-142 E O1	3.7005	Adder	4.35
939981	AE1-238 C	97.0830	50/50	97.0830
939982	AE1-238 E	255.0047	50/50	255.0047
940161	AE2-000 C O1	65.7434	Adder	77.35
940162	AE2-000 E O1	168.2148	Adder	197.9
940401	AE2-024 C O1	55.1150	Adder	64.84
940402	AE2-024 E O1	258.0422	Adder	303.58
940411	AE2-025 C	27.8220	Adder	32.73
940412	AE2-025 E	130.2478	Adder	153.23
940691	AE2-056 C	0.5784	Adder	0.68
940692	AE2-056 E	0.5061	Adder	0.6
940701	AE2-057 C	0.1819	Adder	0.21
940702	AE2-057 E	0.1819	Adder	0.21
940921	AE2-081 C	0.6145	Adder	0.72
940922	AE2-081 E	0.6326	Adder	0.74
940931	AE2-082 C	1.0974	Adder	1.29
940932	AE2-082 E	1.1619	Adder	1.37
943521	AF1-023	3.6130	50/50	3.6130
943561	AF1-027	0.3288	Adder	0.39
944332	AF1-101 E O1	164.9176	Adder	194.02
944611	AF1-126 C O1	9.7967	Adder	11.53
944612	AF1-126 E O1	39.1867	Adder	46.1
945201	AF1-185 1	1.6440	Adder	1.93
945211	AF1-185 2	0.3653	Adder	0.43
945571	AF1-222 C	60.5151	50/50	60.5151
945572	AF1-222 E	159.5397	50/50	159.5397
945951	AF1-260	0.3465	Adder	0.41
NEWTON	NEWTON	0.7339	Confirmed LTF	0.7339
FARMERCITY	FARMERCITY	0.0382	Confirmed LTF	0.0382
G-007A	G-007A	103.0469	Confirmed LTF	103.0469
NY	NY	0.7233	Confirmed LTF	0.7233
PRAIRIE	PRAIRIE	1.7616	Confirmed LTF	1.7616
O-066	O-066	11.7331	Confirmed LTF	11.7331
COFFEEN	COFFEEN	0.1363	Confirmed LTF	0.1363
CHEOAH	CHEOAH	0.3383	Confirmed LTF	0.3383
EDWARDS	EDWARDS	0.2394	Confirmed LTF	0.2394
TILTON	TILTON	0.4309	Confirmed LTF	0.4309
MADISON	MADISON	0.0040	Confirmed LTF	0.0040
GIBSON	GIBSON	0.3735	Confirmed LTF	0.3735
CALDERWOOD	CALDERWOOD	0.3360	Confirmed LTF	0.3360
BLUEG	BLUEG	1.1874	Confirmed LTF	1.1874
TRIMBLE	TRIMBLE	0.3806	Confirmed LTF	0.3806

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
CATAWBA	CATAWBA	0.2345	Confirmed LTF	0.2345

11.7.12 Index 12

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42236930	206314	28RED OAKA	JCP&L	206305	28RAR RVR	JCP&L	1	JC-P2-3-JCC-230-26B	breaker	869.0	164.83	182.26	AC	151.97

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206201	28JERSVGEN	0.0656	50/50	0.0656
206271	28MCRC/REC	1.8971	50/50	1.8971
206325	28O C GEN (Deactivation : 17/09/2018)	29.8295	Adder	35.09
206327	28S RIV G1	7.9615	50/50	7.9615
206328	28S RIV G2	7.9615	50/50	7.9615
206329	28S RIV G3	10.6406	50/50	10.6406
206358	28PARLN1&2	4.8084	50/50	4.8084
206359	28PARLN3&4	5.8128	50/50	5.8128
206363	28RDOAKCT2	19.4644	50/50	19.4644
206364	28RDOAKCT3	19.4644	50/50	19.4644
206366	28LKWD CT1	5.3948	50/50	5.3948
206367	28LKWD CT2	5.3948	50/50	5.3948
206370	LKWD CT3	30.8040	Adder	36.24
206412	28R11	269.7376	50/50	269.7376
207143	28BYD_X4-031	0.4965	50/50	0.4965
207144	28HOW_X1-037	0.2330	50/50	0.2330
207145	28FRN_Y2-051	0.0662	50/50	0.0662
207148	28MDF_W3-079	0.0944	50/50	0.0944
207204	28HOL_W1-112	0.0820	50/50	0.0820
207206	28TIN_W1-124	0.3339	50/50	0.3339
207400	28AA1-060 E	1.6342	Adder	1.92
901032	W1-024E OP1	0.1991	Adder	0.23
901112	W1-032 E OP1	0.1689	Adder	0.2
901912	W1-112E OP1	0.7551	50/50	0.7551
901982	W1-119 E	0.5986	Adder	0.7
901992	W1-120E	0.5986	Adder	0.7
902032	W1-124E	3.0927	50/50	3.0927
902082	W1-129E	0.3216	Adder	0.38
902322	W2-019 E	0.3807	Adder	0.45
902952	W2-082 E OP1	1.1000	Adder	1.29
903982	W3-079 E	0.8815	50/50	0.8815
905252	W4-025 E	0.4564	Adder	0.54
907082	X1-037 E	2.1464	50/50	2.1464
907272	X1-085 E	0.2553	Adder	0.3
912102	X4-015 E	0.3229	Adder	0.38
914092	Y2-051 E	0.6101	50/50	0.6101
917612	Z2-102 E	0.8827	Adder	1.04
919662	AA2-048 E	1.4025	50/50	1.4025

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
920732	AA2-184 E	3.4247	50/50	3.4247
923293	AB1-138 E	0.8749	Adder	1.03
923463	AB1-163 E	0.7101	Adder	0.84
939121	AE1-142 C O1	1.1778	Adder	1.39
939122	AE1-142 E O1	1.6949	Adder	1.99
939981	AE1-238 C	67.0455	50/50	67.0455
939982	AE1-238 E	176.1062	50/50	176.1062
940161	AE2-000 C O1	30.4266	Adder	35.8
940162	AE2-000 E O1	77.8513	Adder	91.59
940401	AE2-024 C O1	29.1506	50/50	29.1506
940402	AE2-024 E O1	136.4801	50/50	136.4801
940411	AE2-025 C	14.7152	50/50	14.7152
940412	AE2-025 E	68.8889	50/50	68.8889
940691	AE2-056 C	0.3230	50/50	0.3230
940692	AE2-056 E	0.2826	50/50	0.2826
940701	AE2-057 C	0.0926	50/50	0.0926
940702	AE2-057 E	0.0926	50/50	0.0926
940921	AE2-081 C	0.3432	50/50	0.3432
940922	AE2-081 E	0.3533	50/50	0.3533
943521	AF1-023	2.4799	50/50	2.4799
943561	AF1-027	0.1816	Adder	0.21
944332	AF1-101 E O1	76.3253	Adder	89.79
944611	AF1-126 C O1	4.8454	Adder	5.7
944612	AF1-126 E O1	19.3817	Adder	22.8
945201	AF1-185 1	-1.0984	Adder	-1.29
945211	AF1-185 2	-0.2441	Adder	-0.29
945571	AF1-222 C	41.7917	50/50	41.7917
945572	AF1-222 E	110.1781	50/50	110.1781
945951	AF1-260	0.2004	50/50	0.2004
NEWTON	NEWTON	0.2772	Confirmed LTF	0.2772
FARMERCITY	FARMERCITY	0.0143	Confirmed LTF	0.0143
NY	NY	1.5412	Confirmed LTF	1.5412
PRAIRIE	PRAIRIE	0.6587	Confirmed LTF	0.6587
O-066	O-066	33.6941	Confirmed LTF	33.6941
COFFEEN	COFFEEN	0.0517	Confirmed LTF	0.0517
CHEOAH	CHEOAH	0.1176	Confirmed LTF	0.1176
EDWARDS	EDWARDS	0.0917	Confirmed LTF	0.0917
TILTON	TILTON	0.1644	Confirmed LTF	0.1644
G-007	G-007	29.8657	Confirmed LTF	29.8657
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.1414	Confirmed LTF	0.1414
CALDERWOOD	CALDERWOOD	0.1173	Confirmed LTF	0.1173
BLUEG	BLUEG	0.4496	Confirmed LTF	0.4496
TRIMBLE	TRIMBLE	0.1441	Confirmed LTF	0.1441
CATAWBA	CATAWBA	0.0756	Confirmed LTF	0.0756

11.7.13 Index 13

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42236925	206315	28RED OAKB	JCP&L	206305	28RAR RVR	JCP&L	1	JC-P2-3-JCC-230-26D	breaker	869.0	168.48	185.84	AC	151.26

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206201	28JERSVGEN	0.0655	50/50	0.0655
206271	28MCRC/REC	1.8910	50/50	1.8910
206325	28O C GEN (Deactivation : 17/09/2018)	29.6728	Adder	34.91
206327	28S RIV G1	7.7962	50/50	7.7962
206328	28S RIV G2	7.7962	50/50	7.7962
206329	28S RIV G3	10.4196	50/50	10.4196
206358	28PARLN1&2	4.8745	50/50	4.8745
206359	28PARLN3&4	5.8926	50/50	5.8926
206362	28RDOAKCT1	19.5060	50/50	19.5060
206365	28RDOAKST1	27.3533	50/50	27.3533
206366	28LKWD CT1	5.3674	50/50	5.3674
206367	28LKWD CT2	5.3674	50/50	5.3674
206370	LKWD CT3	30.6476	Adder	36.06
206412	28R11	264.1364	50/50	264.1364
207143	28BYD_X4-031	0.4948	50/50	0.4948
207144	28HOW_X1-037	0.2324	50/50	0.2324
207145	28FRN_Y2-051	0.0659	50/50	0.0659
207148	28MDF_W3-079	0.0940	50/50	0.0940
207204	28HOL_W1-112	0.0821	50/50	0.0821
207206	28TIN_W1-124	0.3327	50/50	0.3327
207400	28AA1-060 E	1.6247	Adder	1.91
901032	W1-024E OP1	0.1981	Adder	0.23
901112	W1-032 E OP1	0.1679	Adder	0.2
901912	W1-112E OP1	0.7567	50/50	0.7567
901982	W1-119 E	0.5953	Adder	0.7
901992	W1-120E	0.5953	Adder	0.7
902032	W1-124E	3.0823	50/50	3.0823
902082	W1-129E	0.3198	Adder	0.38
902322	W2-019 E	0.3785	Adder	0.45
902952	W2-082 E OP1	1.0938	Adder	1.29
903982	W3-079 E	0.8775	50/50	0.8775
905252	W4-025 E	0.4538	Adder	0.53
907082	X1-037 E	2.1407	50/50	2.1407
907272	X1-085 E	0.2538	Adder	0.3
912102	X4-015 E	0.3211	Adder	0.38
914092	Y2-051 E	0.6073	50/50	0.6073
917612	Z2-102 E	0.8778	Adder	1.03
919662	AA2-048 E	1.3960	50/50	1.3960

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
920732	AA2-184 E	3.4139	50/50	3.4139
923293	AB1-138 E	0.8704	Adder	1.02
923463	AB1-163 E	0.7065	Adder	0.83
939121	AE1-142 C O1	1.1717	Adder	1.38
939122	AE1-142 E O1	1.6860	Adder	1.98
939981	AE1-238 C	66.7305	50/50	66.7305
939982	AE1-238 E	175.2788	50/50	175.2788
940161	AE2-000 C O1	30.2668	Adder	35.61
940162	AE2-000 E O1	77.4423	Adder	91.11
940401	AE2-024 C O1	29.0032	50/50	29.0032
940402	AE2-024 E O1	135.7897	50/50	135.7897
940411	AE2-025 C	14.6408	50/50	14.6408
940412	AE2-025 E	68.5404	50/50	68.5404
940691	AE2-056 C	0.3221	50/50	0.3221
940692	AE2-056 E	0.2819	50/50	0.2819
940701	AE2-057 C	0.0924	50/50	0.0924
940702	AE2-057 E	0.0924	50/50	0.0924
940921	AE2-081 C	0.3423	50/50	0.3423
940922	AE2-081 E	0.3523	50/50	0.3523
943521	AF1-023	2.4721	50/50	2.4721
943561	AF1-027	0.1805	Adder	0.21
944332	AF1-101 E O1	75.9244	Adder	89.32
944611	AF1-126 C O1	4.8169	Adder	5.67
944612	AF1-126 E O1	19.2678	Adder	22.67
945201	AF1-185 1	-1.0951	Adder	-1.29
945211	AF1-185 2	-0.2434	Adder	-0.29
945571	AF1-222 C	41.5953	50/50	41.5953
945572	AF1-222 E	109.6605	50/50	109.6605
945951	AF1-260	0.1994	50/50	0.1994
NEWTON	NEWTON	0.3009	Confirmed LTF	0.3009
FARMERCITY	FARMERCITY	0.0156	Confirmed LTF	0.0156
NY	NY	1.5478	Confirmed LTF	1.5478
PRAIRIE	PRAIRIE	0.7155	Confirmed LTF	0.7155
O-066	O-066	33.7210	Confirmed LTF	33.7210
COFFEEN	COFFEEN	0.0559	Confirmed LTF	0.0559
CHEOAH	CHEOAH	0.1286	Confirmed LTF	0.1286
EDWARDS	EDWARDS	0.0994	Confirmed LTF	0.0994
TILTON	TILTON	0.1783	Confirmed LTF	0.1783
G-007	G-007	29.7762	Confirmed LTF	29.7762
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.1534	Confirmed LTF	0.1534
CALDERWOOD	CALDERWOOD	0.1282	Confirmed LTF	0.1282
BLUEG	BLUEG	0.4878	Confirmed LTF	0.4878
TRIMBLE	TRIMBLE	0.1564	Confirmed LTF	0.1564
CATAWBA	CATAWBA	0.0833	Confirmed LTF	0.0833

11.7.14 Index 14

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41034357	206316	28WINDSOR	JCP&L	219752	CLRKSVLL_1	PSE&G	1	PS_P7-1_1LINE+2LINE	tower	813.0	126.48	136.74	AC	97.95

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206201	28JERSVGEN	0.0713	50/50	0.0713
206271	28MCRC/REC	1.0626	Adder	1.25
206325	28O C GEN (Deactivation : 17/09/2018)	28.6639	Adder	33.72
206350	28RRCT1&2	2.8687	50/50	2.8687
206351	28RRCT3&4	2.6862	50/50	2.6862
206362	28RDOAKCT1	6.0788	50/50	6.0788
206365	28RDOAKST1	8.5243	50/50	8.5243
206368	28MDLSEXCO	0.7120	50/50	0.7120
206370	LKWD CT3	30.0220	Adder	35.32
206403	28W4-009	25.3899	50/50	25.3899
206412	28R11	73.7191	Adder	86.73
206414	28X4-005E	3.9546	50/50	3.9546
206416	28MLH_W1-032	0.0416	50/50	0.0416
207143	28BYD_X4-031	0.3252	Adder	0.38
207152	28NPK_W1-113	0.1326	50/50	0.1326
207162	28APLW2-078	0.1484	50/50	0.1484
207169	28W4-060COL	0.1341	50/50	0.1341
207205	28MAN_W1-024	0.0582	50/50	0.0582
207400	28AA1-060 E	2.4094	Adder	2.83
207419	AB2-139 E	0.0979	50/50	0.0979
207420	AC1-207 C	1.1006	50/50	1.1006
219679	NRGCENTER E	-0.3193	Adder	-0.38
901032	W1-024E OP1	0.5401	50/50	0.5401
901112	W1-032 E OP1	0.3973	50/50	0.3973
901422	W1-113 E	1.2314	50/50	1.2314
901912	W1-112E OP1	0.4117	Adder	0.48
901982	W1-119 E	0.6146	Adder	0.72
901992	W1-120E	0.6146	Adder	0.72
902032	W1-124E	1.7178	Adder	2.02
902082	W1-129E	0.3351	Adder	0.39
902322	W2-019 E	0.3977	Adder	0.47
902912	W2-078 E	1.3794	50/50	1.3794
902952	W2-082 E OP1	1.1433	Adder	1.35
903982	W3-079 E	0.6788	Adder	0.8
905252	W4-025 E	0.4757	Adder	0.56
905502	W4-060 E OP1	1.2352	50/50	1.2352
907082	X1-037 E	1.7604	Adder	2.07
907272	X1-085 E	0.4739	Adder	0.56
912102	X4-015 E	0.3361	Adder	0.4

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
914092	Y2-051 E	0.4813	Adder	0.57
917612	Z2-102 E	0.9064	Adder	1.07
919662	AA2-048 E	1.0799	Adder	1.27
919712	AA2-058 E	-0.1713	Adder	-0.2
919802	AA2-066 E	-0.3193	Adder	-0.38
920732	AA2-184 E	1.9000	Adder	2.24
923293	AB1-138 E	0.8425	Adder	0.99
923463	AB1-163 E	0.6828	Adder	0.8
927132	AC1-207 E	1.8419	50/50	1.8419
934841	AD1-113 (Suspended)	14.8298	50/50	14.8298
937523	AD2-213 BAT	1.8022	Merchant Transmission	1.8022
939121	AE1-142 C O1	1.1390	Adder	1.34
939122	AE1-142 E O1	1.6390	Adder	1.93
939981	AE1-238 C	36.7315	Adder	43.21
939982	AE1-238 E	96.4813	Adder	113.51
940161	AE2-000 C O1	29.2377	Adder	34.4
940162	AE2-000 E O1	74.8092	Adder	88.01
940401	AE2-024 C O1	24.2437	Adder	28.52
940402	AE2-024 E O1	113.5062	Adder	133.54
940411	AE2-025 C	12.2382	Adder	14.4
940412	AE2-025 E	57.2927	Adder	67.4
940691	AE2-056 C	0.2675	Adder	0.31
940692	AE2-056 E	0.2341	Adder	0.28
940701	AE2-057 C	0.1007	50/50	0.1007
940702	AE2-057 E	0.1007	50/50	0.1007
940921	AE2-081 C	0.2842	Adder	0.33
940922	AE2-081 E	0.2926	Adder	0.34
940931	AE2-082 C	0.7623	50/50	0.7623
940932	AE2-082 E	0.8072	50/50	0.8072
943521	AF1-023	1.3759	Adder	1.62
943561	AF1-027	0.2677	Adder	0.31
944332	AF1-101 E O1	73.3429	Adder	86.29
944432	AF1-108 BAT	1.8618	Merchant Transmission	1.8618
944442	AF1-109 BAT	5.4178	50/50	5.4178
944611	AF1-126 C O1	4.2215	Adder	4.97
944612	AF1-126 E O1	16.8861	Adder	19.87
945201	AF1-185 1	0.8898	50/50	0.8898
945211	AF1-185 2	0.1977	50/50	0.1977
945571	AF1-222 C	22.8960	Adder	26.94
945572	AF1-222 E	60.3621	Adder	71.01
945723	AF1-237 BAT	38.8140	Merchant Transmission	38.8140
945951	AF1-260	0.1543	Adder	0.18
NEWTON	NEWTON	0.2579	Confirmed LTF	0.2579
FARMERCITY	FARMERCITY	0.0134	Confirmed LTF	0.0134
G-007A	G-007A	47.4058	Confirmed LTF	47.4058
NY	NY	0.2124	Confirmed LTF	0.2124
PRAIRIE	PRAIRIE	0.6199	Confirmed LTF	0.6199
O-066	O-066	7.3315	Confirmed LTF	7.3315
COFFEEN	COFFEEN	0.0479	Confirmed LTF	0.0479
CHEOAH	CHEOAH	0.1196	Confirmed LTF	0.1196
EDWARDS	EDWARDS	0.0840	Confirmed LTF	0.0840
TILTON	TILTON	0.1512	Confirmed LTF	0.1512

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
GIBSON	GIBSON	0.1310	Confirmed LTF	0.1310
CALDERWOOD	CALDERWOOD	0.1188	Confirmed LTF	0.1188
BLUEG	BLUEG	0.4166	Confirmed LTF	0.4166
TRIMBLE	TRIMBLE	0.1336	Confirmed LTF	0.1336
CATAWBA	CATAWBA	0.0833	Confirmed LTF	0.0833

11.7.15 Index 15

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42768367	206326	28E WINDSR	JCP&L	206316	28WINDSOR	JCP&L	1	PS_P7-1_1LINE+2LINE	tower	869.0	126.0	135.89	AC	87.12

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206271	28MCRC/REC	1.0956	50/50	1.0956
206325	28O C GEN (Deactivation : 17/09/2018)	25.9792	Adder	30.56
206327	28S RIV G1	2.1249	50/50	2.1249
206328	28S RIV G2	2.1249	50/50	2.1249
206329	28S RIV G3	2.8400	50/50	2.8400
206358	28PARLN1&2	1.3584	50/50	1.3584
206359	28PARLN3&4	1.6421	50/50	1.6421
206362	28RDOAKCT1	5.0341	50/50	5.0341
206363	28RDOAKCT2	5.0353	50/50	5.0353
206364	28RDOAKCT3	5.0353	50/50	5.0353
206365	28RDOAKST1	7.0593	50/50	7.0593
206370	LKWD CT3	27.3292	Adder	32.15
206403	28W4-009	20.8507	50/50	20.8507
206412	28R11	71.9928	50/50	71.9928
206414	28X4-005E	2.7605	Adder	3.25
206416	28MLH_W1-032	0.0381	50/50	0.0381
207143	28BYD_X4-031	0.3287	50/50	0.3287
207145	28FRN_Y2-051	0.0551	50/50	0.0551
207204	28HOL_W1-112	0.0449	50/50	0.0449
207206	28TIN_W1-124	0.1915	50/50	0.1915
219679	NRG CENTER E	-0.3114	Adder	-0.37
901032	W1-024E OP1	0.3138	Adder	0.37
901112	W1-032 E OP1	0.3634	50/50	0.3634
901912	W1-112E OP1	0.4135	50/50	0.4135
902032	W1-124E	1.7739	50/50	1.7739
903982	W3-079 E	0.6002	Adder	0.71
905502	W4-060 E OP1	0.8376	Adder	0.99
907082	X1-037 E	1.4308	Adder	1.68
914092	Y2-051 E	0.5075	50/50	0.5075
919662	AA2-048 E	0.9548	Adder	1.12
919712	AA2-058 E	-0.1674	Adder	-0.2
919802	AA2-066 E	-0.3114	Adder	-0.37
920732	AA2-184 E	1.9576	50/50	1.9576
923293	AB1-138 E	0.7324	Adder	0.86
923463	AB1-163 E	0.5447	Adder	0.64
934841	AD1-113 (Suspended)	10.3517	Adder	12.18
937523	AD2-213 BAT	1.7466	Merchant Transmission	1.7466
939121	AE1-142 C O1	1.0267	Adder	1.21
939122	AE1-142 E O1	1.4774	Adder	1.74

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939981	AE1-238 C	38.4345	50/50	38.4345
939982	AE1-238 E	100.9546	50/50	100.9546
940161	AE2-000 C O1	26.4993	Adder	31.18
940162	AE2-000 E O1	67.8026	Adder	79.77
940401	AE2-024 C O1	26.0228	50/50	26.0228
940402	AE2-024 E O1	121.8357	50/50	121.8357
940411	AE2-025 C	13.1363	50/50	13.1363
940412	AE2-025 E	61.4971	50/50	61.4971
940691	AE2-056 C	0.2169	Adder	0.26
940692	AE2-056 E	0.1898	Adder	0.22
940701	AE2-057 C	0.0666	Adder	0.08
940702	AE2-057 E	0.0666	Adder	0.08
940921	AE2-081 C	0.2305	Adder	0.27
940922	AE2-081 E	0.2373	Adder	0.28
940931	AE2-082 C	0.2438	Adder	0.29
940932	AE2-082 E	0.2581	Adder	0.3
943521	AF1-023	1.4176	50/50	1.4176
944332	AF1-101 E O1	66.4736	Adder	78.2
944432	AF1-108 BAT	1.8050	Merchant Transmission	1.8050
944442	AF1-109 BAT	5.2966	50/50	5.2966
944611	AF1-126 C O1	3.4650	Adder	4.08
944612	AF1-126 E O1	13.8601	Adder	16.31
945201	AF1-185 1	0.6211	Adder	0.73
945211	AF1-185 2	0.1380	Adder	0.16
945571	AF1-222 C	23.9575	50/50	23.9575
945572	AF1-222 E	63.1607	50/50	63.1607
945723	AF1-237 BAT	37.9620	50/50	37.9620
945951	AF1-260	0.1364	Adder	0.16
NEWTON	NEWTON	0.2020	Confirmed LTF	0.2020
FARMERCITY	FARMERCITY	0.0105	Confirmed LTF	0.0105
G-007A	G-007A	38.9330	Confirmed LTF	38.9330
NY	NY	0.1488	Confirmed LTF	0.1488
PRAIRIE	PRAIRIE	0.4856	Confirmed LTF	0.4856
O-066	O-066	6.3235	Confirmed LTF	6.3235
COFFEEN	COFFEEN	0.0375	Confirmed LTF	0.0375
CHEOAH	CHEOAH	0.0941	Confirmed LTF	0.0941
EDWARDS	EDWARDS	0.0658	Confirmed LTF	0.0658
TILTON	TILTON	0.1184	Confirmed LTF	0.1184
GIBSON	GIBSON	0.1026	Confirmed LTF	0.1026
CALDERWOOD	CALDERWOOD	0.0934	Confirmed LTF	0.0934
BLUEG	BLUEG	0.3264	Confirmed LTF	0.3264
TRIMBLE	TRIMBLE	0.1046	Confirmed LTF	0.1046
CATAWBA	CATAWBA	0.0658	Confirmed LTF	0.0658

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237050	206410	28R11RINGB	JCP&L	206315	28RED OAKB	JCP&L	1	JC-P2-3-JCC-230-26D	breaker	869.0	124.97	142.46	AC	152.36

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206201	28JERSVGEN	0.0662	50/50	0.0662
206271	28MCRC/REC	1.9051	50/50	1.9051
206306	28LKWD G1	2.6950	50/50	2.6950
206308	28LKWD G2	2.6950	50/50	2.6950
206325	28O C GEN (Deactivation : 17/09/2018)	30.0875	Adder	35.4
206327	28S RIV G1	7.8244	50/50	7.8244
206328	28S RIV G2	7.8244	50/50	7.8244
206329	28S RIV G3	10.4573	50/50	10.4573
206358	28PARLN1&2	4.8924	50/50	4.8924
206359	28PARLN3&4	5.9144	50/50	5.9144
206366	28LKWD CT1	5.4320	50/50	5.4320
206367	28LKWD CT2	5.4320	50/50	5.4320
206370	LKWD CT3	31.0165	Adder	36.49
206412	28R11	265.0912	50/50	265.0912
207143	28BYD_X4-031	0.4992	50/50	0.4992
207144	28HOW_X1-037	0.2349	50/50	0.2349
207145	28FRN_Y2-051	0.0667	50/50	0.0667
207148	28MDF_W3-079	0.0950	50/50	0.0950
207204	28HOL_W1-112	0.0827	50/50	0.0827
207206	28TIN_W1-124	0.3352	50/50	0.3352
207400	28AA1-060 E	1.6579	Adder	1.95
901032	W1-024E OP1	0.2027	Adder	0.24
901112	W1-032 E OP1	0.1713	Adder	0.2
901912	W1-112E OP1	0.7621	50/50	0.7621
901982	W1-119 E	0.6054	Adder	0.71
901992	W1-120E	0.6054	Adder	0.71
902032	W1-124E	3.1051	50/50	3.1051
902082	W1-129E	0.3255	Adder	0.38
902322	W2-019 E	0.3854	Adder	0.45
902952	W2-082 E OP1	1.1131	Adder	1.31
903982	W3-079 E	0.8870	50/50	0.8870
905252	W4-025 E	0.4620	Adder	0.54
907082	X1-037 E	2.1635	50/50	2.1635
907272	X1-085 E	0.2595	Adder	0.31
912102	X4-015 E	0.3268	Adder	0.38
914092	Y2-051 E	0.6140	50/50	0.6140
917612	Z2-102 E	0.8926	Adder	1.05
919662	AA2-048 E	1.4112	50/50	1.4112

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
920732	AA2-184 E	3.4391	50/50	3.4391
923293	AB1-138 E	0.8815	Adder	1.04
923463	AB1-163 E	0.7157	Adder	0.84
939121	AE1-142 C O1	1.1868	Adder	1.4
939122	AE1-142 E O1	1.7078	Adder	2.01
939981	AE1-238 C	67.2188	50/50	67.2188
939982	AE1-238 E	176.5613	50/50	176.5613
940161	AE2-000 C O1	30.6897	Adder	36.11
940162	AE2-000 E O1	78.5245	Adder	92.38
940401	AE2-024 C O1	29.3385	50/50	29.3385
940402	AE2-024 E O1	137.3595	50/50	137.3595
940411	AE2-025 C	14.8100	50/50	14.8100
940412	AE2-025 E	69.3328	50/50	69.3328
940691	AE2-056 C	0.3256	50/50	0.3256
940692	AE2-056 E	0.2849	50/50	0.2849
940701	AE2-057 C	0.0935	50/50	0.0935
940702	AE2-057 E	0.0935	50/50	0.0935
940921	AE2-081 C	0.3460	50/50	0.3460
940922	AE2-081 E	0.3561	50/50	0.3561
943521	AF1-023	2.4903	50/50	2.4903
943561	AF1-027	0.1842	Adder	0.22
944332	AF1-101 E O1	76.9853	Adder	90.57
944611	AF1-126 C O1	4.9092	Adder	5.78
944612	AF1-126 E O1	19.6367	Adder	23.1
945201	AF1-185 1	-1.0868	Adder	-1.28
945211	AF1-185 2	-0.2415	Adder	-0.28
945571	AF1-222 C	41.8997	50/50	41.8997
945572	AF1-222 E	110.4628	50/50	110.4628
945951	AF1-260	0.2016	50/50	0.2016
NEWTON	NEWTON	0.0677	Confirmed LTF	0.0677
FARMERCITY	FARMERCITY	0.0034	Confirmed LTF	0.0034
NY	NY	1.4267	Confirmed LTF	1.4267
PRAIRIE	PRAIRIE	0.1550	Confirmed LTF	0.1550
O-066	O-066	32.2627	Confirmed LTF	32.2627
COFFEEN	COFFEEN	0.0126	Confirmed LTF	0.0126
CHEOAH	CHEOAH	0.0200	Confirmed LTF	0.0200
EDWARDS	EDWARDS	0.0234	Confirmed LTF	0.0234
TILTON	TILTON	0.0416	Confirmed LTF	0.0416
G-007	G-007	29.5506	Confirmed LTF	29.5506
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.0349	Confirmed LTF	0.0349
CALDERWOOD	CALDERWOOD	0.0204	Confirmed LTF	0.0204
BLUEG	BLUEG	0.1111	Confirmed LTF	0.1111
TRIMBLE	TRIMBLE	0.0356	Confirmed LTF	0.0356
CATAWBA	CATAWBA	0.0073	Confirmed LTF	0.0073

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237041	206411	28R11RINGA	JCP&L	206314	28RED OAKA	JCP&L	1	JC-P2-3-JCC-230-26B	breaker	869.0	129.08	146.65	AC	152.89

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206201	28JERSVGEN	0.0662	50/50	0.0662
206271	28MCRC/REC	1.9089	50/50	1.9089
206306	28LKWD G1	2.7033	50/50	2.7033
206308	28LKWD G2	2.7033	50/50	2.7033
206325	28O C GEN (Deactivation : 17/09/2018)	30.1754	Adder	35.5
206327	28S RIV G1	7.9849	50/50	7.9849
206328	28S RIV G2	7.9849	50/50	7.9849
206329	28S RIV G3	10.6718	50/50	10.6718
206358	28PARLN1&2	4.8234	50/50	4.8234
206359	28PARLN3&4	5.8309	50/50	5.8309
206366	28LKWD CT1	5.4487	50/50	5.4487
206367	28LKWD CT2	5.4487	50/50	5.4487
206370	LKWD CT3	31.1117	Adder	36.6
206412	28R11	270.5296	50/50	270.5296
207143	28BYD_X4-031	0.5001	50/50	0.5001
207144	28HOW_X1-037	0.2351	50/50	0.2351
207145	28FRN_Y2-051	0.0668	50/50	0.0668
207148	28MDF_W3-079	0.0952	50/50	0.0952
207204	28HOL_W1-112	0.0824	50/50	0.0824
207206	28TIN_W1-124	0.3359	50/50	0.3359
207400	28AA1-060 E	1.6619	Adder	1.96
901032	W1-024E OP1	0.2029	Adder	0.24
901112	W1-032 E OP1	0.1717	Adder	0.2
901912	W1-112E OP1	0.7595	50/50	0.7595
901982	W1-119 E	0.6070	Adder	0.71
901992	W1-120E	0.6070	Adder	0.71
902032	W1-124E	3.1117	50/50	3.1117
902082	W1-129E	0.3263	Adder	0.38
902322	W2-019 E	0.3864	Adder	0.45
902952	W2-082 E OP1	1.1161	Adder	1.31
903982	W3-079 E	0.8895	50/50	0.8895
905252	W4-025 E	0.4632	Adder	0.54
907082	X1-037 E	2.1654	50/50	2.1654
907272	X1-085 E	0.2600	Adder	0.31
912102	X4-015 E	0.3277	Adder	0.39
914092	Y2-051 E	0.6158	50/50	0.6158
917612	Z2-102 E	0.8950	Adder	1.05
919662	AA2-048 E	1.4151	50/50	1.4151

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
920732	AA2-184 E	3.4457	50/50	3.4457
923293	AB1-138 E	0.8841	Adder	1.04
923463	AB1-163 E	0.7177	Adder	0.84
939121	AE1-142 C O1	1.1904	Adder	1.4
939122	AE1-142 E O1	1.7130	Adder	2.02
939981	AE1-238 C	67.4527	50/50	67.4527
939982	AE1-238 E	177.1759	50/50	177.1759
940161	AE2-000 C O1	30.7794	Adder	36.21
940162	AE2-000 E O1	78.7539	Adder	92.65
940401	AE2-024 C O1	29.4301	50/50	29.4301
940402	AE2-024 E O1	137.7883	50/50	137.7883
940411	AE2-025 C	14.8563	50/50	14.8563
940412	AE2-025 E	69.5492	50/50	69.5492
940691	AE2-056 C	0.3259	50/50	0.3259
940692	AE2-056 E	0.2851	50/50	0.2851
940701	AE2-057 C	0.0935	50/50	0.0935
940702	AE2-057 E	0.0935	50/50	0.0935
940921	AE2-081 C	0.3462	50/50	0.3462
940922	AE2-081 E	0.3564	50/50	0.3564
943521	AF1-023	2.4951	50/50	2.4951
943561	AF1-027	0.1847	Adder	0.22
944332	AF1-101 E O1	77.2102	Adder	90.84
944611	AF1-126 C O1	4.9219	Adder	5.79
944612	AF1-126 E O1	19.6877	Adder	23.16
945201	AF1-185 1	-1.0915	Adder	-1.28
945211	AF1-185 2	-0.2426	Adder	-0.29
945571	AF1-222 C	42.0455	50/50	42.0455
945572	AF1-222 E	110.8474	50/50	110.8474
945951	AF1-260	0.2022	50/50	0.2022
NEWTON	NEWTON	0.0827	Confirmed LTF	0.0827
FARMERCITY	FARMERCITY	0.0042	Confirmed LTF	0.0042
NY	NY	1.4400	Confirmed LTF	1.4400
PRAIRIE	PRAIRIE	0.1911	Confirmed LTF	0.1911
O-066	O-066	32.4778	Confirmed LTF	32.4778
COFFEEN	COFFEEN	0.0156	Confirmed LTF	0.0156
CHEOAH	CHEOAH	0.0270	Confirmed LTF	0.0270
EDWARDS	EDWARDS	0.0283	Confirmed LTF	0.0283
TILTON	TILTON	0.0504	Confirmed LTF	0.0504
G-007	G-007	29.6774	Confirmed LTF	29.6774
MADISON	MADISON	0.0101	Confirmed LTF	0.0101
GIBSON	GIBSON	0.0426	Confirmed LTF	0.0426
CALDERWOOD	CALDERWOOD	0.0273	Confirmed LTF	0.0273
BLUEG	BLUEG	0.1354	Confirmed LTF	0.1354
TRIMBLE	TRIMBLE	0.0434	Confirmed LTF	0.0434
CATAWBA	CATAWBA	0.0122	Confirmed LTF	0.0122

11.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AA1-060	Great Adventure 34.5kV	In Service
AA2-048	Allenwood-Larrabee 34kV	Engineering and Procurement
AA2-058	Mount Rose 13kV	Partially in Service - Under Construction
AA2-066	Penns Neck 13kV	In Service
AA2-184	Atlantic-Red Bank 34kV	In Service
AB1-001	Absecon 12 kV	Engineering and Procurement
AB1-138	Navy 34.5kV	In Service
AB1-163	Glidden-Van Hiseville 34.5 kV	Engineering and Procurement
AB2-014	Lakewood 230kV	Partially in Service - Under Construction
AB2-122	Egg Harbor 12kV	Engineering and Procurement
AB2-139	Raceway Mall 34.5kV	In Service
AC1-207	Monroe 34.5kV	Engineering and Procurement
AD1-019	Ontario 23 kV	Active
AD1-059	Lakewood 230 kV	In Service
AD1-113	Raritan River 230 kV	Suspended
AD2-210	Cedar Knolls 12.5 kV	Suspended
AD2-213	East Flemington-Lebanon 34.5 kV	Active
AE1-061	Minotola 12 kV	Active
AE1-104	BL England 138 kV	Active
AE1-142	Manitou-Pleasant Plains 34.5 kV	Engineering and Procurement
AE1-161	Landis 138 kV	Active
AE1-238	Oceanview Wind 230 kV	Active
AE2-000	N/A	N/A
AE2-020	Cardiff 230 kV I	Active
AE2-021	Cardiff 230 kV II	Active
AE2-022	Cardiff 230 kV III	Active
AE2-024	Larrabee 230 kV I	Active
AE2-025	Larrabee 230 kV II	Active
AE2-056	Howell 12.47 kV Solar I	Engineering and Procurement
AE2-057	Jerseyville 12.47 kV Solar	Engineering and Procurement
AE2-081	Howell 12.47 kV Solar II	Engineering and Procurement
AE2-082	Englishtown-Wyckoff 34.5 kV Solar	Active
AE2-222	Higbee 69 kV	Active
AE2-251	Cardiff 230 kV	Active
AF1-023	Atlantic-Red Bank 34 kV II	Active
AF1-027	Plumsted 537 Energy Storage (CIRs)	Active
AF1-041	Absecon 12.47 kV	In Service
AF1-101	Oyster Creek 230 kV III	Active
AF1-108	East Flemington 34.5 kV	Active
AF1-109	Pleasant Valley 230 kV	Active
AF1-126	Cookstown 230 kV	Active

Queue Number	Project Name	Status
AF1-185	Sayreville 1-2-3 230 kV	In Service
AF1-222	Oceanview Wind 2 230 kV	Active
AF1-237	Mercer 230 kV	Active
AF1-238	Sherman Ave. 69 kV	Active
AF1-239	Sherman Ave-Vineland 69 kV	Active
AF1-260	Allenwood-Larrabee 2 34.5 kV	Active
V4-054	Fairfield Township 12kV	In Service
V4-067	Cates Road Egg Harbor Township 12kV	In Service
W1-024	Manalapan 12.5kV	In Service
W1-032	Millhurst 12.5kV	In Service
W1-112	Holmdel	In Service
W1-113	Millstone 2	In Service
W1-119	Pemberton Township 1 34.5 kV	In Service
W1-120	Pemberton Township 2 34.5 kV	In Service
W1-124	Tinton Falls 34.5kV	In Service
W1-129	Cookstown 34.5kV	In Service
W2-019	Wrightstown 34.5kV	In Service
W2-030	Egg Harbor Township	In Service
W2-078	Applegarth 12.5kV	In Service
W2-082	Fort Dix-McGuire 34.5kV	In Service
W3-079	Allenwood-Larabee 34.5kV	In Service
W4-009	Raritan River 230kV	In Service
W4-025	Cookstown-Fort Dix 34.5kV	In Service
W4-060	Midland-Werner 34.5kV	In Service
X1-037	Howell	In Service
X1-085	Hornerstown-Windsor 34kV	In Service
X4-005	Raritan River 230kV	In Service
X4-015	Cookstown-Fort Dix 34kV	In Service
X4-031	Bennett-Farmingdale 34kV	In Service
Y2-051	Brick-Lane Mill 34.5kV	In Service
Z2-102	Argonne-New Lisbon 34.5kV	In Service

11.9 Contingency Descriptions

Contingency Name	Contingency Definition
JC-P2-3-JCC-230-85	CONTINGENCY 'JC-P2-3-JCC-230-85' /* SOUTH RIVER JUNCTION PARLIN- G1047 RARITAN RIVER LINES STUCK BREAKER DISCONNECT BRANCH FROM BUS 206410 TO BUS 206322 CKT 1 /* SOUTH RIVER JUNCTION - PARLIN 230 LINE DISCONNECT BUS 206314 /* RED OAK 230 KV A TAP DISCONNECT BUS 206363 /* RED OAK 230 KV #2 DISCONNECT BUS 206364 /* RED OAK 230 KV #3 END
JC-P2-3-JCC-230-80	CONTINGENCY 'JC-P2-3-JCC-230-80' /* SOUTH RIVER JUNCTION G-T (RARITAN RIVER) LINES STUCK BREAKER DISCONNECT BUS 206314 /* RED OAK 230 KV A TAP DISCONNECT BUS 206363 /* RED OAK 230 KV #2 DISCONNECT BUS 206364 /* RED OAK 230 KV #3 DISCONNECT BUS 206315 /* RED OAK 230 KV B TAP DISCONNECT BUS 206362 /* RED OAK 230 KV #1 DISCONNECT BUS 206365 /* RED OAK 230 KV #4 END
PS_P1-2_#1LINE	CONTINGENCY 'PS_P1-2_#1LINE' /* LAKE NELSON TO RARITAN RIVER DISCONNECT BUS 218331 /* KILMER #1 CLOSE LINE FROM BUS 218384 TO BUS 218387 CKT Z /* KILMER CLOSE LINE FROM BUS 218385 TO BUS 218386 CKT Z /* KILMER END

Contingency Name	Contingency Definition
PS_P2-3_LNEPS_P1-2_1_LT	CONTINGENCY 'PS_P2-3_LNEPS_P1-2_1_LT' /* STUCK BREAKER ON LAKE NELSON I LINE DISCONNECT BUS 218312 /* GREEN BROOK STATION DISCONNECT BUS 218301 /* MIDDLE LOSS OF BUS DISCONNECT BUS 218333 /* LAKE NELSON BUS I DISCONNECT BUS 218331 /* REMOVE KILMER LINE #1 CLOSE LINE FROM BUS 218384 TO BUS 218387 CKT Z /* KILMER CLOSE LINE FROM BUS 218385 TO BUS 218386 CKT Z /* KILMER MOVE 8 MW LOAD FROM BUS 218387 TO BUS 218399 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO NEWDOVR T2 MOVE 8 MW LOAD FROM BUS 218385 TO BUS 218393 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO MDWRD T2 MOVE 8 MW LOAD FROM BUS 218386 TO BUS 218347 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO BNNTLANE T2 CLOSE LINE FROM BUS 218388 TO BUS 218389 CKT Z /* LAKE NELSON MOVE 8 MW LOAD FROM BUS 218389 TO BUS 218384 /* INTERSTATION TIE TRANSFER LOAD FROM LNELSN TO KILMER T2 MOVE 8 MW LOAD FROM BUS 218388 TO BUS 218385 /* INTERSTATION TIE TRANSFER LOAD FROM LNELSN TO KILMER T4 CLOSE LINE FROM BUS 218382 TO BUS 218383 CKT Z /* GREENBROOK MOVE 8 MW LOAD FROM BUS 218382 TO BUS 218406 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO SOMRVLE T2 MOVE 8 MW LOAD FROM BUS 218382 TO BUS 218386 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO KILMER T1 MOVE 8 MW LOAD FROM BUS 218383 TO BUS 218395 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO SOUTH SECOND ST.(METUCHEN 26KV IM) END
JC-P7-1-JCC-230-7A	CONTINGENCY 'JC-P7-1-JCC-230-7A' /* SMITHBURG - LARRABEE 230 LINES DISCONNECT BRANCH FROM BUS 206309 TO BUS 206294 CKT 1 DISCONNECT BRANCH FROM BUS 206309 TO BUS 206294 CKT 2 END
JC-P1-2-JCC-230-031	CONTINGENCY 'JC-P1-2-JCC-230-031' /* VAN HISEVILLE TAP (B1042) 230 KV LINES DISCONNECT BRANCH FROM BUS 206318 TO BUS 206287 CKT 1 DISCONNECT BRANCH FROM BUS 206318 TO BUS 206720 CKT 1 DISCONNECT BRANCH FROM BUS 206720 TO BUS 206319 CKT 1 DISCONNECT BRANCH FROM BUS 206318 TO BUS 206294 CKT 1 DISCONNECT BUS 206318 DISCONNECT BUS 206720 END
JC-P1-2-JCC-230-018	CONTINGENCY 'JC-P1-2-JCC-230-018' /* LARRABEE - SMITHBURG (H2008) 230 KV DISCONNECT BRANCH FROM BUS 206309 TO BUS 206294 CKT 2 DISCONNECT BRANCH FROM BUS 200017 TO BUS 206309 CKT 4 /* SMITHBURG SMITHBURG 500 230 END

Contingency Name	Contingency Definition
JC-P1-2-JCC-230-043T	CONTINGENCY 'JC-P1-2-JCC-230-043T' /** RARITAN RIVER - LAKE NELSON (W1037) 230 KV LINE DISCONNECT BRANCH FROM BUS 206305 TO BUS 218332 CKT 1 /* 28RAR RVR - KILMER_W 230.00 LINE DISCONNECT BRANCH FROM BUS 218332 TO BUS 218334 CKT 1 /* KILMER_W - LNELSN_W 230.00 LINE DISCONNECT BRANCH FROM BUS 218386 TO BUS 218332 CKT 1 /* KILMER_L1 13.20 - KILMER_W 230.00 XFMR DISCONNECT BRANCH FROM BUS 218387 TO BUS 218332 CKT 1 /* KILMER_L3 13.20 - KILMER_W 230.00 XFMR END
JC-P1-2-JCC-230-017	CONTINGENCY 'JC-P1-2-JCC-230-017' /* FUTURE BREAKERS LARRABEE - SMITHBURG (D2004) 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 1 END
PJM500_PS_P1-2_5022	CONTINGENCY 'PJM500_PS_P1-2_5022' /* DEANS TO WINDSOR TRIP LINE FROM BUS 200006 TO BUS 200028 CKT 1 /* DEANS TO WINDSOR END
JC-P2-3-JCC-230-13A	CONTINGENCY 'JC-P2-3-JCC-230-13A' /* LARRABEE - SMITHBURG (D2004) & LAKEWOOD (Z2026) 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 1 /* LARRABEE- SMITHBURG (D2004) DISCONNECT BRANCH FROM BUS 206294 TO BUS 206323 CKT 2 /* LARRABEE- LAKEWOOD (Z2026) SET BUS 206294 LOAD TO 0 MW /* LARRABEE BK8 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 12 /* LARRABEE BK12 END
JC-P1-2-JCC-230-004	CONTINGENCY 'JC-P1-2-JCC-230-004' /* FUTURE ATLANTIC - OCEANVIEW (Y2025) 230 KV DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT Y /* ATLANTIC 230.00 OCEANV 230.00 END
JC-P2-3-JCC-230-47D	CONTINGENCY 'JC-P2-3-JCC-230-47D' /* ATLANTIC - OCEANVIEW (Y2025) & OCEANVIEW BK1 230 KV FUTURE OCEANVIEW RING BUS DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT Y /* ATLANTIC - OCEANVIEW (Y2025) DISCONNECT BRANCH FROM BUS 206300 TO BUS 206273 CKT 1 /* OCEANVIEW BK1 REDUCE BUS 206273 SHUNT BY 50 PERCENT /* OCEANVIEW CAPS ON 34.5 KV BUS A DISCONNECT BRANCH FROM BUS 206273 TO BUS 206929 CKT 2 /* OCEANVIEW - ALLENHURST (X102) 34.5KV END

Contingency Name	Contingency Definition
JC-P2-3-JCC-230-47E	CONTINGENCY 'JC-P2-3-JCC-230-47E' /* OCEANVIEW BK2 230 KV FUTURE OCEANVIEW RING BUS & ATLANTIC - OCEANVIEW (Y2025) DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT Y /* ATLANTIC - OCEANVIEW (Y2025) DISCONNECT BRANCH FROM BUS 206300 TO BUS 206273 CKT 2 /* OCEANVIEW BK2 REDUCE BUS 206273 SHUNT BY 50 PERCENT /* OCEANVIEW CAPS ON 34.5 KV BUS B END
JC-P2-3-JCC-230-47B	CONTINGENCY 'JC-P2-3-JCC-230-47B' /* LARRABEE - OCEANVIEW (NEW) & ATLANTIC - OCEANVIEW (X2024) DISCONNECT BRANCH FROM BUS 206294 TO BUS 206300 CKT 1 /* LARRABEE - OCEANVIEW (NEW) DISCONNECT BRANCH FROM BUS 206294 TO BUS 206274 CKT 3 /* LARRABEE BK3 DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT X /* ATLANTIC - OCEANVIEW (X2024) END
JC-P2-3-JCC-230-47C	CONTINGENCY 'JC-P2-3-JCC-230-47C' /* ATLANTIC - OCEANVIEW (X2024) & OCEANVIEW BK2 230 KV FUTURE OCEANVIEW RING BUS DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT X /* ATLANTIC - OCEANVIEW (X2024) DISCONNECT BRANCH FROM BUS 206300 TO BUS 206273 CKT 2 /* OCEANVIEW BK2 REDUCE BUS 206273 SHUNT BY 50 PERCENT /* OCEANVIEW CAPS ON 34.5 KV BUS B DISCONNECT BRANCH FROM BUS 206273 TO BUS 206929 CKT 1 /* OCEANVIEW - ALLENHURST (H216) 34.5KV END
JC-P2-3-JCC-230-15E	CONTINGENCY 'JC-P2-3-JCC-230-15E' /* LARRABEE - VAN HISEVILLE JUNCTION (B1042) & LARRABEE NW BUS 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206318 CKT 1 /* LARRABEE - VAN HISVILLE JUNCTION DISCONNECT BRANCH FROM BUS 206287 TO BUS 206318 CKT 1 /* COOKSTOWN - VAN HISVILLE JUNCTION DISCONNECT BUS 206720 /* MANCHESTER 230 KV BUS DISCONNECT BRANCH FROM BUS 206319 TO BUS 206281 CKT 5 /* WHITINGS #5 230/34.5 KV REMOVE LOAD 7 FROM BUS 206294 /* LARRABEE BK7 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 9 /* LARRABEE BK9 END
JC-P2-3-JCC-230-15G	CONTINGENCY 'JC-P2-3-JCC-230-15G' /* LARRABEE-SMITHBURG (H2008) 230 KV & LARRABEE NW BUS 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 2 /* LARRABEE- SMITHBURG (H2008) REMOVE LOAD 7 FROM BUS 206294 /* LARRABEE BK7 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 9 /* LARRABEE BK9 END

Contingency Name	Contingency Definition
JC-P2-3-JCC-230-26D	CONTINGENCY 'JC-P2-3-JCC-230-26D' /* RARITAN RIVER B146 STUCK BREAKER DISCONNECT BUS 206314 /* RED OAK 230 KV A TAP DISCONNECT BUS 206363 /* RED OAK 230 KV #2 DISCONNECT BUS 206364 /* RED OAK 230 KV #3 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 13 /* RARITAN RIVER #13 230/115 KV TRANSFORMER DISCONNECT BUS 206403 /* WOODBRIDGE GENERATOR END
JC-P2-3-JCC-230-15A	CONTINGENCY 'JC-P2-3-JCC-230-15A' /* LAKEWOOD-LARRABEE (K2011) & LARRABEE-SMITHBURG (H2008) 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206323 CKT 1 /* LARRABEE - LAKEWOOD (K2011) DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 2 /* LARRABEE- SMITHBURG (H2008) END
JC-P2-3-JCC-230-26B	CONTINGENCY 'JC-P2-3-JCC-230-26B' /* RARITAN RIVER B144 STUCK BREAKER DISCONNECT BUS 206315 /* RED OAK 230 KV B TAP DISCONNECT BUS 206362 /* RED OAK 230 KV #1 DISCONNECT BUS 206365 /* RED OAK 230 KV #4 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 13 /* RARITAN RIVER #13 230/115 KV TRANSFORMER DISCONNECT BUS 206403 /* WOODBRIDGE GENERATOR END
JC-P7-1-JCC-230-15	CONTINGENCY 'JC-P7-1-JCC-230-15' /* ATLANTIC - OCEANVIEW X2024 & Y2025 DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT X /* ATLANTIC- OCEANVIEW 230 DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT Y /* ATLANTIC- OCEANVIEW 230 END
PS_P7-1_1LINE+2LINE	CONTINGENCY 'PS_P7-1_1LINE+2LINE' /* LAKE NELSON - RARITAN X2 DISCONNECT BUS 218331 /* REMOVE KILMER 1 DISCONNECT BUS 218332 /* REMOVE KILMER 2 MOVE 8 MW LOAD FROM BUS 218384 TO BUS 218383 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO GREENBK T1 MOVE 8 MW LOAD FROM BUS 218387 TO BUS 218399 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO NEWDOVR T2 MOVE 8 MW LOAD FROM BUS 218385 TO BUS 218393 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO MDWRD T2 MOVE 8 MW LOAD FROM BUS 218386 TO BUS 218347 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO BNNTLANE T2 END

Contingency Name	Contingency Definition
PJM500_PS_P2-3_DEAN5_5-6	CONTINGENCY 'PJM500_PS_P2-3_DEAN5_5-6' DISCONNECT BRANCH FROM BUS 200006 TO BUS 218306 CKT 2/* BRANCHBURG TX -2 TRIP LINE FROM BUS 200006 TO BUS 200028 CKT 1 /* DEANS TO WINSOR END
PS_P1-2_#2LINE	CONTINGENCY 'PS_P1-2_#2LINE' /* LAKE NELSON TO RARITAN RIVER DISCONNECT BUS 218332 /* KILMER LINE #2 CLOSE LINE FROM BUS 218384 TO BUS 218387 CKT Z /* KILMER CLOSE LINE FROM BUS 218385 TO BUS 218386 CKT Z /* KILMER END
PS_P7-1_I1023+GBK-LN_LT	CONTINGENCY 'PS_P7-1_I1023+GBK-LN_LT' /* LAKE NELSON - GILLETE BRIDGEWATER & LAKE NELSON -GREENBROOK DISCONNECT BUS 218312 /* GREENBROOK I DISCONNECT BUS 218301 /* MIDDLESEX I DISCONNECT BUS 218333 /* LAKE NELSON I DISCONNECT BUS 218334 /* LAKE NELSON W DISCONNECT BUS 218522 /* MIDDLESEX W CLOSE LINE FROM BUS 218382 TO BUS 218383 CKT Z /* GREENBROOK MOVE 8 MW LOAD FROM BUS 218382 TO BUS 218406 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO SOMRVLE T2 MOVE 8 MW LOAD FROM BUS 218382 TO BUS 218386 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO KILMER T1 MOVE 8 MW LOAD FROM BUS 218383 TO BUS 218384 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO KILMER T2 MOVE 8 MW LOAD FROM BUS 218383 TO BUS 218395 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO SOUTH SECOND ST.(METUCHEN 26KV IM) END
JC-P1-2-JCC-230-027	CONTINGENCY 'JC-P1-2-JCC-230-027' /* RARITAN RIVER - RED OAK - R11 (G1047) 230 KV LINE DISCONNECT BRANCH FROM BUS 206305 TO BUS 206315 CKT 1 DISCONNECT BRANCH FROM BUS 206315 TO BUS 206410 CKT 1 DISCONNECT BUS 206315 DISCONNECT BUS 206362 DISCONNECT BUS 206365 END
PS_P1-2_I-1023	CONTINGENCY 'PS_P1-2_I-1023' /* TIE LINE GILLET I TO BRIDGEWATER THROUGH MIDDLESEX DISCONNECT BUS 218312 /* GREEN BROOK STATION DISCONNECT BUS 218301 /* MIDDLESEX I DISCONNECT BUS 218333 /* LAKE NELSON BUS I CLOSE LINE FROM BUS 218388 TO BUS 218389 CKT Z /* LAKE NELSON CLOSE LINE FROM BUS 218382 TO BUS 218383 CKT Z /* GREENBROOK END

Contingency Name	Contingency Definition
JC-P1-2-JCC-230-042T	CONTINGENCY 'JC-P1-2-JCC-230-042T' /* RARITAN RIVER - LAKE NELSON (I1023) 230 KV LINE DISCONNECT BRANCH FROM BUS 206305 TO BUS 218331 CKT 1 /* 28RAR RVR - KILMER_I 230.00 LINE DISCONNECT BRANCH FROM BUS 218331 TO BUS 218333 CKT 1 /* KILMER_I - LNELSN_I 230.00 LINE DISCONNECT BRANCH FROM BUS 218384 TO BUS 218331 CKT 1 /* KILMER_L2 13.20 - KILMER_I 230.00 XFMR DISCONNECT BRANCH FROM BUS 218385 TO BUS 218331 CKT 1 /* KILMER_L4 13.20 - KILMER_I 230.00 XFMR DISCONNECT BRANCH FROM BUS 218333 TO BUS 218427 CKT 1 /* L.NELSN3 230 L.NELSN3 69 END
Base Case	
JC-P1-2-JCC-230-003	CONTINGENCY 'JC-P1-2-JCC-230-003' /* FUTURE ATLANTIC - OCEANVIEW (X2024) 230 KV DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT X /* ATLANTIC 230.00 OCEANV 230.00 END
JC-P2-3-JCC-230-17A	CONTINGENCY 'JC-P2-3-JCC-230-17A' /* LARRABEE-COOKSTOWN- WHITINGS (B1042) & WHITINGS-MANITOU (V2022) 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206318 CKT 1 /* LARRABEE-VAN HISEVILLE TAP (B1042) DISCONNECT BRANCH FROM BUS 206318 TO BUS 206287 CKT 1 /* VAN HISEVILLE TAP - COOKSTOWN (B1042) DISCONNECT BRANCH FROM BUS 206318 TO BUS 206720 CKT 1 /* VAN HISEVILLE TAP - MANCHESTER (B1042) DISCONNECT BUS 206720 /* ISOLATE MANCHESTER DISCONNECT BRANCH FROM BUS 206319 TO BUS 206281 CKT 5 /* WHITINGS BK5 DISCONNECT BRANCH FROM BUS 206319 TO BUS 206297 CKT 1 /* WHITINGS - MANITOU (V2022) DISCONNECT BRANCH FROM BUS 206319 TO BUS 206281 CKT 4 /* WHITINGS BK4 END
JC-P1-2-JCC-230-028	CONTINGENCY 'JC-P1-2-JCC-230-028' /* RARITAN RIVER - RED OAK - SOUTH RIVER JCT(T1034) 230 KV DISCONNECT BRANCH FROM BUS 206305 TO BUS 206314 CKT 1 /* 28RAR RVR 230.00 28RED OAKA 230.00 DISCONNECT BRANCH FROM BUS 206314 TO BUS 206411 CKT 1 /* 28RED OAKA 230.00 28R11RINGA 230.00 . DISCONNECT BUS 206314 /* 28RED OAKA 230.00 DISCONNECT BUS 206363 /* 28RDOAKCT2 18.00 DISCONNECT BUS 206364 /* 28RDOAKCT3 18.00 END

Contingency Name	Contingency Definition
JC-P2-3-JCC-230-15F	CONTINGENCY 'JC-P2-3-JCC-230-15F' /* LARRABEE - SMITHBURG (D2004) & LARRABEE NW BUS 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 1 /* LARRABEE- SMITHBURG (D2004) REMOVE LOAD 7 FROM BUS 206294 /* LARRABEE BK7 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 9 /* LARRABEE BK9 END
JC-P7-1-JCC-230-11A	CONTINGENCY 'JC-P7-1-JCC-230-11A' /* LAKEWOOD - LEISURE V 230 KV LINES DISCONNECT BRANCH FROM BUS 206323 TO BUS 206295 CKT 1 DISCONNECT BRANCH FROM BUS 206323 TO BUS 206296 CKT 1 DISCONNECT BRANCH FROM BUS 206323 TO BUS 206282 CKT 6 DISCONNECT BRANCH FROM BUS 206282 TO BUS 206999 CKT 1 SET BUS 206295 LOAD TO 0 MW DISCONNECT BUS 206999 DISCONNECT BUS 207110 DISCONNECT BUS 207010 DISCONNECT BUS 207111 END
PJM_P4_P484B	CONTINGENCY 'PJM_P4_P484B' /* VALID FROM 2009 DISCONNECT BRANCH FROM BUS 200028 TO BUS 200006 CKT 1 /* WINDSOR DEANS 500500 DISCONNECT BRANCH FROM BUS 200006 TO BUS 218306 CKT 1 /* DEANS XF 2 500230 END
JC-P2-3-JCC-230-026A	CONTINGENCY 'JC-P2-3-JCC-230-026A' /* RARITAN RIVER B143 STUCK BREAKER DISCONNECT BUS 206315 /* RED OAK 230 KV B TAP DISCONNECT BUS 206362 /* RED OAK 230 KV #1 DISCONNECT BUS 206365 /* RED OAK 230 KV #4 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 17 /* RARITAN RIVER #17 230/115 KV TRANSFORMER DISCONNECT BUS 206350 /* SAYREVILLE GENERATOR DISCONNECT BUS 206351 /* SAYREVILLE GENERATOR END
JC-P2-3-JCC-230-026C	CONTINGENCY 'JC-P2-3-JCC-230-026C' /* RARITAN RIVER B145 STUCK BREAKER DISCONNECT BUS 206314 /* RED OAK 230 KV A TAP DISCONNECT BUS 206363 /* RED OAK 230 KV #2 DISCONNECT BUS 206364 /* RED OAK 230 KV #3 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 17 /* RARITAN RIVER #17 230/115 KV TRANSFORMER DISCONNECT BUS 206350 /* SAYREVILLE GENERATOR DISCONNECT BUS 206351 /* SAYREVILLE GENERATOR END

12 Light Load Analysis

The Queue Project AF1-222 was evaluated as a 510.0 MW injection as an uprate to AE1-238 at the Oceanview 230 kV substation in the JCPL area. Project AF1-222 was evaluated for compliance with applicable reliability planning criteria (PJM, NERC, NERC Regional Reliability Councils, and Transmission Owners). Project AF1-222 was studied with a commercial probability of 100.0 %. Potential network impacts were as follows:

12.1 Generation Deliverability

(Single or N-1 contingencies)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS LOADIN G %	POST PROJECTIONS LOADIN G %	AC/D C	MW IMPACT
71746204	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBRG	230.0	JCP&L	1	Base Case	single	650.0	93.09	101.42	AC	55.57
71746717	206316	28WINDSOR	230.0	JCP&L	219752	CLRKSULL_1	230.0	PSE&G	1	PJM500_PS_P1-2_5022	single	813.0	94.04	100.83	AC	56.11
71746744	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSOR	230.0	JCP&L	1	PJM500_PS_P1-2_5022	single	869.0	94.12	100.08	AC	52.48
71746538	206410	28R11RING B	230.0	JCP&L	206315	28RED OAKB	230.0	JCP&L	1	JC-P1-2-JCC-230-028	single	869.0	90.65	104.37	AC	122.23
71746522	206411	28R11RING A	230.0	JCP&L	206314	28RED OAKA	230.0	JCP&L	1	JC-P1-2-JCC-230-027	single	869.0	92.63	106.48	AC	122.68

12.2 Multiple Facility Contingency

(Double Circuit Tower Line, Fault with a Stuck Breaker, and Bus Fault contingencies)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS LOADIN G %	POST PROJECTIONS LOADIN G %	AC/D C	MW IMPACT
42237121	206300	28OCEANV W	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	Y	JC-P2-3-JCC-230-47B	breaker	869.0	58.91	102.82	AC	392.55
42237122	206300	28OCEANV W	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	Y	JC-P2-3-JCC-230-47C	breaker	869.0	70.12	100.56	AC	271.86
42237180	206300	28OCEANV W	230.0	JCP&L	206286	28ATLANTIC	230.0	JCP&L	X	JC-P2-3-JCC-230-47E	breaker	869.0	70.0	100.42	AC	271.67
41034357	206316	28WINDSOR	230.0	JCP&L	219752	CLRKSULL_1	230.0	PSE&G	1	PS_P7-1_1LINE+2LINE	tower	813.0	90.84	100.23	AC	78.44
42237050	206410	28R11RING B	230.0	JCP&L	206315	28RED OAKB	230.0	JCP&L	1	JC-P2-3-JCC-230-26D	breaker	869.0	94.48	108.27	AC	122.15
42237051	206410	28R11RING B	230.0	JCP&L	206315	28RED OAKB	230.0	JCP&L	1	JC-P2-3-JCC-230-026C	breaker	869.0	90.38	104.16	AC	122.35
42237041	206411	28R11RING A	230.0	JCP&L	206314	28RED OAKA	230.0	JCP&L	1	JC-P2-3-JCC-230-26B	breaker	869.0	96.31	110.15	AC	122.59
42237042	206411	28R11RING A	230.0	JCP&L	206314	28RED OAKA	230.0	JCP&L	1	JC-P2-3-JCC-230-026A	breaker	869.0	92.2	106.11	AC	122.8

12.3 Contribution to Previously Identified Overloads

(This project contributes to the following contingency overloads, i.e. "Network Impacts", identified for earlier generation or transmission interconnection projects in the PJM Queue)

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADIN G %	POST PROJECT LOADIN G %	AC/D C	MW IMPACT
42236940	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	2	JC-P2-3-JCC-230-15F	breaker	813.0	114.11	124.25	AC	84.97
42236945	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	1	JC-P2-3-JCC-230-15G	breaker	817.0	113.55	123.64	AC	84.97
42236946	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	1	JC-P2-3-JCC-230-15A	breaker	817.0	108.32	118.35	AC	85.51
71746180	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	2	JC-P1-2-JCC-230-017	single	813.0	113.54	123.71	AC	84.98
71746201	206294	28LARRABE E	230.0	JCP&L	206309	28SMITHBR G	230.0	JCP&L	1	JC-P1-2-JCC-230-018	single	817.0	106.39	115.12	AC	73.43
42236930	206314	28RED OAKA	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P2-3-JCC-230-26B	breaker	869.0	107.21	121.06	AC	122.09
42236931	206314	28RED OAKA	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P2-3-JCC-230-026A	breaker	869.0	102.93	116.87	AC	122.3
42541413	206314	28RED OAKA	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P1-2-JCC-230-027	single	869.0	103.3	117.23	AC	122.18
42236925	206315	28RED OAKB	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P2-3-JCC-230-26D	breaker	869.0	108.33	121.96	AC	121.55
42236926	206315	28RED OAKB	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P2-3-JCC-230-026C	breaker	869.0	104.73	118.4	AC	121.75
42541377	206315	28RED OAKB	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P1-2-JCC-230-028	single	869.0	104.59	118.21	AC	121.63
49598302	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	Base Case	single	760.0	101.29	106.87	AC	33.04

12.4 Steady-State Voltage Requirements

To be determined in Facilities Study phase.

12.5 Potential Congestion due to Local Energy Deliverability

PJM also studied the delivery of the energy portion of this interconnection request. Any problems identified below are likely to result in operational restrictions to the project under study. The developer can proceed with network upgrades to eliminate the operational restriction at their discretion by submitting a Merchant Transmission Interconnection request.

Note: Only the most severely overloaded conditions are listed below. There is no guarantee of full delivery of energy for this project by fixing only the conditions listed in this section. With a Transmission Interconnection

Request, a subsequent analysis will be performed which shall study all overload conditions associated with the overloaded element(s) identified.

ID	FROM BUS#	FROM BUS	kV	FROM BUS AREA	TO BUS#	TO BUS	kV	TO BUS AREA	CK T ID	CONT NAME	Type	Rating MVA	PRE PROJECTIONS LOADIN G %	POST PROJECTIONS LOADIN G %	AC/D C	MW IMPACT
42541346	206294	28LARRABEE	230.0	JCP&L	206309	28SMITHBRG	230.0	JCP&L	2	JC-P1-2-JCC-230-017	operation	813.0	112.98	123.12	AC	84.98
42541472	206294	28LARRABEE	230.0	JCP&L	206309	28SMITHBRG	230.0	JCP&L	1	JC-P1-2-JCC-230-018	operation	817.0	103.48	112.12	AC	73.43
42541408	206314	28RED OAKA	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P1-2-JCC-230-027	operation	869.0	100.72	114.65	AC	122.18
42541372	206315	28RED OAKB	230.0	JCP&L	206305	28RAR RVR	230.0	JCP&L	1	JC-P1-2-JCC-230-028	operation	869.0	102.48	116.09	AC	121.63
71746720	206316	28WINDSOR	230.0	JCP&L	219752	CLRKSVLL_1	230.0	PSE&G	1	PJM500_PS_P1-2_5022	operation	813.0	90.64	97.38	AC	56.11
49598742	206326	28E WINDSR	230.0	JCP&L	206316	28WINDSOR	230.0	JCP&L	1	PJM500_PS_P1-2_5022	operation	869.0	90.94	96.85	AC	52.48
42541727	206410	28R11RINGB	230.0	JCP&L	206315	28RED OAKB	230.0	JCP&L	1	JC-P1-2-JCC-230-028	operation	869.0	88.18	101.91	AC	122.23
42541679	206411	28R11RINGA	230.0	JCP&L	206314	28RED OAKA	230.0	JCP&L	1	JC-P1-2-JCC-230-027	operation	869.0	89.96	103.88	AC	122.68
42541625	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	Base Case	operation	760.0	99.36	104.12	AC	33.04
49598299	213922	RICHMOND	230.0	PECO	214012	WANEETA3	230.0	PECO	1	PJM500_PS_P1-2_5015	operation	1180.0	91.56	95.22	AC	43.99

12.6 System Reinforcements

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number																												
49598302	6	RICHMOND 230.0 kV - WANEETA3 230.0 kV Ckt 1	Add a second pair of underground cables in parallel with the existing underground portion of the Richmond - Waneeta 230 kV line. Cost: \$19,000,000 Time: 84 Months New Ratings: 1026/1247/1337 The cost allocation table is below: <table><tr><th>Queue</th><th>MW contribution</th><th>% of Cost</th><th>Cost (\$19,000,000)</th></tr><tr><td>AE2-021</td><td>102.2</td><td>19%</td><td>3,684,152</td></tr><tr><td>AE2-022</td><td>64.8</td><td>12%</td><td>2,337,153</td></tr><tr><td>AE2-222</td><td>56.6</td><td>11%</td><td>2,040,143</td></tr><tr><td>AE2-251</td><td>220.6</td><td>42%</td><td>7,950,068</td></tr><tr><td>AF1-126</td><td>49.9</td><td>9%</td><td>1,797,560</td></tr><tr><td>AF1-222</td><td>33.0</td><td>6%</td><td>1,190,924</td></tr></table>	Queue	MW contribution	% of Cost	Cost (\$19,000,000)	AE2-021	102.2	19%	3,684,152	AE2-022	64.8	12%	2,337,153	AE2-222	56.6	11%	2,040,143	AE2-251	220.6	42%	7,950,068	AF1-126	49.9	9%	1,797,560	AF1-222	33.0	6%	1,190,924	\$19,000,000	\$1,190,924	N6477
Queue	MW contribution	% of Cost	Cost (\$19,000,000)																															
AE2-021	102.2	19%	3,684,152																															
AE2-022	64.8	12%	2,337,153																															
AE2-222	56.6	11%	2,040,143																															
AE2-251	220.6	42%	7,950,068																															
AF1-126	49.9	9%	1,797,560																															
AF1-222	33.0	6%	1,190,924																															
42236940, 71746180	10	28LARRABE 230.0 kV - 28SMITHBRG 230.0 kV Ckt 2	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-																												
71746201, 71746204, 42236946, 42236945	1	28LARRABE 230.0 kV - 28SMITHBRG 230.0 kV Ckt 1	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-																												
71746717, 41034357	2	28WINDSOR 230.0 kV - CLRKSVLL_1 230.0 kV Ckt 1	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-																												
42237051, 42237050, 71746538	4	28R11RINGB 230.0 kV - 28RED OAKB 230.0 kV Ckt 1	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-																												
42541377, 42236926, 42236925	12	28RED OAKB 230.0 kV - 28RAR RVR 230.0 kV Ckt 1	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-																												

ID	Idx	Facility	Upgrade Description	Cost	Cost Allocated to AF1-222	Upgrade Number
71746522, 42237042, 42237041	5	28R11RING A 230.0 kV - 28RED OAKA 230.0 kV Ckt 1	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-
71746744	3	28E WINDSR 230.0 kV - 28WINDSOR 230.0 kV Ckt 1	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-
42237180	9	28OCEANV W 230.0 kV - 28ATLANTIC 230.0 kV Ckt X	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-
42236931, 42236930, 42541413	11	28RED OAKA 230.0 kV - 28RAR RVR 230.0 kV Ckt 1	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-
42237121, 42237122	8	28OCEANV W 230.0 kV - 28ATLANTIC 230.0 kV Ckt Y	AF1-222 Summer Peak has higher facility loading than Light Load. Refer to AF1-222 Summer Peak Load Flow for Mitigation.	\$0	\$0	-
			TOTAL COST	\$19,000,000	\$1,190,924	

Note : For customers with System Reinforcements listed: If your present cost allocation to a System Reinforcement indicates \$0, then please be aware that as changes to the interconnection process occur, such as prior queued projects withdrawing from the queue, reducing in size, etc, the cost responsibilities can change and a cost allocation may be assigned to your project. In addition, although your present cost allocation to a System Reinforcement is presently \$0, your project may need this system reinforcement completed to be deliverable to the PJM system. If your project comes into service prior to completion of the system reinforcement, an interim deliverability study for your project will be required.

12.7 Flow Gate Details

The following indices contain additional information about each facility presented in the body of the report. For each index, a description of the flowgate and its contingency was included for convenience. The intent of the indices is to provide more details on which projects/generators have contributions to the flowgate in question. All New Service Queue Requests, through the end of the Queue under study, that are contributors to a flowgate will be listed in the indices. Please note that there may be contributors that are subsequently queued after the queue under study that are not listed in the indices. Although this information is not used "as is" for cost allocation purposes, it can be used to gage the impact of other projects/generators. It should be noted the project/generator MW contributions presented in the body of the report are Full MW Impact contributions which are also noted in the indices column named "Full MW Impact", whereas the loading percentages reported in the body of the report, take into consideration the PJM Generator Deliverability Test rules such as commercial probability of each project as well as the ramping impact of "Adder" contributions. The MW Impact found and used in the analysis is shown in the indices column named "Gendeliv MW Impact".

12.7.1 Index 1

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42236945	206294	28LARRABEE	JCP&L	206309	28SMITHBRG	JCP&L	1	JC-P2-3-JCC-230-15G	breaker	817.0	113.55	123.64	AC	84.97

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	67.8581	50/50	67.8581
938421	AE1-061 C	0.1293	Adder	0.15
938422	AE1-061 E	0.1293	Adder	0.15
938781	AE1-104 C O1	6.5257	Adder	7.68
938782	AE1-104 E O1	16.6960	Adder	19.64
939981	AE1-238 C	37.4850	50/50	37.4850
939982	AE1-238 E	98.4606	50/50	98.4606
940161	AE2-000 C O1	55.3732	50/50	55.3732
940162	AE2-000 E O1	141.6810	50/50	141.6810
940361	AE2-020 C	8.5442	50/50	8.5442
940362	AE2-020 E	40.0044	50/50	40.0044
940371	AE2-021 C	8.5442	50/50	8.5442
940372	AE2-021 E	40.0044	50/50	40.0044
940381	AE2-022 C	4.9841	50/50	4.9841
940382	AE2-022 E	23.3359	50/50	23.3359
940401	AE2-024 C O1	46.2374	50/50	46.2374
940402	AE2-024 E O1	216.4786	50/50	216.4786
940411	AE2-025 C	23.3406	50/50	23.3406
940412	AE2-025 E	109.2684	50/50	109.2684
942101	AE2-222 C O1	5.4223	Adder	6.38
942102	AE2-222 E O1	13.6211	Adder	16.02
942381	AE2-251 C	27.0677	50/50	27.0677
942382	AE2-251 E	69.2587	50/50	69.2587
943561	AF1-027	0.3099	50/50	0.3099
944332	AF1-101 E O1	138.9039	50/50	138.9039
944611	AF1-126 C O1	10.4795	50/50	10.4795
944612	AF1-126 E O1	41.9180	50/50	41.9180
945201	AF1-185 1	0.2617	Adder	0.31
945211	AF1-185 2	0.0581	Adder	0.07
945571	AF1-222 C	23.3657	50/50	23.3657
945572	AF1-222 E	61.6004	50/50	61.6004
945731	AF1-238 C	2.1835	Adder	2.57
945732	AF1-238 E	3.2753	Adder	3.85
945741	AF1-239 C	0.5138	Adder	0.6
945742	AF1-239 E	0.7707	Adder	0.91

12.7.2 Index 2

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
41034357	206316	28WINDSOR	JCP&L	219752	CLRKSVLL_1	PSE&G	1	PS_P7-1_1LINE+2LINE	tower	813.0	90.84	100.23	AC	78.44

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	280 C GEN (Deactivation : 17/09/2018)	33.7672	50/50	33.7672
938231	AE1-037 FTIR (Withdrawn : 01/24/2020)	77.2800	Merchant Transmission	77.2800
939981	AE1-238 C	34.6068	50/50	34.6068
939982	AE1-238 E	90.9005	50/50	90.9005
940161	AE2-000 C O1	27.5545	50/50	27.5545
940162	AE2-000 E O1	70.5026	50/50	70.5026
940361	AE2-020 C	3.8412	Adder	4.52
940362	AE2-020 E	17.9846	Adder	21.16
940371	AE2-021 C	3.8412	Adder	4.52
940372	AE2-021 E	17.9846	Adder	21.16
940381	AE2-022 C	2.2407	Adder	2.64
940382	AE2-022 E	10.4910	Adder	12.34
940401	AE2-024 C O1	22.8424	50/50	22.8424
940402	AE2-024 E O1	106.9457	50/50	106.9457
940411	AE2-025 C	11.5308	50/50	11.5308
940412	AE2-025 E	53.9812	50/50	53.9812
942381	AE2-251 C	12.1687	Adder	14.32
942382	AE2-251 E	31.1364	Adder	36.63
943561	AF1-027	0.3154	50/50	0.3154
944332	AF1-101 E O1	69.1206	50/50	69.1206
944442	AF1-109 BAT	5.4138	50/50	5.4138
944611	AF1-126 C O1	4.2300	Adder	4.98
944612	AF1-126 E O1	16.9201	Adder	19.91
945201	AF1-185 1	0.8907	50/50	0.8907
945211	AF1-185 2	0.1979	50/50	0.1979
945571	AF1-222 C	21.5716	50/50	21.5716
945572	AF1-222 E	56.8705	50/50	56.8705
945723	AF1-237 BAT	38.7740	50/50	38.7740

12.7.3 Index 3

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71746744	206326	28E WINDSR	JCP&L	206316	28WINDSOR	JCP&L	1	PJM500_PS_P1-2_5022	single	869.0	94.12	100.08	AC	52.48

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	28.3877	80/20	28.3877
937523	AD2-213 BAT	1.7910	80/20	1.7910
938781	AE1-104 C O1	5.6650	80/20	5.6650
938782	AE1-104 E O1	14.4938	80/20	14.4938
939981	AE1-238 C	23.1534	80/20	23.1534
939982	AE1-238 E	60.8163	80/20	60.8163
940161	AE2-000 C O1	23.1648	80/20	23.1648
940162	AE2-000 E O1	59.2708	80/20	59.2708
940361	AE2-020 C	5.5843	80/20	5.5843
940362	AE2-020 E	26.1460	80/20	26.1460
940371	AE2-021 C	5.5843	80/20	5.5843
940372	AE2-021 E	26.1460	80/20	26.1460
940381	AE2-022 C	3.2575	80/20	3.2575
940382	AE2-022 E	15.2518	80/20	15.2518
940401	AE2-024 C O1	18.4997	80/20	18.4997
940402	AE2-024 E O1	86.6135	80/20	86.6135
940411	AE2-025 C	9.3386	80/20	9.3386
940412	AE2-025 E	43.7185	80/20	43.7185
942101	AE2-222 C O1	4.2601	80/20	4.2601
942102	AE2-222 E O1	10.7015	80/20	10.7015
942381	AE2-251 C	17.6909	80/20	17.6909
942382	AE2-251 E	45.2659	80/20	45.2659
944332	AF1-101 E O1	58.1090	80/20	58.1090
944432	AF1-108 BAT	1.8470	80/20	1.8470
944442	AF1-109 BAT	5.0882	80/20	5.0882
944611	AF1-126 C O1	4.0240	80/20	4.0240
944612	AF1-126 E O1	16.0960	80/20	16.0960
945201	AF1-185 1	0.2324	80/20	0.2324
945211	AF1-185 2	0.0516	80/20	0.0516
945571	AF1-222 C	14.4323	80/20	14.4323
945572	AF1-222 E	38.0488	80/20	38.0488
945723	AF1-237 BAT	37.0800	80/20	37.0800

12.7.4 Index 4

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237050	206410	28R11RINGB	JCP&L	206315	28RED OAKB	JCP&L	1	JC-P2-3-JCC-230-26D	breaker	869.0	94.48	108.27	AC	122.15

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	35.5409	50/50	35.5409
938781	AE1-104 C O1	4.5981	Adder	5.41
938782	AE1-104 E O1	11.7643	Adder	13.84
939981	AE1-238 C	53.8902	50/50	53.8902
939982	AE1-238 E	141.5516	50/50	141.5516
940161	AE2-000 C O1	29.0019	50/50	29.0019
940162	AE2-000 E O1	74.2058	50/50	74.2058
940361	AE2-020 C	4.7524	Adder	5.59
940362	AE2-020 E	22.2512	Adder	26.18
940371	AE2-021 C	4.7524	Adder	5.59
940372	AE2-021 E	22.2512	Adder	26.18
940381	AE2-022 C	2.7722	Adder	3.26
940382	AE2-022 E	12.9799	Adder	15.27
940401	AE2-024 C O1	23.5503	50/50	23.5503
940402	AE2-024 E O1	110.2597	50/50	110.2597
940411	AE2-025 C	11.8882	50/50	11.8882
940412	AE2-025 E	55.6540	50/50	55.6540
942101	AE2-222 C O1	3.6199	Adder	4.26
942102	AE2-222 E O1	9.0934	Adder	10.7
942381	AE2-251 C	15.0556	Adder	17.71
942382	AE2-251 E	38.5230	Adder	45.32
943561	AF1-027	0.2180	50/50	0.2180
944332	AF1-101 E O1	72.7513	50/50	72.7513
944611	AF1-126 C O1	5.8080	50/50	5.8080
944612	AF1-126 E O1	23.2320	50/50	23.2320
945571	AF1-222 C	33.5916	50/50	33.5916
945572	AF1-222 E	88.5596	50/50	88.5596

12.7.5 Index 5

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237041	206411	28R11RINGA	JCP&L	206314	28RED OAKA	JCP&L	1	JC-P2-3-JCC-230-26B	breaker	869.0	96.31	110.15	AC	122.59

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	35.6555	50/50	35.6555
938781	AE1-104 C O1	4.6081	Adder	5.42
938782	AE1-104 E O1	11.7896	Adder	13.87
939981	AE1-238 C	54.0846	50/50	54.0846
939982	AE1-238 E	142.0622	50/50	142.0622
940161	AE2-000 C O1	29.0954	50/50	29.0954
940162	AE2-000 E O1	74.4452	50/50	74.4452
940361	AE2-020 C	4.7633	Adder	5.6
940362	AE2-020 E	22.3020	Adder	26.24
940371	AE2-021 C	4.7633	Adder	5.6
940372	AE2-021 E	22.3020	Adder	26.24
940381	AE2-022 C	2.7786	Adder	3.27
940382	AE2-022 E	13.0095	Adder	15.31
940401	AE2-024 C O1	23.6297	50/50	23.6297
940402	AE2-024 E O1	110.6318	50/50	110.6318
940411	AE2-025 C	11.9283	50/50	11.9283
940412	AE2-025 E	55.8419	50/50	55.8419
942101	AE2-222 C O1	3.6280	Adder	4.27
942102	AE2-222 E O1	9.1138	Adder	10.72
942381	AE2-251 C	15.0900	Adder	17.75
942382	AE2-251 E	38.6110	Adder	45.42
943561	AF1-027	0.2186	50/50	0.2186
944332	AF1-101 E O1	72.9860	50/50	72.9860
944611	AF1-126 C O1	5.8250	50/50	5.8250
944612	AF1-126 E O1	23.3000	50/50	23.3000
945571	AF1-222 C	33.7127	50/50	33.7127
945572	AF1-222 E	88.8790	50/50	88.8790

12.7.6 Index 6

ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC/DC	MW IMPACT
49598302	213922	RICHMOND	PECO	214012	WANEETA3	PECO	1	Base Case	single	760.0	101.29	106.87	AC	33.04

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	280 C GEN (Deactivation : 17/09/2018)	25.5822	80/20	25.5822
937563	AD2-077 BAT	14.0360	80/20	14.0360
938231	AE1-037 FTIR (Withdrawn : 01/24/2020)	59.4960	Merchant Transmission	59.4960
938421	AE1-061 C	0.5515	80/20	0.5515
938422	AE1-061 E	0.5515	80/20	0.5515
938431	AE1-062 C	2.1187	80/20	2.1187
938432	AE1-062 E	2.1187	80/20	2.1187
938781	AE1-104 C O1	16.8756	80/20	16.8756
938782	AE1-104 E O1	43.1759	80/20	43.1759
938871	AE1-115 C	2.1471	80/20	2.1471
938872	AE1-115 E	2.1471	80/20	2.1471
939301	AE1-161 C	4.1554	80/20	4.1554
939302	AE1-161 E	6.2331	80/20	6.2331
939981	AE1-238 C	14.5782	80/20	14.5782
939982	AE1-238 E	38.2921	80/20	38.2921
940161	AE2-000 C O1	20.8755	80/20	20.8755
940162	AE2-000 E O1	53.4132	80/20	53.4132
940361	AE2-020 C	14.3413	80/20	14.3413
940362	AE2-020 E	67.1470	80/20	67.1470
940371	AE2-021 C	14.3413	80/20	14.3413
940372	AE2-021 E	67.1470	80/20	67.1470
940381	AE2-022 C	8.3658	80/20	8.3658
940382	AE2-022 E	39.1691	80/20	39.1691
940401	AE2-024 C O1	11.5603	80/20	11.5603
940402	AE2-024 E O1	54.1240	80/20	54.1240
940411	AE2-025 C	5.8356	80/20	5.8356
940412	AE2-025 E	27.3193	80/20	27.3193
942101	AE2-222 C O1	11.7914	80/20	11.7914
942102	AE2-222 E O1	29.6206	80/20	29.6206
942381	AE2-251 C	45.4330	80/20	45.4330
942382	AE2-251 E	116.2502	80/20	116.2502
943561	AF1-027	0.2650	80/20	0.2650
944332	AF1-101 E O1	52.3662	80/20	52.3662
944611	AF1-126 C O1	8.1965	80/20	8.1965
944612	AF1-126 E O1	32.7860	80/20	32.7860
944951	AF1-160 C	2.1187	80/20	2.1187
944952	AF1-160 E	2.1187	80/20	2.1187
945201	AF1-185 1	0.2623	80/20	0.2623

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945211	AF1-185 2	0.0583	80/20	0.0583
945431	AF1-208 C O1	0.0002	80/20	0.0002
945432	AF1-208 E O1	2.0203	80/20	2.0203
945571	AF1-222 C	9.0871	80/20	9.0871
945572	AF1-222 E	23.9568	80/20	23.9568
945721	AF1-237 C	10.6780	80/20	10.6780
945722	AF1-237 E	0.0001	80/20	0.0001
945731	AF1-238 C	9.5788	80/20	9.5788
945732	AF1-238 E	14.3682	80/20	14.3682
945741	AF1-239 C	2.2538	80/20	2.2538
945742	AF1-239 E	3.3808	80/20	3.3808

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
71746710	214010	WANEETA2	PECO	213817	N PHILA	PECO	1	PJM500_PS_P1-2_5015	single	621.0	88.87	94.93	AC	25.81

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	280 C GEN (Deactivation : 17/09/2018)	18.0357	80/20	18.0357
938231	AE1-037 FTIR (Withdrawn : 01/24/2020)	55.7760	Merchant Transmission	55.7760
938421	AE1-061 C	0.3302	80/20	0.3302
938422	AE1-061 E	0.3302	80/20	0.3302
938431	AE1-062 C	1.2866	80/20	1.2866
938432	AE1-062 E	1.2866	80/20	1.2866
938781	AE1-104 C O1	10.6793	80/20	10.6793
938782	AE1-104 E O1	27.3229	80/20	27.3229
938871	AE1-115 C	1.2386	80/20	1.2386
938872	AE1-115 E	1.2386	80/20	1.2386
939301	AE1-161 C	2.4566	80/20	2.4566
939302	AE1-161 E	3.6849	80/20	3.6849
939981	AE1-238 C	11.3886	80/20	11.3886
939982	AE1-238 E	29.9141	80/20	29.9141
940161	AE2-000 C O1	14.7174	80/20	14.7174
940162	AE2-000 E O1	37.6568	80/20	37.6568
940361	AE2-020 C	9.1768	80/20	9.1768
940362	AE2-020 E	42.9666	80/20	42.9666
940371	AE2-021 C	9.1768	80/20	9.1768
940372	AE2-021 E	42.9666	80/20	42.9666
940381	AE2-022 C	5.3532	80/20	5.3532
940382	AE2-022 E	25.0639	80/20	25.0639
940401	AE2-024 C O1	8.6631	80/20	8.6631
940402	AE2-024 E O1	40.5596	80/20	40.5596
940411	AE2-025 C	4.3731	80/20	4.3731
940412	AE2-025 E	20.4726	80/20	20.4726
942101	AE2-222 C O1	7.4657	80/20	7.4657
942102	AE2-222 E O1	18.7543	80/20	18.7543
942381	AE2-251 C	29.0720	80/20	29.0720
942382	AE2-251 E	74.3872	80/20	74.3872
943561	AF1-027	0.1803	80/20	0.1803
944332	AF1-101 E O1	36.9186	80/20	36.9186
944611	AF1-126 C O1	5.2980	80/20	5.2980
944612	AF1-126 E O1	21.1920	80/20	21.1920
944951	AF1-160 C	1.2866	80/20	1.2866
944952	AF1-160 E	1.2866	80/20	1.2866
945201	AF1-185 1	0.2280	80/20	0.2280
945211	AF1-185 2	0.0507	80/20	0.0507

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
945431	AF1-208 C O1	0.0001	80/20	0.0001
945432	AF1-208 E O1	1.2001	80/20	1.2001
945571	AF1-222 C	7.0989	80/20	7.0989
945572	AF1-222 E	18.7153	80/20	18.7153
945721	AF1-237 C	8.1480	80/20	8.1480
945722	AF1-237 E	0.0001	80/20	0.0001
945731	AF1-238 C	5.8946	80/20	5.8946
945732	AF1-238 E	8.8419	80/20	8.8419
945741	AF1-239 C	1.3870	80/20	1.3870
945742	AF1-239 E	2.0804	80/20	2.0804

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237121	206300	28OCEANVW	JCP&L	206286	28ATLANTIC	JCP&L	Y	JC-P2-3-JCC-230-47B	breaker	869.0	58.91	102.82	AC	392.55

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
939981	AE1-238 C	173.1834	50/50	173.1834
939982	AE1-238 E	454.8951	50/50	454.8951
945571	AF1-222 C	107.9510	50/50	107.9510
945572	AF1-222 E	284.5981	50/50	284.5981

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42237180	206300	28OCEANVW	JCP&L	206286	28ATLANTIC	JCP&L	X	JC-P2-3-JCC-230-47E	breaker	869.0	70.0	100.42	AC	271.67

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	14.0482	Adder	16.53
939981	AE1-238 C	119.8548	50/50	119.8548
939982	AE1-238 E	314.8186	50/50	314.8186
940161	AE2-000 C O1	11.4635	Adder	13.49
940162	AE2-000 E O1	29.3312	Adder	34.51
940401	AE2-024 C O1	9.4125	Adder	11.07
940402	AE2-024 E O1	44.0681	Adder	51.84
940411	AE2-025 C	4.7514	Adder	5.59
940412	AE2-025 E	22.2436	Adder	26.17
944332	AF1-101 E O1	28.7563	Adder	33.83
944611	AF1-126 C O1	2.2431	Adder	2.64
944612	AF1-126 E O1	8.9726	Adder	10.56
945571	AF1-222 C	74.7095	50/50	74.7095
945572	AF1-222 E	196.9614	50/50	196.9614

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42236940	206294	28LARRABEE	JCP&L	206309	28SMITHBRG	JCP&L	2	JC-P2-3-JCC-230-15F	breaker	813.0	114.11	124.25	AC	84.97

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	280 C GEN (Deactivation : 17/09/2018)	67.8581	50/50	67.8581
938421	AE1-061 C	0.1293	Adder	0.15
938422	AE1-061 E	0.1293	Adder	0.15
938781	AE1-104 C O1	6.5257	Adder	7.68
938782	AE1-104 E O1	16.6960	Adder	19.64
939981	AE1-238 C	37.4850	50/50	37.4850
939982	AE1-238 E	98.4606	50/50	98.4606
940161	AE2-000 C O1	55.3732	50/50	55.3732
940162	AE2-000 E O1	141.6810	50/50	141.6810
940361	AE2-020 C	8.5442	50/50	8.5442
940362	AE2-020 E	40.0044	50/50	40.0044
940371	AE2-021 C	8.5442	50/50	8.5442
940372	AE2-021 E	40.0044	50/50	40.0044
940381	AE2-022 C	4.9841	50/50	4.9841
940382	AE2-022 E	23.3359	50/50	23.3359
940401	AE2-024 C O1	46.2374	50/50	46.2374
940402	AE2-024 E O1	216.4786	50/50	216.4786
940411	AE2-025 C	23.3406	50/50	23.3406
940412	AE2-025 E	109.2684	50/50	109.2684
942101	AE2-222 C O1	5.4223	Adder	6.38
942102	AE2-222 E O1	13.6211	Adder	16.02
942381	AE2-251 C	27.0677	50/50	27.0677
942382	AE2-251 E	69.2587	50/50	69.2587
943561	AF1-027	0.3099	50/50	0.3099
944332	AF1-101 E O1	138.9039	50/50	138.9039
944611	AF1-126 C O1	10.4795	50/50	10.4795
944612	AF1-126 E O1	41.9180	50/50	41.9180
945201	AF1-185 1	0.2617	Adder	0.31
945211	AF1-185 2	0.0581	Adder	0.07
945571	AF1-222 C	23.3657	50/50	23.3657
945572	AF1-222 E	61.6004	50/50	61.6004
945731	AF1-238 C	2.1835	Adder	2.57
945732	AF1-238 E	3.2753	Adder	3.85
945741	AF1-239 C	0.5138	Adder	0.6
945742	AF1-239 E	0.7707	Adder	0.91

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42236930	206314	28RED OAKA	JCP&L	206305	28RAR RVR	JCP&L	1	JC-P2-3-JCC-230-26B	breaker	869.0	107.21	121.06	AC	122.09

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	35.3813	50/50	35.3813
938781	AE1-104 C O1	4.5073	Adder	5.3
938782	AE1-104 E O1	11.5320	Adder	13.57
939981	AE1-238 C	53.8650	50/50	53.8650
939982	AE1-238 E	141.4854	50/50	141.4854
940161	AE2-000 C O1	28.8716	50/50	28.8716
940162	AE2-000 E O1	73.8726	50/50	73.8726
940361	AE2-020 C	4.6750	Adder	5.5
940362	AE2-020 E	21.8886	Adder	25.75
940371	AE2-021 C	4.6750	Adder	5.5
940372	AE2-021 E	21.8886	Adder	25.75
940381	AE2-022 C	2.7271	Adder	3.21
940382	AE2-022 E	12.7683	Adder	15.02
940401	AE2-024 C O1	23.4782	50/50	23.4782
940402	AE2-024 E O1	109.9225	50/50	109.9225
940411	AE2-025 C	11.8518	50/50	11.8518
940412	AE2-025 E	55.4838	50/50	55.4838
942101	AE2-222 C O1	3.5572	Adder	4.18
942102	AE2-222 E O1	8.9358	Adder	10.51
942381	AE2-251 C	14.8102	Adder	17.42
942382	AE2-251 E	37.8952	Adder	44.58
943561	AF1-027	0.2162	50/50	0.2162
944332	AF1-101 E O1	72.4246	50/50	72.4246
944611	AF1-126 C O1	5.7640	50/50	5.7640
944612	AF1-126 E O1	23.0560	50/50	23.0560
945571	AF1-222 C	33.5759	50/50	33.5759
945572	AF1-222 E	88.5182	50/50	88.5182

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ID	FROM BUS#	FROM BUS	FROM BUS AREA	TO BUS#	TO BUS	TO BUS AREA	CKT ID	CONT NAME	Type	Rating MVA	PRE PROJECT LOADING %	POST PROJECT LOADING %	AC DC	MW IMPACT
42236925	206315	28RED OAKB	JCP&L	206305	28RAR RVR	JCP&L	1	JC-P2-3-JCC-230-26D	breaker	869.0	108.33	121.96	AC	121.55

Bus #	Bus	Gendeliv MW Impact	Type	Full MW Impact
206325	28O C GEN (Deactivation : 17/09/2018)	35.2104	50/50	35.2104
938781	AE1-104 C O1	4.4776	Adder	5.27
938782	AE1-104 E O1	11.4559	Adder	13.48
939981	AE1-238 C	53.6256	50/50	53.6256
939982	AE1-238 E	140.8566	50/50	140.8566
940161	AE2-000 C O1	28.7322	50/50	28.7322
940162	AE2-000 E O1	73.5159	50/50	73.5159
940361	AE2-020 C	4.6460	Adder	5.47
940362	AE2-020 E	21.7530	Adder	25.59
940371	AE2-021 C	4.6460	Adder	5.47
940372	AE2-021 E	21.7530	Adder	25.59
940381	AE2-022 C	2.7102	Adder	3.19
940382	AE2-022 E	12.6893	Adder	14.93
940401	AE2-024 C O1	23.3689	50/50	23.3689
940402	AE2-024 E O1	109.4109	50/50	109.4109
940411	AE2-025 C	11.7966	50/50	11.7966
940412	AE2-025 E	55.2256	50/50	55.2256
942101	AE2-222 C O1	3.5345	Adder	4.16
942102	AE2-222 E O1	8.8789	Adder	10.45
942381	AE2-251 C	14.7185	Adder	17.32
942382	AE2-251 E	37.6605	Adder	44.31
943561	AF1-027	0.2151	50/50	0.2151
944332	AF1-101 E O1	72.0749	50/50	72.0749
944611	AF1-126 C O1	5.7345	50/50	5.7345
944612	AF1-126 E O1	22.9380	50/50	22.9380
945571	AF1-222 C	33.4266	50/50	33.4266
945572	AF1-222 E	88.1247	50/50	88.1247

12.8 Queue Dependencies

The Queue Projects below are listed in one or more indices for the overloads identified in your report. These projects contribute to the loading of the overloaded facilities identified in your report. The percent overload of a facility and cost allocation you may have towards a particular reinforcement could vary depending on the action of these earlier projects. The status of each project at the time of the analysis is presented in the table. This list may change as earlier projects withdraw or modify their requests.

Queue Number	Project Name	Status
AD2-077	Buxmont 69 kV	Active
AD2-213	East Flemington-Lebanon 34.5 kV	Active
AE1-037	Deans 500kV	Withdrawn
AE1-061	Minotola 12 kV	Active
AE1-062	Silver Lake 69 kV	Active
AE1-104	BL England 138 kV	Active
AE1-115	Churchtown 69 kV	Active
AE1-161	Landis 138 kV	Active
AE1-238	Oceanview Wind 230 kV	Active
AE2-000	N/A	N/A
AE2-020	Cardiff 230 kV I	Active
AE2-021	Cardiff 230 kV II	Active
AE2-022	Cardiff 230 kV III	Active
AE2-024	Larrabee 230 kV I	Active
AE2-025	Larrabee 230 kV II	Active
AE2-222	Higbee 69 kV	Active
AE2-251	Cardiff 230 kV	Active
AF1-027	Plumsted 537 Energy Storage (CIRs)	Active
AF1-101	Oyster Creek 230 kV III	Active
AF1-108	East Flemington 34.5 kV	Active
AF1-109	Pleasant Valley 230 kV	Active
AF1-126	Cookstown 230 kV	Active
AF1-160	Silver Lake 69 kV	Active
AF1-185	Sayreville 1-2-3 230 kV	In Service
AF1-208	Quinton-Roadstown 69 kV	Active
AF1-222	Oceanview Wind 2 230 kV	Active
AF1-237	Mercer 230 kV	Active
AF1-238	Sherman Ave. 69 kV	Active
AF1-239	Sherman Ave-Vineland 69 kV	Active

12.9 Contingency Descriptions

Contingency Name	Contingency Definition
JC-P2-3-JCC-230-15F	CONTINGENCY 'JC-P2-3-JCC-230-15F' /* LARRABEE - SMITHBURG (D2004) & LARRABEE NW BUS 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 1 /* LARRABEE- SMITHBURG (D2004) REMOVE LOAD 7 FROM BUS 206294 /* LARRABEE BK7 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 9 /* LARRABEE BK9 END
PS_P7-1_1LINE+2LINE	CONTINGENCY 'PS_P7-1_1LINE+2LINE' /* LAKE NELSON - RARITIAN X2 DISCONNECT BUS 218331 /* REMOVE KILMER 1 DISCONNECT BUS 218332 /* REMOVE KILMER 2 MOVE 8 MW LOAD FROM BUS 218384 TO BUS 218383 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO GREENBK T1 MOVE 8 MW LOAD FROM BUS 218387 TO BUS 218399 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO NEWDOVR T2 MOVE 8 MW LOAD FROM BUS 218385 TO BUS 218393 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO MDWRD T2 MOVE 8 MW LOAD FROM BUS 218386 TO BUS 218347 /* INTERSTATION TIE TRANSFER LOAD FROM KILMER TO BNNTLANE T2 END
PJM500_PS_P2-3_DEAN5_5-6	CONTINGENCY 'PJM500_PS_P2-3_DEAN5_5-6' DISCONNECT BRANCH FROM BUS 200006 TO BUS 218306 CKT 2/* BRANCHBURG TX -2 TRIP LINE FROM BUS 200006 TO BUS 200028 CKT 1 /* DEANS TO WINSOR END
JC-P2-3-JCC-230-15A	CONTINGENCY 'JC-P2-3-JCC-230-15A' /* LAKEWOOD-LARRABEE (K2011) & LARRABEE-SMITHBURG (H2008) 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206323 CKT 1 /* LARRABEE - LAKEWOOD (K2011) DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 2 /* LARRABEE- SMITHBURG (H2008) END

Contingency Name	Contingency Definition
PS_P7-1_I1023+GBK-LN_LT	CONTINGENCY 'PS_P7-1_I1023+GBK-LN_LT' /* LAKE NELSON - GILLETE BRIDGEWATER & LAKE NELSON -GREENBROOK DISCONNECT BUS 218312 /* GREENBROOK I DISCONNECT BUS 218301 /* MIDDLESEX I DISCONNECT BUS 218333 /* LAKE NELSON I DISCONNECT BUS 218334 /* LAKE NELSON W DISCONNECT BUS 218522 /* MIDDLESEX W CLOSE LINE FROM BUS 218382 TO BUS 218383 CKT Z /* GREENBROOK MOVE 8 MW LOAD FROM BUS 218382 TO BUS 218406 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO SOMRVLE T2 MOVE 8 MW LOAD FROM BUS 218382 TO BUS 218386 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO KILMER T1 MOVE 8 MW LOAD FROM BUS 218383 TO BUS 218384 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO KILMER T2 MOVE 8 MW LOAD FROM BUS 218383 TO BUS 218395 /* INTERSTATION TIE TRANSFER LOAD FROM GREENBK TO SOUTH SECOND ST.(METUCHEN 26KV IM) END
JC-P2-3-JCC-230-26B	CONTINGENCY 'JC-P2-3-JCC-230-26B' /* RARITAN RIVER B144 STUCK BREAKER DISCONNECT BUS 206315 /* RED OAK 230 KV B TAP DISCONNECT BUS 206362 /* RED OAK 230 KV #1 DISCONNECT BUS 206365 /* RED OAK 230 KV #4 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 13 /* RARITAN RIVER #13 230/115 KV TRANSFORMER DISCONNECT BUS 206403 /* WOODBRIDGE GENERATOR END
JC-P1-2-JCC-230-027	CONTINGENCY 'JC-P1-2-JCC-230-027' /* RARITAN RIVER - RED OAK - R11 (G1047) 230 KV LINE DISCONNECT BRANCH FROM BUS 206305 TO BUS 206315 CKT 1 DISCONNECT BRANCH FROM BUS 206315 TO BUS 206410 CKT 1 DISCONNECT BUS 206315 DISCONNECT BUS 206362 DISCONNECT BUS 206365 END
PJM500_PS_P1-2_5015	CONTINGENCY 'PJM500_PS_P1-2_5015' /* HOPE CREEK TO RED LION TRIP LINE FROM BUS 200027 TO BUS 200029 CKT 1 /* HOPE CREEK TO RED LION END
JC-P2-3-JCC-230-026A	CONTINGENCY 'JC-P2-3-JCC-230-026A' /* RARITAN RIVER B143 STUCK BREAKER DISCONNECT BUS 206315 /* RED OAK 230 KV B TAP DISCONNECT BUS 206362 /* RED OAK 230 KV #1 DISCONNECT BUS 206365 /* RED OAK 230 KV #4 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 17 /* RARITAN RIVER #17 230/115 KV TRANSFORMER DISCONNECT BUS 206350 /* SAYREVILLE GENERATOR DISCONNECT BUS 206351 /* SAYREVILLE GENERATOR END

Contingency Name	Contingency Definition
JC-P2-3-JCC-230-47E	CONTINGENCY 'JC-P2-3-JCC-230-47E' /* OCEANVIEW BK2 230 KV FUTURE OCEANVIEW RING BUS & ATLANTIC - OCEANVIEW (Y2025) DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT Y /* ATLANTIC - OCEANVIEW (Y2025) DISCONNECT BRANCH FROM BUS 206300 TO BUS 206273 CKT 2 /* OCEANVIEW BK2 REDUCE BUS 206273 SHUNT BY 50 PERCENT /* OCEANVIEW CAPS ON 34.5 KV BUS B END
JC-P2-3-JCC-230-47B	CONTINGENCY 'JC-P2-3-JCC-230-47B' /* LARRABEE - OCEANVIEW (NEW) & ATLANTIC - OCEANVIEW (X2024) DISCONNECT BRANCH FROM BUS 206294 TO BUS 206300 CKT 1 /* LARRABEE - OCEANVIEW (NEW) DISCONNECT BRANCH FROM BUS 206294 TO BUS 206274 CKT 3 /* LARRABEE BK3 DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT X /* ATLANTIC - OCEANVIEW (X2024) END
JC-P2-3-JCC-230-47C	CONTINGENCY 'JC-P2-3-JCC-230-47C' /* ATLANTIC - OCEANVIEW (X2024) & OCEANVIEW BK2 230 KV FUTURE OCEANVIEW RING BUS DISCONNECT BRANCH FROM BUS 206286 TO BUS 206300 CKT X /* ATLANTIC - OCEANVIEW (X2024) DISCONNECT BRANCH FROM BUS 206300 TO BUS 206273 CKT 2 /* OCEANVIEW BK2 REDUCE BUS 206273 SHUNT BY 50 PERCENT /* OCEANVIEW CAPS ON 34.5 KV BUS B DISCONNECT BRANCH FROM BUS 206273 TO BUS 206929 CKT 1 /* OCEANVIEW - ALLENHURST (H216) 34.5KV END
JC-P1-2-JCC-230-018	CONTINGENCY 'JC-P1-2-JCC-230-018' /* LARRABEE - SMITHBURG (H2008) 230 KV DISCONNECT BRANCH FROM BUS 206309 TO BUS 206294 CKT 2 DISCONNECT BRANCH FROM BUS 200017 TO BUS 206309 CKT 4 /* SMITHBURG SMITHBURG 500 230 END
Base Case	
JC-P1-2-JCC-230-017	CONTINGENCY 'JC-P1-2-JCC-230-017' /* FUTURE BREAKERS LARRABEE - SMITHBURG (D2004) 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 1 END
PJM500_PS_P1-2_5022	CONTINGENCY 'PJM500_PS_P1-2_5022' /* DEANS TO WINDSOR TRIP LINE FROM BUS 200006 TO BUS 200028 CKT 1 /* DEANS TO WINDSOR END

Contingency Name	Contingency Definition
JC-P2-3-JCC-230-26D	CONTINGENCY 'JC-P2-3-JCC-230-26D' /* RARITAN RIVER B146 STUCK BREAKER DISCONNECT BUS 206314 /* RED OAK 230 KV A TAP DISCONNECT BUS 206363 /* RED OAK 230 KV #2 DISCONNECT BUS 206364 /* RED OAK 230 KV #3 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 13 /* RARITAN RIVER #13 230/115 KV TRANSFORMER DISCONNECT BUS 206403 /* WOODBRIDGE GENERATOR END
JC-P2-3-JCC-230-15G	CONTINGENCY 'JC-P2-3-JCC-230-15G' /* LARRABEE-SMITHBURG (H2008) 230 KV & LARRABEE NW BUS 230 KV DISCONNECT BRANCH FROM BUS 206294 TO BUS 206309 CKT 2 /* LARRABEE- SMITHBURG (H2008) REMOVE LOAD 7 FROM BUS 206294 /* LARRABEE BK7 DISCONNECT BRANCH FROM BUS 206294 TO BUS 206275 CKT 9 /* LARRABEE BK9 END
PJM_P4_P484B	CONTINGENCY 'PJM_P4_P484B' /* VALID FROM 2009 DISCONNECT BRANCH FROM BUS 200028 TO BUS 200006 CKT 1 /* WINDSOR DEANS 500500 DISCONNECT BRANCH FROM BUS 200006 TO BUS 218306 CKT 1 /* DEANS XF 2 500230 END
JC-P2-3-JCC-230-026C	CONTINGENCY 'JC-P2-3-JCC-230-026C' /* RARITAN RIVER B145 STUCK BREAKER DISCONNECT BUS 206314 /* RED OAK 230 KV A TAP DISCONNECT BUS 206363 /* RED OAK 230 KV #2 DISCONNECT BUS 206364 /* RED OAK 230 KV #3 DISCONNECT BRANCH FROM BUS 206305 TO BUS 206303 CKT 17 /* RARITAN RIVER #17 230/115 KV TRANSFORMER DISCONNECT BUS 206350 /* SAYREVILLE GENERATOR DISCONNECT BUS 206351 /* SAYREVILLE GENERATOR END
JC-P1-2-JCC-230-028	CONTINGENCY 'JC-P1-2-JCC-230-028' /* RARITAN RIVER - RED OAK - SOUTH RIVER JCT(T1034) 230 KV DISCONNECT BRANCH FROM BUS 206305 TO BUS 206314 CKT 1 /* 28RAR RVR 230.00 28RED OAKA 230.00 DISCONNECT BRANCH FROM BUS 206314 TO BUS 206411 CKT 1 /* 28RED OAKA 230.00 28R11RINGA 230.00 . DISCONNECT BUS 206314 /* 28RED OAKA 230.00 DISCONNECT BUS 206363 /* 28RDOAKCT2 18.00 DISCONNECT BUS 206364 /* 28RDOAKCT3 18.00 END

13 Short Circuit Analysis

The following Breakers are overdutied:

None

14 Stability and Reactive Power

(Summary of the VAR requirements based upon the results of the dynamic studies)

To be determined in the Facilities Study Phase.

15 Affected Systems

Not required.

16 Attachment 1: One Line Diagram

